

Facing Unequal Opportunities: Does Experience Shape Redistributive Preferences?

Martin Brun & Andrea Pogliano*

SEPTEMBER, 2026

We study whether firsthand experience of unequal opportunity changes how people judge merit and redistribute income. In the main experiment, third-party spectators are randomly assigned to complete either an easy or a hard version of a real-effort task, then decide how to divide a bonus between two workers, one of whom faced each version. By holding the workers' circumstances fixed while varying spectators' own experience, the design isolates the causal effect of experiencing advantage versus disadvantage. Spectators who complete the hard task attribute performance more to unequal opportunity, view the initial allocation as less fair, and redistribute more toward the disadvantaged worker, even when doing so is personally costly. An extension compares one-sided experience with experiencing both tasks and with receiving comparable information through reports or statistics. Experiencing both sides produces the strongest redistributive response, while information alone has weaker effects. Redistributive preferences thus depend not only on what people know about unequal opportunities, but also on how they encounter them.

JEL Classification: D91, D63, D83, D84.

Keywords: experience, preferences for redistribution, fairness, inequality of opportunities.

***Affiliations:** Brun: Tampere University, Finnish Centre of Excellence in Tax Systems Research, and EQUALITAS. Pogliano: Erasmus University Rotterdam and Tinbergen Institute. **Acknowledgements:** We thank Georg Granic, Stefan Lipman, Yiming Liu, Topi Miettinen, Juho Nuutinen, Kirsten Rohde, Matthew Robson, Morten N. Støstad, Bertil Tungodden, Andreas Ziegler, Chen Li and seminar participants at 1st EER Summer School in Experimental and Behavior Economics, Helsinki GSE Behavioral Brown Bag, Norwegian School of Economics, 41st Summer Seminar of Finnish Economists, EUR BOM Seminar, and ASFEE 2026 for helpful comments and discussions. All remaining errors are our own. **Funding:** Brun acknowledges support from the Research Council of Finland (project 364188). Pogliano acknowledges support from the Smarter Choice for Better Health Initiative and ERIM. **Ethics:** The procedures for this study received approval from Erasmus School of Economics previous to its implementation, with reference number ETH2425-0577. **Research transparency:** The study was preregistered at the AEA RCT Registry as #AEARCTR-0017214.

1 Introduction

Fairness views shape whether people regard inequalities as acceptable and how they want such inequalities to be addressed (Cappelen et al., 2007, 2013; Konow, 2000; Stantcheva, 2021). In modern societies, the dominant view is meritocratic. Inequalities are more accepted when they result from individual effort and choices, but considered less fair when they arise from circumstances beyond individual control, such as random luck (Almås et al., 2020; Almås et al., 2025). Yet meritocratic judgments become difficult when individuals face unequal opportunities. In such settings, observed performance may no longer reveal effort or ability. The same effort can produce very different outcomes depending on the conditions individuals face and effort itself is often shaped by opportunities (Altmejd et al., 2021; Bursztyn et al., 2017; Falk et al., 2026; Glover et al., 2017; Parsons et al., 2011). Poor access to training, discrimination, or a weak labor market may both make success harder to attain and discourage hard work, whereas advantageous opportunities can make success easier and effort more rewarding. This creates a fundamental dilemma for societies that aspire to be meritocratic. How should outcomes be rewarded when performance reflects not only effort, but also the unequal opportunities through which effort is translated into success?

Recent experimental work has shown that individuals exhibit a puzzling insensitivity to unequal opportunities (Andre, 2024; Bhattacharya and Mollerstrom, 2025; Brun, 2026; Cappelen et al., 2024; Dong et al., 2024; Pogliano, 2024; Pruess et al., 2025). This pattern reflects not only fairness views, but also imperfect beliefs about how opportunities shape performance. Individuals tend to underweight the constraints faced by others and over-attribute outcomes to effort, consistent with attribution bias and misspecified mental models of what determines outcomes (Jones and Harris, 1967). Personal experiences may provide a significant source of information that could potentially affect these beliefs. In other settings, experiences have been shown to shape beliefs about important economic outcomes, such as inflation, unemployment, or the performance of financial markets (Busse et al., 2015; Conlin et al., 2007; Kuchler and Zafar, 2019; Malmendier and Nagel, 2016; Malmendier, 2021a). Whether experience with unequal opportunity environments affects beliefs about the role of opportunities in performance and shifts redistributive decisions remains an open question.

In this paper, we address these questions by implementing a two-phase spectator game experiment (Almås et al., 2020; Cappelen et al., 2007, 2013). In the first phase of the experiment, participants we call “agents” complete a visual counting task under unequal opportunities. Agents are randomly assigned to one of two versions of the task. In both versions, they count a target letter in a grid of similar-looking characters. The versions differ only in the target letter. In the hard version, the target is more similar to the surrounding characters, making counting more difficult. We deliberately design the difference between the two versions to be difficult to infer from the instructions and example alone, while making its effects cumulative and consequential for performance (DiPrete and Eirich, 2006). On average, agents assigned to the easy version outperform those assigned to the hard version by 72%. We then match agents in pairs, featuring one agent assigned to each version of the task, and assign the best performing agent with a bonus payment. In more than 90% of the cases, the bonus is assigned to the agent completing the easy

version of the task.

In the second phase of our experiment, a different set of participants we call ‘spectators’ learn about both versions of the counting task, report their beliefs about the determinants of performance in the task, observe a pair of agents, and decide whether and to what extent to redistribute earnings. In our main analysis, we randomly assign spectators to experience either the easy version (**EASY** treatment, also referred to as the experience of advantage) or hard version of the task (**HARD** treatment, also referred to as the experience of disadvantage), in an incentivized way for four minutes. This intervention takes place after spectators learn about the different versions of the task. We elicit beliefs about task performance before and after the experience. This allows us to test whether experience shifts beliefs on the determinants of task success. We find that before experiencing the task, all spectators acknowledge the procedural differences between versions but only around 15% expect any impact on performance due to those differences. After the experience, these beliefs change with very different intensities. We find that the share of spectators anticipating an advantage for agents assigned to the easy version climbs by 22 p.p. in **EASY** and by 41 p.p. in **HARD**.

We then explore whether experiencing unequal opportunities changes redistributive preferences. In our experiment, each respondent makes two consequential redistribution decisions, one that entails no personal cost (which we call ‘costless’), and one that carries a fixed cost (which we call ‘costly’). We start by focusing on the costless decisions, which feature first in our experiment. We find that spectators who experience the easy version of the task assign \$1.53 to the disadvantaged agent (i.e., the agent that completed the hard version), while those who experience the hard version assign \$1.80 to the disadvantaged agent. This treatment effect amounts to a 18% difference in our main specification, which accounts for socio-demographic covariates. Our experimental design allows us to explore different plausible explanations for the higher redistribution observed among spectators assigned to **HARD**. We show that self-serving biases are unlikely to drive this result. Instead, the evidence points to changes in how spectators interpret the source of performance differences. In particular, beliefs about the relevance of luck and unequal opportunities in explaining agents’ performance explain a large share of the treatment effect.

Next, we explore whether redistribution decisions persist if they carry a fixed cost for the decision-maker. We find that around half of the spectators keep redistributing when it entails a personal cost to do so. However, those experiencing disadvantageous opportunities themselves are 9 p.p. more likely to keep redistributing, and assign 37% more to the disadvantaged agent. We exploit the information collected for both the costless and costly decisions to estimate a structural model and assess the weight assigned to fairness among spectators in each treatment arm. We find that spectators who experience advantageous opportunities care little about fairness if it entails a cost on themselves. In turn, spectators assigned to **HARD** are willing to pay \$0.21 for each \$1.00 reduction in an inequality they perceive as unfair.

Our main analysis shows spectators assigned to disadvantageous opportunities redistribute more and are more willing to incur a cost than spectators assigned to advantageous opportunities. We then assess which

aspects of the experience drive this result. Specifically, we study whether receiving information through other channels is enough or if the experience *per se* is necessary for learning and for behavioral change. To do so, we run an extension with three additional treatment arms. First, we assign spectators to complete both versions of the task. This intervention is the most complete source of information we test, as it provides a direct exposure to the opportunity gap. We find that the amount redistributed is substantially higher than in the previous treatments. On average, spectators transfer \$2.13 to the disadvantaged agent after experiencing both sides of the opportunity structure. This is 40% above the mean in **EASY** and 22% above that in **HARD**. We use this treatment arm as benchmark to compare two additional interventions that provide direct information on the impacts of the opportunity gap. We present the spectators with qualitative and quantitative information about agents' experiences. Redistribution decisions among both treatment arms fall short when compared with what happened for spectators who experience both sides of the opportunity structure. Still, spectators receiving qualitative information transfer 24% more than those experiencing only advantaged opportunities. In turn, transfers by spectators receiving quantitative information are similar to the ones by spectators who experience advantaged opportunities.

Related Literature. This paper builds on and contributes to several strands of the literature. First, we add to the expanding literature on fairness and inequality acceptance under unequal opportunities (Andre, 2024; Bhattacharya and Mollerstrom, 2025; Brun, 2026; Cappelen et al., 2024; Dong et al., 2024; Pogliano, 2024; Pruess et al., 2025). Across this literature, a common design feature is that studies vary the opportunity structure workers face, or the information spectators receive about it, while holding spectators' own exposure to those opportunities fixed. The resulting evidence shows that spectators only partially account for unequal opportunities when evaluating merit and correcting inequalities. Dong et al. (2024) show that redistribution responds to the nature and magnitude of opportunity differences, although this response remains incomplete. Cappelen et al. (2024) trace this incompleteness to two distinct sources: a failure to recognize how opportunities shape performance and a continued willingness to reward performance even when this relationship is recognized. Andre (2024) shows that spectators continue to reward choices shaped by unequal circumstances partly because they remain uncertain about the relevant equal opportunity counterfactual. Bhattacharya and Mollerstrom (2025) similarly show that people reward work even when luck determines who receives the opportunity to work in the first place. Brun (2026) shows that accounting for the relevant counterfactual is cognitively demanding, with more cognitively able individuals better equipped to make the inferences needed to account for unequal opportunities. Pogliano (2024) shows that these judgments are also shaped by beliefs about the origins of ability, with beliefs about the importance of genetic endowments affecting redistributive decisions. Pruess et al. (2025) show that spectators underestimate how small relative advantages compound over the course of a task, leading them to redistribute less when inequality stems from unequal opportunities than when luck acts directly on outcomes. We complement this literature by varying spectators' own exposure to the opportunity structure while holding the workers and distributive problem fixed. We show that imperfect beliefs help explain this unresponsiveness to opportunity gaps, as direct experience changes how spectators interpret performance differences and increases compensation of the disadvantaged worker.

Moreover, experiencing both sides of the opportunity gap produces stronger redistributive responses than receiving qualitative or quantitative information about it. This sheds light on how beliefs enter redistributive decisions, and on how task-specific fairness judgments are formed.

We also contribute to the emerging literature on costly fairness and the willingness to implement redistribution (Dimant et al., 2024; Dwenger et al., 2024; Piraino and Ryan, 2024). Our magnitudes are comparable to Piraino and Ryan (2024): for a similarly modest personal cost, roughly half to two-thirds of those who redistribute when it is free continue to do so once redistribution becomes costly. Our contribution is to show that this willingness to act on fairness views is itself malleable. This speaks directly to the recent call to distinguish the fairness ideal individuals hold from the behavioral weight they place on that ideal when acting on it entails personal costs (?). To our knowledge, we are the first to empirically examine how this willingness changes due to interventions and to explore its magnitude in a setting featuring unequal opportunities.

Second, we contribute to the literature focusing on how experiences shape meritocratic beliefs (Amasino et al., 2024; Cassar and Klein, 2019; Deffains et al., 2016; Espinosa et al., 2020; Fehr and Vollmann, 2025; Jia, 2022). Previous findings have focused on self-serving biases and motivated reasoning that arise from personal experience of success or failure. Our study departs from this research by isolating the effect of imperfect beliefs. We show that even with no self-interested motives in play, experiencing disadvantageous opportunities can change beliefs and drive people towards more redistributive decisions.

Finally, we connect to the literature on experience effects and belief formation (Kuchler and Zafar, 2019; Malmendier and Nagel, 2016; Malmendier, 2021a; Malmendier and Wachter, 2024). We extend this literature from the domain of macroeconomic expectations to moral reasoning, showing that firsthand experiences can reshape related behaviors. Moreover, our comparison between experiences and information provision relates to Graeber et al. (2024), who document that qualitative information is more impactful than similar quantitative information. We replicate that finding and extend it by showing that direct personal experience is even stronger than both, consistent with evidence on the neural mechanisms that underlie learning (Chen et al., 2026).

The paper is structured as follows. Section 2 presents the experimental design and procedure. Sections 3, 4, and 5 present the results of our study, analyzing treatment differences in beliefs and behaviors. Section 6 discusses the main findings, relates them to the literature, and Section 7 concludes.

2 Experimental design

We adopt the spectator design by Cappelen et al. (2007), which comprises an Agent and a Spectator Phase. This design takes place in two phases. In the first, two participants, called ‘agents’, perform a certain task and are assigned initial rewards based on their performance. In the second, a third subject, called ‘spectator’, is informed about the agents’ task and initial reward allocation, then given an opportunity to reallocate the earnings between the two agents. The primary variable of interest is the spectator’s

redistribution decision. We now discuss both phases in detail.¹

2.1 Agents

Agents complete an effort task with different levels of difficulty (as in Deffains et al., 2016; Fehr and Vollmann, 2025). The effort task consists of a counting task (as introduced in Abeler et al., 2011). Agents count how many times a character appears in a table like the one below (see Figure 1). After receiving general instructions, we assign each agent to perform either an easy version of the task or a hard version of the task. In the instructions shown to participants, we refer to the two task versions only as Version A and Version B, rather than as the easy and hard versions. For the easy version, we assign agents to count letters As. For the hard version, we assign agents to count letters Bs. We make counting Bs relatively harder by both including decoys that can be confused with Bs and slightly increasing the size of As.² This feature of our task is not immediately apparent, mimicking the hidden nature of structural barriers and privileges in society.

¹ We implement the experiment using the oTree platform (Chen et al., 2016).

² We also impose a penalty for incorrect responses within a given table. After the first incorrect attempt, agents must wait 20 seconds before submitting again. After the second incorrect attempt, agents receive a new table. The purpose of these penalties is to discourage guessing. We expect the penalties to fall largely on agents assigned to the HARD version. We find that that is indeed the case. In Version A, participants complete correctly 86% of the tasks they attempted, in Version B 49%.

Figure 1: Counting Task Example

B	B	β	a	a	A	β	I	s	I
I	I	I	s	a	A	A	a	s	I
B	I	β	β	A	β	β	β	A	a
I	β	s	a	s	I	β	β	I	a
s	β	a	a	β	β	s	B	A	β
β	s	β	s	β	s	β	I	β	a
s	B	β	β	s	A	I	β	A	s
β	B	A	A	A	β	β	s	A	I
A	β	s	A	I	A	B	A	A	s
β	a	B	I	β	B	I	A	A	s

Notes: This figure depicts an example of the table used in the counting task. The task is to count how often a specific character appears in the table. The goal is to solve as many tables as possible: the final score is the total number of tables solved correctly. Each table is filled with characters. The participants must count how many times a target character appears and submit the count. When the answer is correct, a new table appears with new characters in shuffled positions. There are two versions of the task. The type of tables are the same in both versions, only the target character differs. In Version A, the participant counts how many times uppercase “A” appears in the table (e.g., 17 in the example below). In Version B, the participant count how many times uppercase “B” appears in the table (e.g., 14 in the example below).

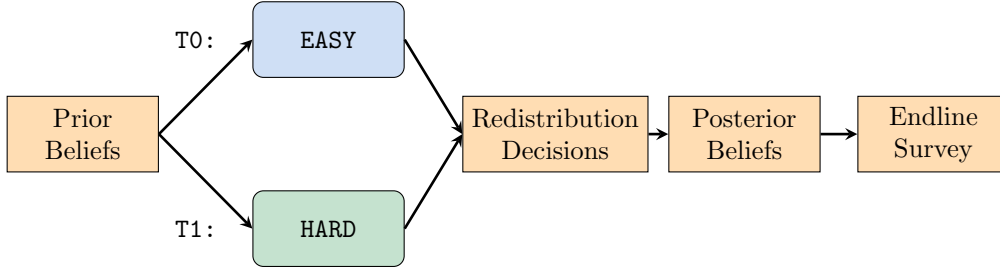
We inform agents that better performance can increase their chances of getting an additional payment. Agents have 4 minutes to complete as many tables as possible. Agents receive a fixed participation fee and can receive a bonus payment determined by the spectators from Section 2.2.

2.2 Spectators

The spectator phase consists of three main parts. In the first (experience) part, spectators work on effort tasks. In the second part, participants act as third party spectators: we elicit their beliefs about agents and decide how to redistribute the agents’ income. In the third phase, participants answer socio-demographics questions. Figure 2 shows the complete experiment flow. See Table A2 for the full set of questions included.

First, we explain spectators the same counting task as the one completed by the agents (see Figure 1). The versions of the tasks are always referred as version A (i.e., the easy version) and version B (i.e., the

Figure 2: Experiment Flow



Notes: This figure depicts the experiment flow for the Spectator phase.

hard version). The explanation does not disclose which version of the task each spectator will practice with. We require all spectators to respond correctly to a set of comprehension questions before advancing.

Next, we ask spectators three questions to measure their prior beliefs. We focus on perceived differences between the versions of the task and on determinants of success in a competition where individuals are assigned two different versions of the task, asking spectators to rank determinants (effort, ability, version of the task) from most to least important.³

Experience. We inform spectators they will have a chance to get familiar with the task. We refer to this part as the experience. We neither mention nor implement performance incentives. Nevertheless, we invite spectators to try to solve as many tables as possible within 4 minutes (i.e. the same time given to the agents). Because spectators' performance during the experience affects does not affect their own earnings, the design limits incentives to justify a favorable personal outcome and the scope for self-serving beliefs. We randomly assign spectators either to practice with the easy version or with the hard version. This is the main intervention of our study and creates two treatment groups: **EASY** and **HARD**.

We include an attention check in the middle of the experience. We record both the answer submitted and the time taken to answer to it. We also ask spectators to report the version of the task they practiced with at the end of the experience. After the experience is over, we inform spectators that they get a \$1.00 bonus payment to reward their effort so far.

Redistribution Decisions. After the experience, we introduce spectators to agents from Section 2.1. We explain that agents were randomly assigned to either the easy or hard version of the task. To ensure correct understanding, we require correct response of three comprehension questions before proceeding. We elicit spectators' beliefs about agents at group-level (i.e., not about a specific pair of agents).⁴ We then present a pair of agents. The pair is composed of one agent assigned to the easy version of the task

³ The exact wording of these questions can be found in Table A2.

⁴ We ask: (1) the average number of tasks completed by agents assigned to the easy version; (2) the average number of tasks completed by agents assigned to the hard version; and (3) the likelihood that an agent assigned to the easy version outperforms an agent assigned to the hard version. For each belief, we also ask the perceived confidence. We inform spectators that one of the three estimates will be compared to the results of our agent's study, and that they will get a bonus if their estimate is correct.

and one assigned to the hard version. A \$6.00 bonus payment is initially assigned to the agent within the pair that completed more correct tasks. We create pairs in such a way that the agent that completed the easy version is always assigned the initial bonus. We ask spectators to decide whether and how much of the initial bonus to redistribute. We include three comprehension questions and an attention check before the first redistribution decision. We show spectators the conditions to which each agent was randomly assigned, which agent performed better and the initial bonus distribution.⁵

The decision about the redistribution for the pair of agents is our main outcome. It is followed by an open text form, in which we ask spectators to explain their reasoning. We then introduce a second redistribution decision for a new pair of agents. The structure is identical to the first decision: each agent completed a different version of the task, and the bonus is initially assigned to the best-performing agent. The only difference is that changing the initial allocation now entails a private cost for the spectator. Specifically, spectators must pay a fixed cost of \$0.20 (deducted from the \$1.00 bonus received earlier) if they choose any positive redistribution.⁶ If spectators maintain the initial allocation unchanged, they incur no cost. The screen automatically displays the spectator’s resulting bonus while they make the decision. This second decision allows us to distinguish what spectators consider fair from whether they are willing to act on that fairness view when doing so is personally costly.

Posterior Beliefs. We elicit posterior beliefs about the task. We use questions that mirror the prior beliefs elicited before the experience. Specifically, we reiterate questions on perceived differences between the versions of the task and on determinants of success in a competition where individuals are assigned two different versions of the task. The first question is identical to the one used for eliciting prior beliefs, while the second is uses a different elicitation mode to mitigate experiment demand effects and consistency biases, as suggested by (Haaland et al., 2023).

We also ask beliefs about agents at group- and individual-level (i.e., about the specific pair of agents for which the spectators previously decided for)⁷, including counterfactual beliefs about who would perform better if unequal task assignment were removed. These questions capture whether spectators attribute the observed performance difference to merit or to unequal opportunities. Finally, we ask spectators how fair they consider the initial allocation of the full bonus to the higher-performing agent.

Endline Survey. The endline survey is composed of the elicitation of experienced emotions, additional and non-consequential redistribution decisions, and socio-demographic questions.

⁵ These decisions are probabilistically consequential, as we inform the spectators that we will randomly select 1 out of every 10 participants, and, for that person, we will randomly implement one of their two decisions for a pair of real agents.

⁶ This follows a similar cost structure to that implemented in Cappelen et al. (2023), where spectators need to pay to compensate a third party. The costless decision is always elicited first because it replicates the standard spectator game framework and provides a clean measure of the spectator’s fairness view before private costs are introduced. This fixed ordering could induce consistency or status-quo effects, potentially increasing the absolute rate of continued redistribution. Since the order is identical across treatment arms, however, such effects should not mechanically explain treatment differences in costly redistribution.

⁷ We try to differentiate between beliefs about the group-level and the individual-level. It should be noted that our study is mostly suited to capture the former, as we do not provide much information about the specific pair agents.

First, we focus on the ability to imagine the agents’ experience and on spectators’ emotions (anger, compassion, sadness, indifference, irritation, elicited on a 4-point Likert scale) when learning about the initial situation of the two agents. Second, we ask spectators to make four hypothetical redistribution decisions. First, for a pair of agents in which both completed the same version of the task and the bonus is initially assigned to the best performer. Second, for a pair of agents in which both participated in a coin toss and the bonus is initially assigned to the winner of the coin toss. These two settings involve pure merit and pure luck, respectively, and are often used in the literature (see e.g., Almås et al., 2025; Harris and Sterba, 2023). Third, for a pair of agents that completed different versions of the task (i.e., one completed version A and the other completed version B) in which the one completing version B performed better and was initially assigned the bonus. Fourth, for a pair of agents that completed a different task under unequal opportunities: answering questions after reading learning materials of either low or high relevance (as in Dong et al., 2024). After this, we ask spectators to self-report the fairness views they follow out of a close list: libertarian, egalitarian, non-compensating meritocrats, compensating meritocrats, and others. Third, we collect socio-demographic information: sex, age, race, state of residence, education, dyslexia, employment status, personal and household gross annual income, political affiliation, political leaning in social and economic issues (see Annex F for more details).

2.3 Implementation

We collected data in November and December 2025. A total of 811 individuals were recruited via *Prolific*, with 160 agents (i.e., 80 pairs) and 651 spectators (336 in *EASY* and 315 in *HARD*).⁸

Agents received \$1.30 as a flat payment, plus the share allocated to them by the spectators in the redistribution task, which averaged to a total payment of \$4.30 per worker. Spectators received a flat payment of \$3.90, plus a bonus of \$1.30 during the experiment. We removed 1 spectator from the sample due to failing the attention check, as preregistered.⁹ We also take additional measures to limit bot participation and the use of AI tools.¹⁰

The median completion time was about 24 minutes. Tables A3 and A4 in the Annex report spectator de-

⁸ Most participants completed the entire survey (91.1% in *EASY* and 85.9% in *HARD*). The lower completion share in *HARD* than in *EASY* could reflect differential attrition. Most dropouts are in the first screen (7.3% in *EASY* and 9.5% in *HARD*), when participants are still not aware of the treatment assignment. We find a slightly higher attrition in participants in *HARD* immediately after the experience (0.8% in *EASY* and 3.5% in *HARD*), and no difference towards the end of the survey (0.8% in *EASY* and 0.8% in *HARD*). The fact that the hard task is more difficult may cause frustrated participants to abandon the study. Since demographic characteristics are collected at the end of the experiment, we cannot directly assess whether the differential attrition is selective. Among participants who completed the experiment, however, the treatment arms remain balanced on almost all observed demographic characteristics. This provides no indication of strong differential selection on observed characteristics, although it cannot rule out selection on unobserved characteristics such as patience or tolerance for frustration. We believe that selective attrition could, if anything, attenuate our estimate. Participants who are most frustrated or negatively affected by the hard task may be both more likely to abandon the study and, had they remained, more likely to recognize the opportunity gap and compensate the disadvantaged agent.

⁹ The pre-analysis plan specified excluding participants who met at least three of six criteria: failing the mid-experience attention check; taking more than 40 seconds to answer that check; failing the post-experience attention check; taking more than 40 seconds to answer that check; failing to make two attempts during the effort task; and failing the pre-redistribution attention check.

¹⁰ Specifically, the survey includes a CAPTCHA check before participants enter the study. In addition, copy-pasting is disabled in the open-text response boxes, which makes it harder for participants to directly paste AI-generated text into the survey.

mographics and socio-economic characteristics by treatment arm. Most pre-randomization characteristics are balanced across treatment arms, although individual income differs across **EASY** and **HARD**; all main specifications therefore control for baseline characteristics.¹¹

2.4 Hypotheses

We adapt the model from Cappelen et al. (2013). We consider a spectator who observes a pair of agents. We focus on the transfer towards an Agent B, who completed a harder version of the task and is initially assigned no earnings. The spectator decides on a distribution $(1 - y, y)$ in treatment T , where y is the share to Agent B and $1 - y$ is the share to Agent A.

The spectator derives utility from implementing what they perceive as a fair allocation, but may face a fixed cost when doing so:

$$V(y, \cdot) = -\frac{\beta(\cdot)}{2}(y - m(\cdot))^2 - c(k) \mathbb{1}\{y > 0\}, \quad (1)$$

where $m(\cdot) \geq 0$ is what the spectator considers to be the fair transfer to Agent B; $\beta(\cdot) \geq 0$ is the weight attached to fairness relative to spectator's own income; and $c(k)$ is the cost on spectator of the redistribution.

We distinguish two environments k that affect the cost of redistribution ($c(k)$). In the *costless* environment, redistribution has no cost on the spectator ($c = 0$). Then, the spectator optimizes by implementing what they consider to be fair ($y^* = m$). This is the traditional environment explored in the fairness literature (Cappelen et al., 2007), and allows eliciting fairness views. In the *costly* environment, redistribution has a cost on spectator ($c > 0$). We model the cost as independent of the amount transferred ($c \mathbb{1}\{y > 0\}$), as done by recent work (Dwenger et al., 2024; Dimant et al., 2024; Piraino and Ryan, 2024). This captures settings in which taking redistributive action requires incurring a fixed burden, such as making an appeal, filing a claim, intervening in a dispute, or publicly challenging an allocation. In these settings, the main cost is not the marginal amount transferred, but the act of intervening itself. Conditional on paying the fixed cost, the optimal transfer is $y = m(\cdot)$, which minimizes the distance from the spectator's fairness benchmark. The spectator therefore decides whether to leave the initial allocation unchanged or to pay the cost and implement the fair allocation. Redistribution occurs iff the utility gain from moving from $y = 0$ to $y = m(\cdot)$ exceeds the fixed cost:

$$\frac{\beta(\cdot)}{2}m(\cdot)^2 > c(k). \quad (2)$$

The spectator who considers Agent B deserves higher incomes (m) or has a stronger concern for fairness

¹¹ Covariates include sex, age, race, education, employment, individual and household income, and political party. To ensure that the results are not driven by differences in individual income, we reweight **EASY** and **HARD** to a common income distribution and find that the estimate remains unchanged. Moreover, income is balanced across treatment arms when grouped into low-, middle-, and high-income categories, suggesting that the imbalance arises from the fine-grained income classification rather than from a broad income gradient.

(β) is more willing to redistribute. The two parameters m and β play distinct roles. m governs how much the spectator redistributes when redistribution is costless. β governs whether the spectator is willing to pay to act on that view when redistribution is costly. This distinction motivates two separate hypotheses.

Hypothesis 1: Spectators who experience the disadvantaged task consider it fair to transfer more to Agent B than spectators who experience the advantaged task. Our first hypothesis is that $m(\text{HARD}) > m(\text{EASY})$. Since the costless decision directly reveals $y^* = m$, this hypothesis is tested by comparing costless redistribution across treatment arms:

$$\mathbb{E}(y \mid T = \text{HARD}, k = \text{costless}) > \mathbb{E}(y \mid T = \text{EASY}, k = \text{costless}). \quad (3)$$

Hypothesis 2: Spectators who experience the disadvantaged task, compared to spectators who experience the advantaged task, place greater weight on implementing what they consider fair (β). Our second hypothesis is that $\beta(\text{HARD}) > \beta(\text{EASY})$.

Testing this hypothesis requires separating $\beta(\cdot)$ from $m(\cdot)$. Since the costless decision reveals the spectator's preferred transfer, we define spectator i 's observed fairness view as $\hat{m}_i = y_i^0$, where y_i^0 is the transfer to Agent B when redistribution is costless. Let y_i^1 denote the transfer to Agent B when redistribution is costly. Costly redistribution occurs if

$$\beta(\cdot) > \frac{2c}{\hat{m}_i^2}. \quad (4)$$

Starting from this, we use two complementary empirical approaches. First, we estimate treatment-specific values of $\beta(\cdot)$ directly by maximum likelihood. Adding a logistic error term to the threshold rule, for each treatment t we estimate

$$\Pr(y_i^1 > 0 \mid T = t) = \Lambda\left(\beta(t) - \frac{2c}{\hat{m}_i^2}\right), \quad (5)$$

where $\Lambda(\cdot)$ is the logistic cumulative distribution function. Higher values of $\beta(t)$ indicate greater willingness to sustain redistribution when costly, conditional on what the spectator considered fair when redistribution was free.

Second, the same threshold condition yields a reduced-form implication. Under the null hypothesis that $\beta(\text{HARD}) = \beta(\text{EASY})$, treatment should not affect the probability of costly redistribution once $\hat{m}_i$ is held fixed. If instead $\beta(\text{HARD}) > \beta(\text{EASY})$, then

$$\Pr(y_i^1 > 0 \mid \hat{m}_i, T = \text{HARD}) > \Pr(y_i^1 > 0 \mid \hat{m}_i, T = \text{EASY}) \quad \forall \hat{m}_i > 0. \quad (6)$$

We test this implication by restricting the sample to spectators with $\hat{m}_i > 0$ and estimating whether

treatment predicts staying in redistribution after conditioning on fixed effects for the exact costless transfer.

3 Results

3.1 Prior and Posterior Beliefs about Unequal Opportunities

We start by reporting how spectators perceive the environment prior to the intervention (see Panel A in Table 1). We find a high degree of misperception about the role of opportunities in explaining task success. Most spectators do not perceive one version to be more advantageous than the other and state that the two versions are approximately the same (mean=.862, SD=.345). In fact, 81.9% of the participants (SD=.386) rank the version of the task as the least important factor to determine success, stating that individual effort (mean=.252, SD=.434) and, especially, ability (mean=.664, SD=.473) are the most relevant.¹² These beliefs are similar across treatment arms (difference=.016 and p-value=.563 for the version similarity, see row 1 in Table 1; difference=-.006 and p-value=.854 for the importance of the version, see row 3 in Table 1).

Table 1: Beliefs and Task Performance

	Total	EASY	HARD	Diff.
	(1)	(2)	(3)	(4)
Panel A. Prior Beliefs				
Versions are similar	.862	.854	.870	.016 (.027)
Easy version is better	.109	.107	.111	.004 (.024)
Version assigned is least relevant	.819	.821	.816	-.006 (.030)
Panel B. Task Performance				
Tables completed	6.77	10.53	2.75	-7.79*** (.28)
Panel C. Posterior Beliefs				
Versions are similar	.562	.637	.483	-.154*** (.039)
Easy version is better	.415	.345	.489	.144*** (.038)
Version assigned is least relevant	.604	.690	.511	-.179*** (.038)

Notes: This table reports beliefs and performance about the effort task. Columns (1) refers to the whole sample. Columns (2) and (3) distinguishes by treatment arm. The first group (**EASY**) is exposed to the easy version of the task. The second group (**HARD**) is exposed to the hard version of the task. Columns (2) and (3) report statistics in each group. Columns (4) reports the difference between the two groups. Each row reports estimates from a linear model. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

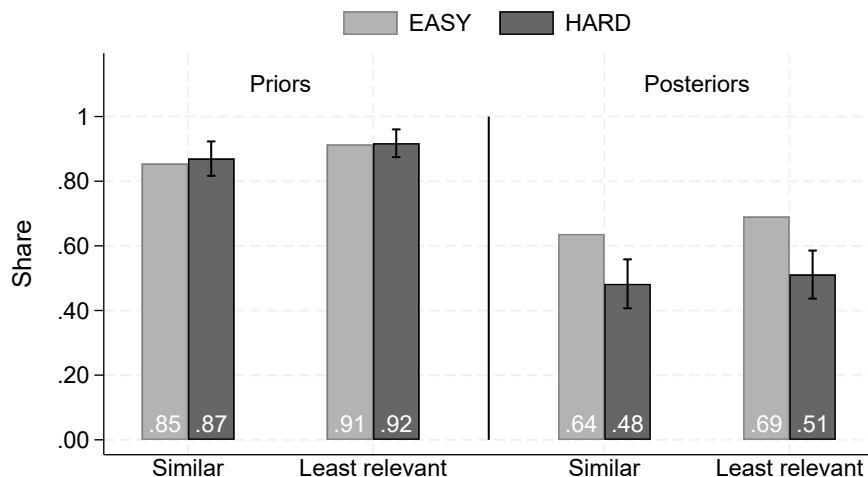
We then focus on actual performance in task completion (see Panel B in Table 1). Participants assigned

¹² It should be noted that these statements are collected following a detailed explanation about the task (including an example image), and after answering three comprehension questions referring to the goal and scoring of the task.

to the easy version outperform those assigned to the hard version, both in the agent (difference=9.01 and p-value<.001) and spectator sample (difference=7.79 and p-value<.001, see row 4 in Table 1). On average, spectators assigned to the easy version complete 10.53 tables (12.53 tables in the agent sample), while those assigned to the hard version complete 2.75 tables (3.52 tables in the agent sample). Figure B1 in the Annex shows completion by version assigned. 98.2% of spectators assigned to the easy version complete more tables than the best performer among those assigned to the hard version, which highlights how determinant the version assigned is.

Finally, we report beliefs elicited after the experience (see Panel C in Table 1). Spectators update their beliefs on the relevance of the version of the task assigned. The share of participants who consider both versions are similar drops to 63.7% of those assigned to **EASY** (difference with prior beliefs=-.217 and p-value<.001) and 48.3% of those assigned to **HARD** (difference with prior beliefs=-.387 and p-value<.001). The difference in posterior beliefs between treatment arms is -15.4 p.p. and is statistically significant (p-value<.001, see row 5 in Table 1).¹³

Figure 3: Prior and Posterior Beliefs about Task Version



Notes: This figure shows prior and posterior beliefs about the task version. The first and second group of bars plot beliefs collected before the experience. The third and fourth group of bars plot beliefs collected after the experience. The first and third group of bars plot the share of participants who state that both version of the task are similar. The second and fourth group of bars plot the share of participants who state that the version of the task is the least relevant factor in determining success. The light bars plot shares among participants in **EASY**. The dark bars plot shares among participants in **HARD**. The error spikes correspond to the 95% confidence interval as estimated via ordinary least squares regression with robust standard errors.

After the experience, spectators state that the easy version yields an advantage (mean=.415 and SD=.493), and this is especially pronounced among those assigned to **HARD** (difference=-.144 and p-value<.001, see row 6 in Table 1). We find this change is also reflected in the factors the spectators believe are relevant for success in the task. The share of participants ranking the version of the task as the least important

¹³ We show the pattern in these prior and posterior beliefs remains stable across several socio-demographic characteristics in the Annex. Table B2 shows prior beliefs on the similarity of the versions of the task are equal in both treatment arms across subgroups determined by socio-demographic characteristics. Table B3 shows how the effect of the intervention is similar across those subgroups. It should be noted that although the statistical significance varies, the treatment effect has the same sign for every subgroup. That is, in all subgroups exposure to the hard version of the task yields less perceptions that the version of the tasks are similar.

factor drops to 60.4% (SD=.490). Although the fall occurs in both treatment arms (difference with prior beliefs=-.223 and p-value<.001 for **EASY**, difference with prior beliefs=-.406 and p-value<.001 for **HARD**), it is stronger for those assigned to the hard version (difference between treatment arms=.179 and p-value<.001, see row 7 in Table 1).

Table B1 in the Annex disentangles how the beliefs we collected only once differ between treatment arms. Around 40% of participants believe that the agent assigned to the easy version would win if assigned the same version than the other agent. Given that participants are aware that version assignment is random, we take this as evidence of existence of attribution bias in our sample. Almost 35% think the same for any agent assigned to the easy version. Both values are correlated ($\rho=.319$ and p-value<.001), suggesting that participants have attribution bias both in the individual and group-level. We find that spectators in **HARD** are 20 p.p. less likely to believe this at the individual-level (p-value<.001, see row 3 in Table B1 in the Annex) and 13 p.p. at the group-level (p-value<.001, see row 4 in Table B1 in the Annex). We also show that spectators are more accurate at predicting performance for the task they are assigned to (see row 7 and 8 in Table B1 in the Annex).¹⁴ We take it as evidence that spectators learn specifically about the version of the task they are assigned to and that they extrapolate that knowledge to other versions of the task.

3.2 Perceived fair allocation: Test of Hypothesis 1

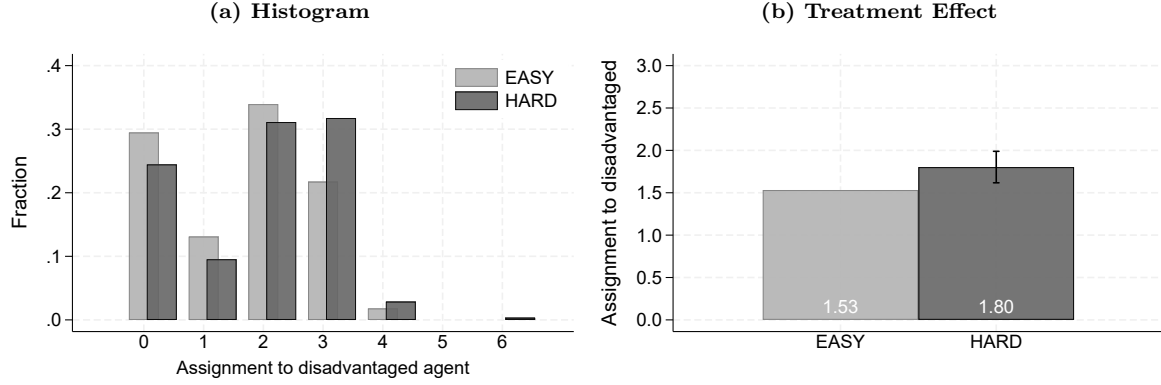
Figure 4 focuses on the redistribution decisions. Subfigure (a) shows histograms of the transfers to the disadvantaged agent by treatment arm. Overall, around half of the spectators assign either nothing to the disadvantaged agent (27.0%) or equalize earning across agents (26.6%). A further 44.4% make a positive transfer but still leave the advantaged agent with a larger share, while only 2.5% give the disadvantaged agent more than the advantaged agent. The distribution of transfers shifts to the right in **HARD**: spectators assigned to **EASY** are more likely to choose low transfers to the disadvantaged agent, while spectators assigned to **HARD** are more likely to choose transfers above \$2.00.

Subfigure (b) compares mean transfers across treatment arms. Spectators in **EASY** assign \$1.53 to the disadvantaged agent who completed the hard version and was outperformed by the agent who completed the easy version. Spectators in **HARD** assign on average \$1.80 to the disadvantaged agent, which is 18% higher than to the **EASY** group (difference=.27 and p-value=.004).

Table 2 reports the treatment effects of our intervention. Column 1 shows the estimated treatment effect in the redistribution decision. Column 2 introduces socio-demographics covariates (our preferred specification), with negligible changes in the estimates: treated spectators assign 18% more to the disadvantaged agent (difference=.27 and p-value=.007). Panel A of Table 5 shows these results alongside a summary of the treatment effects for all interventions in our study.

¹⁴ Remember that on average agents completed 12.53 tables in **EASY** and 3.52 tables in **HARD**. For the easy version, spectators in **EASY** predict 9.55 tables and those in **HARD** predict 5.24 tables (difference=-4.30 and p-value<.001, see row 7 in Table B1 in the Annex). For the hard version, spectators in **EASY** predict 8.41 tables and those in **HARD** predict 4.41 tables (difference=4.00 and p-value<.001, see row 8 in Table B1 in the Annex).

Figure 4: Redistribution Decisions



Notes: These figures show redistribution decisions in a spectator game. Subfigure (a) plots an histogram of the bonus transfer (out of \$6.00) to the disadvantaged agent. Each bar plots the fraction of spectators assigning a discrete value. Subfigure (b) plots the mean bonus transfer (out of \$6.00) to the disadvantaged agent. The light bars plot decisions among participants in EASY. The dark bars plot decisions among participants in HARD. The error spikes correspond to the 95% confidence interval as estimated via ordinary least squares regression with robust standard errors.

Table 2: Redistribution Decisions and Perceived Unfairness

	(1)	(2)	(3)	(4)
Panel A. Total Sample				
	Redistribution		Unfairness	
HARD	.270*** (.095) [.004]	.270*** (.100) [.007]	.077*** (.025) [.002]	.060** (.025) [.018]
Covariates	No	Yes	No	Yes
Effect magn.	18%	18%	25%	19%
Observations	651	651	651	651
R ²	.012	.071	.015	.115
Panel B. By Stated Fairness View				
	Redistribution			
Subsample:	Meritocrat		Non-Meritocrat	
HARD	.307*** (.114) [.007]	.337*** (.121) [.006]	.123 (.155) [.428]	.054 (.175) [.758]
Covariates	No	Yes	No	Yes
Effect magn.	23%	26%	6%	3%
Observations	447	447	204	204
R ²	.016	.115	.003	.177

Notes: This table reports treatment effects on redistribution decisions and the stated unfairness of the initial allocation. Columns (1) and (2) report redistribution decisions for the full sample. The dependent variable is the dollar transfer to the disadvantaged agent. Columns (3) and (4) report the stated unfairness of the initial allocation for the full sample. The dependent variable is a 4-point intensity share ranging from 0 to 1. Columns (5) to (8) report redistribution decisions by self-reported fairness view. Columns (5) and (6) refer to meritocrats. Columns (7) and (8) refer to non-meritocrats, defined as egalitarians, libertarians, and others. The independent variable is spectator's assignment to HARD. Each column reports estimates from a linear model. Columns (2), (4), (6), and (8) include socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Effect magnitude reports the treatment coefficient as a percentage of the mean outcome in EASY within the corresponding estimation sample. Robust standard errors are reported in parentheses. p-values are reported in square brackets.

3.2.1 Heterogeneities and Robustness Checks

We explore whether differences in redistribution decisions differ across socio-demographic characteristics. Table B4 in the Annex shows treatment effects for different subgroups by sex, age, race, educational attainment, employment status, income, and political affiliation. The treatment effect has the same sign for every subgroup. Exposure to the hard version of the task is associated with higher redistribution in every subgroup, though the statistical significance varies (partly due to statistical power).

To check the robustness of our results, we contrast redistribution decisions with stated fairness of the initial allocation. We ask spectators to rate the fairness of the situation in a 4-point Likert scale and we convert the values into a measure of intensity of unfairness that ranges from 0 to 1. We find that spectator in **HARD** perceive the initial allocation as 19% less fair than spectators in **EASY** (p-value=.018, see column 4 of Table 2), which is in line with their higher redistribution.

We also check whether our main findings reflect behavior due to meritocratic views. For doing so, we use self-reported fairness views that we collect in the endline survey. A majority of participants are meritocrats (68.7%). Columns 5 to 8 in Table 2 show treatment effects for the subsample of meritocrats and non-meritocrats. The treatment effect is concentrated among meritocrats, redistribution is 26.0% higher in **HARD** than in **EASY** (p-value=.006, see column 6 of Table 2). In turn, non-meritocrats show no difference in their decisions (p-value=.758, see column 8 of Table 2).

We further analyze the effects of our intervention for decisions under a different context, as a falsification test. We include a non-consequential redistribution decision that also features two agents under unequal opportunities, but differs in the type of task performed. Instead of completing the task featured in the experience intervention, the agents answer a multiple-choice questionnaire after receiving either low or high relevance learning materials.¹⁵ We find a much subdued effect ($\beta=.170$ and p-value=.088, see column 4 of Table B5 in the Annex). These results suggest that experiencing unequal opportunities loses effectiveness when individuals decide outside the exact context of that experience, which aligns with previous findings on experience effects (e.g. see Malmendier, 2021a, for a review).

3.2.2 Mechanism Exploration

We explore several channels that are consistent with the **HARD-EASY** difference in redistribution. We examine whether treatment-induced differences in beliefs, fairness judgments, and emotional responses are related to the observed redistribution gap. Since these measures are collected after treatment, we interpret the results as suggestive evidence on potential mechanisms rather than as a formal causal mediation analysis. Table 3 shows the main findings of separate mediation analysis.

We start by assessing whether a difference in beliefs about the determinants of the initial allocation drives our main findings. We already documented that spectators in **HARD** perceive luck as more relevant for success in the effort task (see Table 1). When spectators' posterior beliefs are added to the redistribution

¹⁵ See Annex F for a complete transcription of the context and decision screen.

regression, the treatment effect falls by 71% and is no longer statistically significant at the 10% confidence level (see column 3 in Table 3). This suggests that a relevant part of our main findings reflect the fact that spectators in **HARD** consider there is less merit in the task performance when there are unequal opportunities.

We also check whether our intervention has shifted what participants consider as fair. We start by exploiting two non-consequential redistribution decisions in contexts of equal opportunities (i.e., initial allocation due to luck and merit, respectively). As these contexts feature agents assigned exactly to the same tasks, there is no room for updated beliefs on the impact of opportunities to impact them. Differences can also reflect evolving perspectives on the merit of performance in the tasks. We find no differences for decisions in contexts in which luck determines the initial allocation (difference=.07 and p-value=.519), but find a slight gap when merit is the reason (difference=.18 and p-value=.064). This result suggests that the intervention may make spectators to assign less premium to merit in the effort task, which can partly account for the difference in redistribution under unequal opportunities.¹⁶ Consistent with this interpretation, controlling for the merit premium reduces the **HARD**--**EASY** redistribution coefficient by 43%. (see column 4 in Table 3).

Lastly, we explore whether explanations related to self-serving biases can account for the difference in redistribution decisions we report. This is particularly relevant, as studies focusing on other types of experiences find them to be important Deffains et al. (2016); Fehr and Vollmann (2025). One of the goals of our intervention is to remove room for motivated beliefs and focus on bounded belief formation instead. First, we assess if our intervention affects spectator's proximity to the agents, as this could make non-moral reasons to affect redistribution decisions. We focus on the disadvantaged agents and proxy this using spectators' self-reported understanding of the disadvantaged agent's experience. Following the intervention, spectators in **HARD** report to understand the disadvantaged agent 17% more than spectators in **EASY** (difference=.127 and p-value<.001). Including this measure leaves the **HARD**--**EASY** redistribution gap largely unchanged (see column 5 in Table 3). This provides little evidence that perceived proximity to the disadvantaged agent is closely associated with the treatment difference in redistribution. Second, we examine whether emotional responses to the agents' initial situation are related to our main findings. Table B6 in the Annex shows that our intervention generated negative emotional responses in spectators. In particular, spectators assigned to the hard version of the task reported more anger (98% above the mean of the **EASY** group and p-value=.001) and irritation (88% above the mean of the **EASY** group and p-value<.001). We test whether each of these responses is associated with the different redistribution decisions by treatment arm (see Table B7 in the Annex for each emotion collected). Including negative emotional responses attenuates the treatment difference by at most 40%, with irritation being most relevant in our sample (see column 6 in Table 3).¹⁷

¹⁶ This might also explain why, in the additional redistribution decision in which the disadvantaged agent won, we find no treatment difference (1.51 and 1.62, p-value=.414). Spectators in **HARD** may be less willing to treat performance differences as a sufficient basis for entitlement, even when the higher performer is the disadvantaged agent.

¹⁷ Although spectators were asked to report the emotions they felt when considering the agents' initial situation, these measures may also capture residual affect generated by the task itself, particularly frustration among spectators assigned to **HARD**.

The results point most strongly to changes in how spectators interpret the role of opportunities and merit in performance. Emotional responses may also accompany these changes, whereas perceived proximity to the disadvantaged agent appears less closely related to the redistribution gap.

Table 3: Mechanism Exploration

	(1)	(2)	(3)	(4)	(5)	(6)
HARD	.270*** (.095)	.270*** (.100)	.078 (.097)	.154* (.083)	.303*** (.105)	.167* (.099)
Relevance of Luck			1.554*** (.170)			
Assignment when Equal Opp.				.581*** (.040)		
Understanding of Agent					-.183 (.189)	
Irritation						1.610*** (.227)
Covariates	No	Yes	Yes	Yes	Yes	Yes
Reduction in TE	-	0%	-71%	-43%	12%	-38%
p-value	.004	.007	.422	.064	.004	.092
Observations	651	651	651	651	651	651
R^2	.012	.071	.194	.403	.072	.137

Notes: This table reports treatment effects on redistribution decisions. The dependent variable is the dollar transfer to the disadvantaged agent. The independent variable is spectator's assignment to **HARD**. Each row reports estimates from a linear model. Column (1) include no covariates. Columns (2) to (6) include a set of socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

3.3 Willingness to implement fairness: Test of Hypothesis 2

Our main finding relates to a costless redistribution decision, aligned with common practice (see Cappelen et al., 2020, for a review on the literature). We further assess a similar decision that implies a cost for the decision-maker. Panel B of Table 5 shows a summary of the treatment effects for the costly redistribution decision. Mean transfers to the disadvantaged agent are lower than in the costless condition (e.g., dropping from \$1.53 to \$0.75 in **EASY**). Spectators in **HARD** assign 47% more to the disadvantaged agent than those in **EASY** (difference=.36 and p-value<.001, see column 2 in Panel B of Table 5).

Comparing decisions with and without a cost, we assess whether our intervention affects spectators' willingness to implement the distribution they consider to be fair.¹⁸ We proceed in two complementary ways. First, we estimate the fairness weight parameter (β) directly using a threshold model. Second, we estimate a reduced-form regression that compares spectators in **EASY** and **HARD** with similar views on what is fair to transfer to the disadvantaged agent ($\hat{m}$).

¹⁸ Our analysis rests on the assumption that redistributive decisions in a costless environment reflect what spectators consider to be fair. The fact that 77.9% of spectators who pay implement exactly the same income distribution as implemented at no cost lends support to the assumption. Allowing for minor deviations of \$1, the share of broadly consistent decisions rises to 94.2%.

Table 4 reports the results for both approaches. Panel A shows that the fairness weight is close to zero and not statistically significant in **EASY** (p-value=.651, see column 1), while it is positive and statistically significant in **HARD** (p-value=.001, see column 2 in Panel A of Table 4). The difference between the two estimates is statistically significant. Table 6 shows these results alongside a summary for all interventions in our study. Panel B shows the corresponding reduced-form result. Conditional on what spectators consider to be fair, those in **HARD** are 9.0 p.p. more likely to continue redistributing when a cost is introduced (p-value=.070, see column 2 in Panel B of Table 4). Together, the two approaches show that spectators in **HARD** are more willing to redistribute at a personal cost costly than those in **EASY**.

Table 4: Willingness to Redistribute

Panel A. Structural Threshold Model			Panel B. Reduced-form Model		
	EASY	HARD		(1)	(2)
$\hat{\beta}$	-.059 (.131)	.417*** (.131)	HARD	.103** (.046)	.089* (.049)
$\hat{\beta}(\text{HARD}) - \hat{\beta}(\text{EASY})$		.476** (.185)	Dep. var. mean	.514	.514
			$\hat{m}_i$	Yes	Yes
Observations	237	238	Covariates	No	Yes
Stay-in rate	.451	.576	Observations	475	475
Log likelihood	-162.084	-161.098	R^2	.037	.134

Notes: Panel A reports treatment-specific estimates of the fairness-weight parameter from the threshold model estimated separately by treatment group. Stars on the $\hat{\beta}(T)$ row indicate whether the treatment-specific estimate differs from zero. Stars on the difference row indicate statistical significance of the test $\hat{\beta}(\text{HARD}) = \hat{\beta}(\text{EASY})$. Panel B reports linear probability model estimates of staying in redistribution, among spectators that redistribute at no cost. The dependent variable equals one if the spectator chose a positive transfer in both the costless and costly decisions. The sample contains only spectators that redistribute under no cost. Panel B specifications control for the exact amount redistributed in the costless decision. Covariates include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

3.3.1 Mechanism Exploration

Similar to the costless redistribution decision, beliefs about the determinants of success may account for a substantial share of the treatment effect on willingness to keep redistributing (see Table B8 in the Annex). In particular, controlling for the perceived relevance of luck produces the largest attenuation of the **HARD** coefficient (-37%, column 3). Emotional responses also matter: anger is strongly associated with staying in redistribution and explains a non-negligible share of the treatment difference (-30%, see column 6 in Table B9 in the Annex). By contrast, self-reported understanding of the disadvantaged agent and the equal-opportunity merit decision account for much less of the effect. Overall, the results suggest that experiencing disadvantage might affect willingness to pay for redistribution partly by changing how spectators interpret the source of the inequality, and partly by increasing the emotional force attached to correcting it.

4 Extension

4.1 Rationale

Our main experiment shows that spectators experiencing the hard version redistribute more than those assigned to the easy version. However, this comparison also raises a natural question: is the effect driven by experiencing disadvantage alone, or by understanding the opportunity gap between advantage and disadvantage? To address this question, we introduce a **MIXED** treatment arm. Unlike spectators in **EASY** or **HARD**, participants in **MIXED** experience both versions of the task. This gives them direct exposure to the opportunity gap, as they can compare the two versions from their own experience.

An additional question is whether direct experience is necessary for learning to occur. Real-world settings rarely expose the same people to different sides of unequal opportunities. Instead, people learn about these inequalities through information (e.g., statistics, reports, personal accounts, media stories, or policy communication). Information may reveal that opportunities matter, but it may be less effective at conveying what those opportunities feel like and how they shape performance. We introduce two further treatment arms with different form of information provision. In **REPORTS**, participants receive qualitative descriptions of agents’ experiences. This captures a narrative form of information, where people learn about unequal opportunities through others’ accounts. In **STATISTICS**, participants receive quantitative information about performance differences across the two task versions. This captures a statistical form of information, where people learn about unequal opportunities through facts about performance gaps.

Together, these treatment arms distinguish between one-sided experience, comparative experience, and information provision. This allows us to ask whether redistributive behavior responds to experiencing one side of unequal opportunities, to experiencing the contrast between advantage and disadvantage, or to being informed about the opportunity gap without experiencing it.

4.2 Experimental Design

We add three additional treatment arms (**MIXED**, **REPORTS** and **STATISTICS**), for which the experimental flow mirrors that of the main experiment. The main difference is in the experience.¹⁹ While **MIXED** consists of a different experience, **REPORTS** and **STATISTICS** replace the experience with information provision, one providing qualitative reports (**REPORTS**) and one providing quantitative statistics (**STATISTICS**).

Mixed. In the **MIXED** treatment arm, we expose participants to both versions of the task. Specifically, spectators spend two minutes working on the hard version and two minutes working on the easy version (in randomized order).

Reports. In the **REPORTS** treatment arm, we provide qualitative information about agents’ experiences instead of requiring task completion. All information is provided on a single screen. At the top of the

¹⁹ Another relevant change is that we remove incentives for belief elicitation in the extension. The main reason for this change is that with information provision, the incentives become unnecessary.

screen, we include a summary of the most recurring themes described by agents who completed Version A, followed by two representative quotes from these agents. Below that, we include a separate summary for agents who completed Version B, again accompanied by two representative quotes. See Figure D3 in the Annex for more details. All of this information is drawn from a pilot study.²⁰ To ensure understanding, we require participants to correctly answer two comprehension questions about the information before proceeding.

Statistics. In the **STATISTICS** treatment arm, we provide quantitative information about agents' experiences instead of requiring task completion. The entire information is provided in one screen. For each type of agent, we details the average number of tasks completed, the maximum and minimum performance, the percentage of correct completions, and the probability that an agent of that type outperforms an agent of the other type. See Figure D4 in the Annex for more details. All of this information is drawn from the data of the agents of the main study. To ensure understanding, we require participants to correctly answer two comprehension questions about the information before proceeding.²¹

Hypotheses. We hypothesize that the new treatments shift both components of redistributive preferences relative to **EASY**. First, we study whether spectators in **MIXED**, **REPORTS**, and **STATISTICS** consider a larger transfer to the disadvantaged agent to be fair, as revealed by costless redistribution. Second, we investigate whether spectators in these treatments arms place greater weight on implementing what they consider fair, β , as revealed by costly redistribution conditional on the costless transfer. We further use the comparison across **MIXED**, **REPORTS**, and **STATISTICS** to explore whether direct comparative experience is associated with stronger responses than information provision.

4.3 Implementation

We collected data in April 2026. A total of 602 individuals were recruited via *Prolific*, with 100 agents (i.e., 50 pairs) and 502 spectators (165 in **MIXED**, 165 in **REPORTS** and 172 in **STATISTICS**).

Tables A5 and A6 in the Annex detail the spectator's demographics and show that the sample is mostly balanced across treatment arms, although sex differs significantly across arms, with a higher share of women in **STATISTICS**. When comparing the sample of our extension to our main study, we find some differences in socio-demographic characteristics. We control for all such characteristics when comparing across samples.

²⁰ The reason for this is that, due to budget constraints, we did not collect qualitative experiences of the agents in the main study.

²¹ Notice that the treatment duration for **REPORTS** and **STATISTICS** is shorter than for the experience treatments. We believe this is a feature that reflects the real world. Visual information is typically consumed more quickly, whereas experiencing something requires more time for engagement and information processing.

4.4 Results

We start by reporting the prior beliefs about the task. We reproduce similar findings as the main study. 76.9% of the participants rank the version of the task as the least important factor to determine success, stating that effort and ability are of greater importance (see row 1 in Table C2 in the Annex). Additionally, 82.9% of participants also state that the two versions are approximately the same, and they do not perceive one of the two as more advantageous (see row 3 in Table C2 in the Annex).

4.4.1 Mixed Experience

First, we examine how the mixed intervention affects beliefs about the tasks (see Table C3 in the Annex). Compared with both **EASY** and **HARD**, spectators in **MIXED** are significantly less likely to report that the two versions of the task are similar (see row 4 in Table C3 in the Annex), more likely to state that the **EASY** version provides an advantage (see row 5 in Table C3 in the Annex), and assign greater importance to luck in determining success (see row 6 in Table C3 in the Annex). These differences suggest that direct exposure to both versions of the task makes the unequal opportunity structure more salient than one-sided experience alone. Comparison with **HARD** shows that experiencing the hard version is not equivalent to experiencing the contrast between the two versions. Spectators in **MIXED** consistently assign more relevance to luck in explaining performance differences.

We then turn to redistribution decisions. Spectators who experience both versions of the task redistribute much more towards the disadvantaged agent. When redistribution is costless they transfer \$0.61 more (equivalent to a 45% rise compared to **EASY** and $p\text{-value} < .001$, see column 4 in Panel A in Table 5), and when it entails a personal cost they transfer \$0.75 more (98% more compared to **EASY**, see column 4 in Panel B in Table 5). We assess whether comparative experience affects redistribution beyond the effect of experiencing disadvantage alone by comparing redistribution in **HARD** and **MIXED**. Spectators in **MIXED** allocate \$0.34 more in the costless condition and about \$0.43 more in the costly condition, that is 22% and 55% higher relative to the corresponding baseline means ($p\text{-value} = .005$ and $p\text{-value} = .002$ respectively, see columns 2 and 4 in Table C4). Our results suggest that the effect of comparative experience is relevant, particularly when redistribution is costly.

Table 6 shows a summary of the structural estimates for all interventions. Estimates point in the same direction as the previous results. The weight assigned to fairness is highest for spectators in **MIXED** ($\beta = .798$, see column 3 in Panel A in Table 6), and the differences with **EASY** and **HARD** are both statistically significant. This suggests that direct comparative experience also affects the weight individuals attach to implementing their fairness views. We also find that the greater willingness to redistribute in **MIXED** persists after accounting for our estimates on what spectators consider to be fair. Spectators in **MIXED** are 20.9 p.p. more likely to keep on redistributing than those in **EASY** ($p\text{-value} < .001$, see column 2 in Table C10 in the Annex), while differences relative to **HARD** are smaller (10.7 p.p. and $p\text{-value} = .052$). Emotional response and beliefs about luck only account partially for the stronger willingness to pay in **MIXED** relative to **EASY** (see Table C11 in the Annex).

The remaining outcomes show a consistent pattern with our previous findings. Spectators in **MIXED** judge the initial allocation as less fair than those in **EASY** and **HARD** (see Table C5 in the Annex). They also report a substantially larger performance gap between the two versions, assign a lower probability for the disadvantaged agent to outperform the advantaged agent, are less likely to believe that the disadvantaged agent could have overcome the performance gap through hard work alone, and revise counterfactual beliefs similarly to those in **HARD** (see Tables (see Tables C6 and C7 in the Annex). Emotional responses are also larger. Spectators assigned to both versions of the task report more anger, irritation, and compassion, and lower indifference to the situation (see Table C8 in Annex). Experiencing both versions reduces the empathy gap, with spectators stating greater ease in imagining the (dis)advantaged agent's experience than those in (**EASY**)**HARD** (see Table C9 in the Annex).

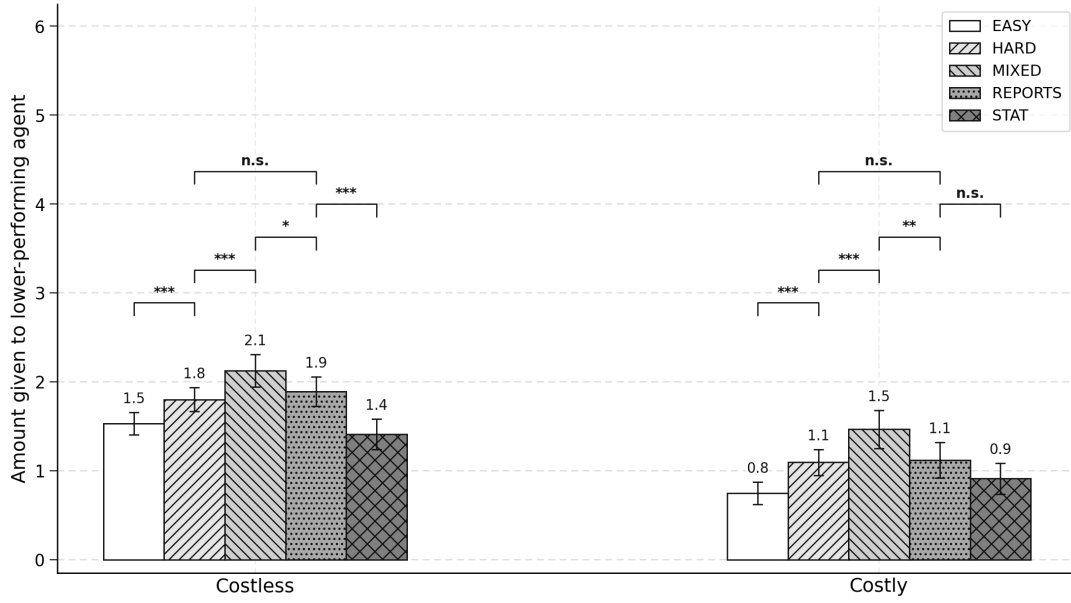
Table 5: Treatment Effects on Redistribution Decisions

Panel A. Costless redistribution								
	HARD		MIXED		REPORTS		STATISTICS	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
TREATED	.270*** (.095)	.270*** (.100)	.595*** (.114)	.596*** (.116)	.358*** (.106)	.375*** (.117)	-.120 (.108)	-.093 (.113)
Covariates	No	Yes	No	Yes	No	Yes	No	Yes
Effect magn.	18%	18%	39%	39%	23%	24%	-8%	-6%
p-value	.004	.007	<.001	<.001	.001	.001	.265	.411
Observations	651	651	501	501	501	501	508	508
R^2	.012	.071	.053	.149	.021	.086	.002	.106

Panel B. Costly redistribution								
	HARD		MIXED		REPORTS		STATISTICS	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
TREATED	.345*** (.099)	.356*** (.104)	.717*** (.126)	.736*** (.130)	.371*** (.121)	.437*** (.136)	.163 (.110)	.221* (.119)
Covariates	No	Yes	No	Yes	No	Yes	No	Yes
Effect magn.	46%	47%	96%	98%	49%	58%	22%	29%
p-value	<.001	<.001	<.001	<.001	.002	.001	.137	.063
Observations	651	651	501	501	501	501	508	508
R^2	.019	.091	.068	.159	.020	.080	.004	.081

Notes: This table reports treatment effects on redistribution decisions. The dependent variable is the dollar transfer to the disadvantaged agent. Panel A reports treatment effects for the costless redistribution decision. Panel B reports treatment effects for the costly redistribution decision, in which spectators must pay a fixed cost to change the initial allocation. The independent variable is spectator's assignment to a treatment arm different from **EASY**. Columns (1) and (2) report the treatment effect of **HARD**. Columns (3) and (4) report the treatment effect of **MIXED**. Columns (5) and (6) report the treatment effect of **REPORTS**. Columns (7) and (8) report the treatment effect of **STATISTICS**. Each row reports estimates from a linear model. Columns (2), (4), (6), and (8) include a set of socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Figure 5: Redistribution Decisions by Treatment Arm



Notes: This figure shows redistribution decisions in each treatment arm. The left group of bars plots the amount assigned to the disadvantaged agent when redistribution is costless. The right group of bars plots the amount assigned to the disadvantaged agent when redistribution is costly. The bars plot decisions among participants in EASY, HARD, MIXED, REPORTS, and STATISTICS. Error spikes correspond to the 95% confidence interval as estimated via ordinary least squares regression with robust standard errors. Brackets report pairwise comparisons between treatment arms. $^*p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 6: Estimates on Fairness Weight by Treatment Arm

Panel A. Treatment-specific estimates					
	EASY	HARD	MIXED	REPORTS	STATS
$\hat{\beta}(T)$	-.059 (.131)	.417*** (.131)	.798*** (.182)	.271 (.171)	.442** (.187)
Observations	237	238	136	138	117
Stay-in rate	.451	.576	.669	.536	.573

Panel B. Pairwise comparisons		
	Diff.	SE
$\hat{\beta}(\text{HARD}) - \hat{\beta}(\text{EASY})$	.476**	(.185)
$\hat{\beta}(\text{MIXED}) - \hat{\beta}(\text{EASY})$	.857***	(.224)
$\hat{\beta}(\text{MIXED}) - \hat{\beta}(\text{HARD})$	.381*	(.225)
$\hat{\beta}(\text{REPORTS}) - \hat{\beta}(\text{MIXED})$	-.527**	(.250)
$\hat{\beta}(\text{STATS}) - \hat{\beta}(\text{MIXED})$	-.356	(.261)
$\hat{\beta}(\text{STATS}) - \hat{\beta}(\text{REPORTS})$	.171	(.254)

Notes: This table shows estimates from the threshold model for each treatment arm. The sample contains only spectators that redistribute under no cost. The threshold term $(2c/\hat{m}_i^2)$ has a fixed -1 coefficient, and c is \$0.20 in our experiment. Standard errors are reported in parentheses. Stars on the $\hat{\beta}(T)$ row indicate whether the treatment-specific estimate differs from zero. Stars in Panel B indicate statistical significance of tests of equality between the corresponding treatment-specific $\hat{\beta}(T)$ estimates. $^*p < .10$, $^{**}p < .05$, $^{***}p < .01$.

4.4.2 Information Provision

We explore whether information about the opportunity gap can reproduce the effect of directly experiencing that gap. To do so, we compare REPORTS and STATISTICS with MIXED. We do so because both REPORTS

and **STATISTICS** describe the contrast between the easy and hard version of the task, as **MIXED** does. We also report some results comparing these treatments with **EASY** and **HARD** to assess how information compares to one-sided experiences.

First, we confirm that information provision affects beliefs about the task. Spectators in both **REPORTS** and **STATISTICS** perceive less advantages of the easy version of the task and assign less importance to luck in determining success than those in **MIXED** (see Table C13 in the Annex). Both forms of information provision affect awareness of unequal opportunities less than direct experience. This suggests that learning about the opportunity gap through either qualitative or quantitative information is not as powerful as personally experiencing the contrast between the two versions.

Participants receiving information on the opportunity gap transfer less to the disadvantaged agent than those who personally experience it (see Figure 5, and Table C12 in the Annex). Qualitative information provision leads to lower redistribution than direct comparative experience. Relative to **MIXED**, spectators in **REPORTS** transfer \$0.23 less when redistribution is costless (p-value=.075, see column 2 of Table C12) and \$0.29 less when redistribution is costly (p-value=.058, see column 4 of Table C12), equivalent to 11% and 20% reductions, respectively. At the same time, **REPORTS** still generates more redistribution than exposure to the easy version of the task (see column 6 of Panels A and B of Table 5). Compared with exposure to the hard version, point estimates in **REPORTS** are slightly higher, but the differences are not statistically significant (difference of \$0.14 for costless decisions and p-value=.209; difference of \$0.09 for costly decisions and p-value=.489, see columns 2 and 4 of Table C20).

Quantitative information is less effective. Relative to **MIXED**, spectators in **STATISTICS** transfer \$0.72 less when redistribution is costless (p-value< .001, see column 2 of Table C12) and \$0.56 less when redistribution is costly (p-value< .001, see column 4 of Table C12), corresponding to reductions of 34% and 38%, respectively. Compared with exposure to the easy version, redistribution in **STATISTICS** is not significantly different in the costless condition, although it is marginally higher when redistribution is costly (see column 8 of Panels A and B of Table 5). Compared with exposure to the hard version, **STATISTICS** leads to significantly lower redistribution in the costless condition (difference=-\$0.33 and p-value=.004, see column 2 of Table C20), while the costly difference is smaller and not statistically significant (difference=-\$0.17 and p-value=.179, see column 4 of Table C20). Overall, statistical information conveys the existence of the opportunity gap, but does not reproduce the redistributive response generated by direct comparative experience.

Qualitative and quantitative information affect redistribution differently. With controls, spectators who receive quantitative information transfer \$0.49 less than those receiving qualitative information when redistribution is costless (p-value< .001, see column 2 of Table C12). When redistribution is costly, the difference is smaller: spectators in **STATISTICS** transfer \$0.27 less than those in **REPORTS** (p-value=.058, see column 4 of Table C12). This suggests that qualitative information is more effective than quantitative information, especially when redistribution is costless.

Evidence on whether information provision affects the willingness to implement what is considered fair is mixed (see Table 6). We estimate a lower fairness weight both in **REPORTS** and **STATISTICS** than in **MIXED**, but the difference is not statistically significant at conventional levels for the latter. Our results suggest that direct comparative experience induces a stronger willingness to implement fairness than receiving information, especially when the information is qualitative. Reduced-form estimates yield similar results (see Table C18 in the Annex). Spectators in **REPORTS** are less likely keep redistributing once a cost is introduced than those in **MIXED** (difference=-10.3 p.p. and p-value=.097, see column 2 in Table C18 in the Annex). The difference between **STATISTICS** and **MIXED** is not statistically significant (difference=-9.2 p.p. and p-value=.176, see column 2 in Table C18 in the Annex). Mechanism exploration sheds little light on the reason for the differences we find (see Table C19 in the Annex). Understanding of the disadvantaged agent attenuates the gap with **MIXED** to some extent, especially for **REPORTS**, but anger, the equal-opportunity merit decision, and beliefs about luck do not consistently account for the lower persistence in the information treatments. Overall, the persistence results suggest that direct comparative experience is associated with greater willingness to pay to implement redistribution, especially relative to qualitative information.

The remaining outcomes are in line with the previous findings. Compared with **MIXED**, both **REPORTS** and **STATISTICS** spectators judge the initial allocation as more fair; these differences are statistically significant at the 1% level (Table C13). For numerical beliefs, **MIXED** spectators are more accurate than those in **REPORTS** and less accurate than those in **STATISTICS** in estimating the proportional performance gap between the two versions,²² while there are no significant differences in the perceived probability that Agent B outperforms Agent A (Table C14). Relative to **MIXED**, both **REPORTS** and **STATISTICS** spectators are less likely to revise hard-work and counterfactual beliefs in the direction observed under direct exposure, with the strongest differences appearing for the hard-work and individual-level counterfactual measures (Table C15). Emotional responses also point to weaker engagement under information provision. Both information treatments are associated with lower anger and irritation, while **STATISTICS** is also associated with greater indifference (Table C16). Finally, perspective-taking measures show that both **REPORTS** and **STATISTICS** spectators report lower ease in imagining the experiences of both agents than spectators in **MIXED**; these differences are statistically significant at the 1% level (Table C17).

5 Fairness Reasoning

The redistribution decisions show whether experience changes the amount spectators allocate to the disadvantaged worker. We next examine whether experience also changes the fairness principles spectators use to justify those decisions. To do so, we classify spectators' open-text explanations into four fairness types and compare their prevalence across treatments. The analysis includes 1,153 explanations, with an average length of 202 characters, or approximately 39 words, and a median length of 172 characters, or 33 words.

²² Notice that, in **STATISTICS**, spectators are actually given the answers to the questions as part of their information provision.

We classify explanations into four pre-specified fairness views: libertarians (who argue that the initial allocation should be left unchanged), non-compensating meritocrats (who reward observed performance as it is), compensating meritocrats (who adjust their judgments of merit to account for unequal circumstances), and egalitarians (who justify the allocation by appealing to equality). We use two large language models, GPT-5.5 Thinking and Claude Sonnet5, independently. Each model was given an identical coding protocol and instructed to classify an explanation using only the written text, never the amount redistributed. When an answer contained multiple principles, the model was instructed to assign the answer to the principle most central to the stated justification. When it was not possible to classify an answer, we instructed the model to classify it as ‘Other’. The exact prompt and example for each category can be found in Appendix G. The two models assign the same code to 979 of 1,153 explanations (84.9%, Cohen’s $\kappa = 0.774$), indicating that the classification is largely robust to the choice of model; we report both classifications in full in Panels A and B of Table 7 and discuss the GPT classification in the text below.

We find that the way spectators reason about unequal opportunities is heterogeneous (see Panel A of Table 7). Overall, we classify 6.9% of spectators as libertarian, 45.4% as non-compensating meritocrats, 32.6% as compensating meritocrats, and 10.1% as egalitarians. Claude’s independent classification is close (5.5%, 47.1%, 30.2%, and 13.1%, respectively; Panel B). The two largest groups are meritocrats, together accounting for 78.1% of explanations under GPT (77.3% under Claude). The distribution is broadly comparable to the estimates in Andre (2024). The share of non-compensating meritocrats is somewhat higher in our data (45.4% vs. 40.0% in Andre, 2024), while compensating meritocrats are also slightly more common (32.6% vs. 28.4% in Andre, 2024). These differences may reflect the more concrete opportunity gap in our design, but also differences in measurement: Andre (2024) infer fairness types structurally from repeated allocation choices, whereas we classify the central principle expressed in participants’ open-text explanations. Relative to Cappelen et al. (2024), we find a larger combined meritocratic share and fewer libertarian explanations. This comparison should likewise be interpreted cautiously, since Cappelen et al. (2024) recover fairness types from treatment differences in redistribution behavior rather than from individual text responses.

The patterns across treatment arms suggest that experience affects not only how much participants redistribute, but also how they think about fairness. Among participants in **EASY**, we classify 8.0% as libertarian, 57.7% as non-compensating meritocrats, 17.0% as compensating meritocrats, and 11.3% as egalitarians (Table 7, Panel A). Among participants in **HARD**, compensating meritocrats are a larger group (34.9%), while non-compensating meritocrats are fewer (43.5%). The difference is largest in **MIXED**, where participants directly experience both versions of the task. More than half of participants (53.3%) are classified as compensating meritocrats, while non-compensating meritocrats are only 23.0%. This pattern suggests that direct comparative experience makes the opportunity gap easier to perceive and harder to ignore. The information treatments display a more nuanced trend. **REPORTS** generates an equally high share of compensating- and non-compensating-meritocratic explanations (40.6% each), suggesting that qualitative accounts of agents’ experiences can make the opportunity gap salient. By contrast, **STATISTICS** looks closer to **EASY**: 51.2% are non-compensating meritocrats and 31.4% are compensating

meritocrats. This pattern is consistent with the idea that quantitative information can communicate the existence of a performance gap, but may be less effective than direct experience or qualitative information at changing the principle participants use to justify redistribution.

We also compare our text-based classification with participants’ self-reported fairness views. The two measures are related, but far from identical: text-coded classifications match self-reports in 44.1% of cases under GPT and 46.6% under Claude. This suggests that fairness views may be partly context-dependent, and that self-classifications do not always fully capture how participants reason when faced with a concrete unequal-opportunity allocation: participants who reason in a straightforwardly performance-based way in their own words are especially likely to select a different label when asked to self-categorize.

Table 7: Fairness views by classification method

	Overall	Mean transfer	EASY	HARD	MIXED	REPORTS	STATISTICS
<i>Panel A. Text classification using GPT-5.5 Thinking</i>							
Libertarian	79 (6.9%)	0.00	27 (8.0%)	16 (5.1%)	12 (7.3%)	7 (4.2%)	17 (9.9%)
Non-compensating meritocrat	524 (45.4%)	1.21	194 (57.7%)	137 (43.5%)	38 (23.0%)	67 (40.6%)	88 (51.2%)
Compensating meritocrat	376 (32.6%)	2.38	57 (17.0%)	110 (34.9%)	88 (53.3%)	67 (40.6%)	54 (31.4%)
Egalitarian	117 (10.1%)	2.89	38 (11.3%)	35 (11.1%)	15 (9.1%)	19 (11.5%)	10 (5.8%)
Other	57 (4.9%)	2.16	20 (6.0%)	17 (5.4%)	12 (7.3%)	5 (3.0%)	3 (1.7%)
<i>Panel B. Text classification using Claude</i>							
Libertarian	63 (5.5%)	0.08	15 (4.5%)	20 (6.3%)	6 (3.6%)	7 (4.2%)	15 (8.7%)
Non-compensating meritocrat	543 (47.1%)	1.07	200 (59.5%)	133 (42.2%)	49 (29.7%)	70 (42.4%)	91 (52.9%)
Compensating meritocrat	348 (30.2%)	2.52	52 (15.5%)	104 (33.0%)	83 (50.3%)	61 (37.0%)	48 (27.9%)
Egalitarian	151 (13.1%)	2.89	51 (15.2%)	47 (14.9%)	21 (12.7%)	21 (12.7%)	11 (6.4%)
Other	48 (4.2%)	1.94	18 (5.4%)	11 (3.5%)	6 (3.6%)	6 (3.6%)	7 (4.1%)
<i>Panel C. participants’ self-classified fairness views</i>							
Libertarian	89 (7.7%)	1.36	26 (7.7%)	26 (8.3%)	9 (5.5%)	18 (10.9%)	10 (5.8%)
Non-compensating meritocrat	278 (24.1%)	1.00	129 (38.4%)	82 (26.0%)	11 (6.7%)	25 (15.2%)	31 (18.0%)
Compensating meritocrat	495 (42.9%)	1.83	107 (31.8%)	129 (41.0%)	89 (53.9%)	80 (48.5%)	90 (52.3%)
Egalitarian	224 (19.4%)	2.59	56 (16.7%)	62 (19.7%)	47 (28.5%)	33 (20.0%)	26 (15.1%)
Other	67 (5.8%)	1.55	18 (5.4%)	16 (5.1%)	9 (5.5%)	9 (5.5%)	15 (8.7%)
Observations	1,153		336	315	165	165	172

Notes: Panels A and B report the distribution of fairness types based on participants’ open-text explanations following the costless redistribution decision. Both text classifications use only the written explanations and do not use the amount redistributed. Panel C reports participants’ self-classified fairness views from the endline survey. The self-classified category corresponding to compensating meritocrats was labeled “meritocratic (comparable)” in the survey. Percentages in the treatment columns are shares within each treatment arm. Mean transfer is the average amount assigned to the disadvantaged agent within each classification category and is calculated only after classification. GPT and Claude assign the same classification to 979 of 1,153 explanations (84.9%). Cohen’s $\kappa = 0.774$, with a bootstrap 95% confidence interval of [0.743, 0.805].

6 Discussion

In this paper, we provide causal evidence that experiencing unequal opportunities reshapes redistributive preferences. Our results speak to a broader question in political economy and behavioral economics: how does fairness impact policy preferences? In our setting, experience affects how people interpret the sources of inequality. Before the intervention, most participants viewed performance differences mainly

through effort and ability. Unequal opportunities were present in the environment, but they were not the main lens through which participants initially interpreted outcomes. Direct exposure to disadvantage changes this interpretation. The opportunity gap becomes more salient, the link between performance and merit weakens, and redistribution shifts toward greater compensation for the disadvantaged worker. In our extension, we show that the effects from direct experience is stronger than information provision. Qualitative and quantitative information affect redistribution, but fall consistently short of the impact of experiencing unequal opportunities. We discuss these findings, examining how experience shifts the fairness views and the beliefs underlying them, how it impacts willingness to pay to act on redistributive judgments, what drives the gap between experience and information, and what the results imply for institutional design and future research.

6.1 Experience, Beliefs, and Fairness Views

Our first main result is that experience affects what spectators consider to be fair. Spectators who experience the hard version of the task redistribute more to the disadvantaged agent than those who experience the easy version. The treatment effect is economically meaningful. Redistribution is 18% larger among participants who experience disadvantage opportunities. This decision reveals participants' fairness view. Experiencing disadvantage shifts views toward greater compensation for the disadvantaged agent.

Our mechanism exploration suggests that this shift is driven by beliefs about the role of circumstances in determining performance. Spectators who experience disadvantage care less about differences in performance, place greater weight on luck and circumstances as determinants of the observed performances, and judge the initial allocation as less fair. In this sense, experience might affect redistribution because it affects how people think about the sources of inequalities.

This result speaks directly to the literature on fairness under unequal opportunities. Recent studies show that redistribution responds to beliefs and information about the role of circumstances in shaping outcomes (Andre, 2024; Bhattacharya and Mollerstrom, 2025; Brun, 2026; Cappelen et al., 2024; Dong et al., 2024; Pogliano, 2024; Pruess et al., 2025). Our results build on this evidence by showing that these beliefs are shaped by experiences. Experiencing disadvantage makes the role of circumstances more visible and reduces the tendency to interpret performance differences as reflecting only effort or ability.

Results from our extension study reinforces this interpretation. Merit evaluation under unequal opportunities needs counterfactual reasoning about what would have happened under different circumstances (Amemiya et al., 2023; Andre, 2024), and that this process is cognitively demanding (Brun, 2026). Direct experience may facilitate it by making the relevant counterfactual more concrete. Consistent with this idea, we find the strongest belief revisions and redistributive responses when participants can directly compare how advantage and disadvantage affect performance. This suggests that experience not only matters because it reveals that circumstances differ, but also because it helps individuals to understand how unequal opportunities affect the translation of effort into performance. Perspective-taking may

reinforce this process (Jia, 2022), but the main pattern is most consistent with experience correcting attribution bias and shifting the fairness view toward greater compensation.

The result is also related to a growing literature showing that personal experiences of success and failure affect redistributive preferences and meritocratic beliefs (Amasino et al., 2024; Cassar and Klein, 2019; Deffains et al., 2016; Espinosa et al., 2020; Fehr and Vollmann, 2025). These studies show that individuals' own outcomes can influence how they interpret inequality, often through self-serving beliefs and motivated reasoning. Our setting is similar in showing that experience can shape fairness views. However, it differs in a key respect: the relevant experience is not winning or losing, but facing advantaged or disadvantaged conditions. The belief formation we document is less likely to be motivated by a need to justify one's own outcome, and more likely to reflect learning about how the environment shapes performance. The extension study further supports this interpretation. If the effect were mainly driven by motivated beliefs or identification with an experiential in-group, we would expect it to be strongest among spectators who experienced the same task version as the disadvantaged agent. Instead, belief revision and redistribution are strongest when participants experience both task versions and therefore have no single task-based in-group by design. This pattern suggests that the effect is better understood as learning about the opportunity structure than as self-serving or in-group motivated reasoning.

More broadly, our findings connect to the literature on experience effects in economic beliefs and behavior. A large body of work shows that individuals often extrapolate from their own experiences when forming beliefs about economic environments (Kuchler and Zafar, 2019; Malmendier and Nagel, 2016; Malmendier, 2021a,b; Malmendier and Wachter, 2024). Experiences may also shape which aspects of an environment become salient and are later retrieved in judgment (Bordalo et al., 2020). Our results extend this logic to fairness views. We show that a short exposure to disadvantage affects not only beliefs about the task, but also the fairness views used to evaluate redistribution between others.

6.2 Experience and Willingness to Pay for Redistribution

The second main result is that experience affects not only what spectators consider fair, but also their willingness to pay to implement that fairness view. This distinction is important because the costless treatment effect could be interpreted as a low-cost expressive response: spectators who experience disadvantage may express sympathy when doing so is free. The costly decision addresses this concern directly. Introducing a personal cost substantially reduces redistribution for all spectators, confirming that self-interest constrains redistributive behavior. Yet the treatment gap persists, as spectators who experienced disadvantage continue to redistribute more toward the disadvantaged agent than spectators who experienced advantage. Thus, the effect of experience is not limited to cheap expressions of fairness; it also affects willingness to bear a private cost to correct an allocation perceived as unfair.

The structural estimates reinforce this interpretation. Experiencing advantage is associated with little willingness to bear a personal cost to correct the allocation, whereas experiencing disadvantage gives the unfairness of the opportunity gap greater behavioral force. In the model, this appears as a fairness

weight that is close to zero and statistically insignificant among spectators who experience the advantaged opportunity set, but positive and significant among those who experience disadvantage; the difference between the two is statistically significant. Interpreted economically, spectators who experience disadvantage are willing to pay about \$0.21 to correct by \$1.00 an inequality they perceive as unfair. The reduced-form analysis points in the same direction. We show that, conditional on the exact amount redistributed when redistribution was costless, spectators who experience disadvantage are more likely to continue redistributing once a cost is introduced compared to those who experienced advantage. In our extension, we show that the willingness to pay is the highest among those who directly experience the opportunity gap. Direct comparative experience therefore affects not only the fairness view, but also the weight attached to acting on that view.

This result contributes to the incipient literature on costly fairness and willingness to pay for redistribution. Most studies using spectator games impose no cost on the decision-makers, preventing researchers from distinguishing fairness views from the willingness to act on them. Recent empirical work shows that redistributive decisions are sensitive to personal costs, but that many individuals remain willing to incur such costs to implement allocations they regard as fair (Dimant et al., 2024; Dwenger et al., 2024; Piraino and Ryan, 2024). Closest to our setting, Dimant et al. (2024) study costly third-party redistribution in a related environment and document willingness-to-pay rates similar to ours. We further show that the willingness to act on fairness views is malleable. This margin is often treated as a stable preference parameter or inferred from cross-sectional differences in costly redistribution. We provide causal evidence that it can be shifted by experience with the opportunity structure that produced the unequal outcome.

6.3 The Role of Direct Experience vs. Information

The information treatments show that information alone does not fully replicate the effect of direct experience. Both qualitative and quantitative information make the opportunity gap explicit, but they generate less redistribution than directly experiencing that gap. The difference is smaller for qualitative accounts of the agents' experiences and substantially larger for quantitative information, suggesting that narratives can shift redistributive behavior but still fall short of reproducing the effect of direct comparative experience.

Redistributive decisions in the costly setting point toward the same direction. Relative to spectators who directly experience the opportunity gap, those who receive information redistribute less when it entails a personal cost. This translates into lower willingness to keep redistributing after receiving qualitative information than after direct comparative experience. The corresponding estimate for quantitative information is negative but not statistically significant, possibly because of limited power. Overall, the results suggest that direct comparative experience affects both fairness views and, to some extent, the willingness to pay to implement those views in a different way than information does.

One interpretation for this asymmetry is that experiential learning differs from information transmission. Personal encounter with an unequal opportunity structure may generate a qualitatively different under-

standing than receiving information about it. A large literature shows that individuals learn differently from experienced evidence than from described or observed evidence, mostly in the context of risk preferences (Hertwig and Erev, 2009; Hertwig, 2015). Recent work by Gonzalez-Jimenez (2026) similarly shows that beliefs and ambiguity attitudes differ when individuals learn from described rather than experienced signals, while Chen et al. (2026) provides related evidence that experience and observation are processed differently. We extend this logic to the domain of fairness preferences by showing that directly encountering an unequal opportunity structure generates different forms of understanding than reading about it.

The asymmetry between experience and information may also reflect differences in emotional engagement and perspective-taking. Information can reveal the existence of unequal opportunities, and narratives can partly convey their subjective meaning. But direct comparative experience appears to combine belief updating, emotional engagement, and perspective-taking in a way that information alone does not fully reproduce. In our data, directly experiencing the opportunity gap is associated with stronger emotional responses and greater perspective-taking than information provision. This matters because emotions such as moral anger and compassion can motivate costly third-party interventions in fairness settings (Chen et al., 2026). Thus, direct experience may generate stronger redistribution not only because it provides information, but because it makes unfairness more emotionally accessible and salient.

The comparison between different type of information provision is also informative. Narrative information generates more redistribution than statistical information in the costless condition. This is consistent with evidence that identifiable or narratively described individuals often elicit stronger prosocial responses than statistical information, both in charitable giving and in attitudes toward immigration and redistribution (Slovic, 2007; Small et al., 2007; ?). More broadly, it is also consistent with the idea that social information is not absorbed mechanically, as people may underuse, reinterpret, or discount information depending on how it is presented (Morin et al., 2021; Graeber et al., 2024; Thaler et al., 2024). In our context, reports may make it easier for spectators to imagine why the harder version matters: they convey frustration, difficulty, and the subjective experience of facing a constraint. Statistics, by contrast, convey the size of the performance gap more precisely, but may do less to make the mechanism behind the gap vivid. At the same time, the advantage of qualitative over quantitative information narrows when redistribution becomes costly. This suggests that narratives may be more effective at shifting costless fairness views than at increasing willingness to pay to implement them.

6.4 Implications

Our findings have implications both for how we understand the support for redistributive and for the design of policies and institutions aimed at promoting awareness of inequality.

Our results suggest that support for redistribution is sensitive to the conditions individuals experience. This has implications for political economy models, in which redistributive preferences are often treated as stable parameters. Structural changes in economic sorting, residential segregation, or occupational

integration could indirectly affect redistribution support through beliefs that are endogenous to experienced circumstances.

For institutional design, our findings highlight the limits of information-based interventions and the comparative advantage of direct experiential exposure. Policies aimed at reducing inequality often rely on communicating facts, for example by publishing statistics on wage gaps, disseminating reports on educational disparities, or sharing personal accounts of disadvantage. Direct experiential exposure may generate stronger belief revision than information-based communication alone. Mandatory rotation schemes, such as those embedded in civil service fast-stream programs (e.g., the UK Civil Service Fast Stream) or medical residencies, expose decision-makers to a range of institutional environments and constituencies. Community Action Poverty Simulations, used widely in municipal and non-profit settings, provide brief structured immersive encounters with resource scarcity. Behavioral simulation programs in organizational settings, including virtual reality experiences designed to convey the experience of disadvantaged groups, all seek to use direct contact to correct perceptual asymmetries and have been found to be relevant in experimental studies (Andries et al., 2024; Piff et al., 2020). Our results provide causal support for this approach: direct comparative exposure to unequal circumstances produces larger belief revisions than providing participants with even detailed statistical or narrative information about the same inequality.

Finally, our results speak to the design of selection and promotion processes in organizations. When evaluators systematically underestimate how much structural conditions shape performance, assessments of merit can be biased against candidates who faced greater barriers. Institutions can respond by exposing evaluators to the environments they assess, and by ensuring that decision-making committees include people with diverse experiences.

6.5 Open Questions

Our findings raise several questions for future research. A first question concerns the persistence of experience effects. Our experiment shows that even a short exposure to unequal opportunities can change beliefs, fairness views, and willingness to pay for redistribution. An important next step is to examine whether these effects persist beyond the immediate decision environment, whether repeated exposure amplifies them, and whether individuals return to prior beliefs once the experience becomes less salient.

A second question concerns external validity across domains. Our design studies unequal opportunities in a controlled effort task, which allows us to isolate the experience of advantage or disadvantage from success and failure. This clean separation is central to identification, but real-world inequalities are more complex. Opportunities differ along multiple dimensions, their effects are harder to observe, and individuals often disagree about which barriers are legitimate. Future research could study whether similar experience effects arise in settings closer to education, labor markets, hiring, or organizational evaluation, where judgments of merit are consequential and where structural barriers are less transparent.

A third question concerns the channels through which experience operates. Our evidence suggests that belief updating plays an important role. Participants who experience disadvantage become less willing to interpret performance as a clean signal of merit. At the same time, direct experience also affects emotional responses and perspective-taking, especially when participants encounter both sides of the opportunity structure. These channels are likely to be mutually reinforcing rather than independent. Future work could experimentally separate information, salience, emotion, and perspective-taking to identify when each channel matters most for redistribution.

A fourth question concerns the relationship between experience and information. Our extension study shows that reports and statistics do not fully reproduce the effect of direct comparative experience. This raises a broader question about what kinds of information can substitute for experience, and when. Narratives may make disadvantage easier to imagine, while statistics may communicate the size of the opportunity gap more precisely. Yet neither fully replicates the effect of direct exposure. Future work could vary the intensity, duration, format, and credibility of information to understand when information affects only beliefs, when it impacts fairness views, and when it also generates willingness to act on those views. A related question is how experience and information interact. Information may be more effective when it is received after experience, because experience can provide a concrete reference point that makes abstract facts easier to interpret. Conversely, information may shape how individuals process subsequent experiences by directing attention to the role of circumstances. Studying these interactions would help clarify whether experience and information are substitutes, complements, or inputs into a sequential process of belief formation.

Finally, our results open questions about institutional design. If experience changes how individuals evaluate merit, then the composition and exposure of decision-makers may matter for redistributive decisions in organizations and policy settings. Evaluators, managers, teachers, and policymakers often judge outcomes without directly observing the constraints under which they were produced. Future work could study whether structured exposure to different opportunity environments improves evaluations of merit, reduces attribution bias, or changes support for compensatory policies. More broadly, our findings suggest that understanding inequality may require not only communicating facts about unequal opportunities, but also designing contexts in which people can better perceive how those opportunities shape performance.

7 Conclusion

This paper studies whether experience with unequal opportunities shapes redistributive preferences. We implement a spectator experiment in which participants first experience either an advantaged or disadvantaged version of an effort task and then decide how to redistribute earnings between two workers whose unequal opportunities arise from differences in task difficulty. We find that experiencing disadvantage affects how participants interpret unequal outcomes: they become more likely to recognize the role of circumstances, less likely to view performance as a pure signal of merit, and more willing to

redistribute toward the disadvantaged worker. This effect persists when redistribution requires paying a personal cost, showing that experience affects not only what spectators regard as fair, but also their willingness to act on that view.

The extension study shows that comparative experience is even more powerful. Participants who experience both the easy and hard versions redistribute more than those who experience only one side of the opportunity structure, suggesting that direct exposure to the contrast between advantage and disadvantage helps individuals understand how circumstances shape performance. Information provision through reports or statistics also conveys unequal opportunities, but produces weaker redistributive responses than direct experience. Taken together, the results suggest that redistributive preferences are shaped not only by ideology or self-interest, but also by the experiences through which individuals learn to interpret inequality. Correcting misperceptions may therefore require more than communicating facts: it may also require encounters through which people experience how unequal opportunities shape economic outcomes.

References

- Abeler, J., Falk, A., Goette, L., and Huffman, D. (2011). “Reference Points and Effort Provision”. *American Economic Review* 101(2): 470–92.
- Almås, I., Cappelen, A. W., Sørensen, E. Ø., and Tungodden, B. (2025). “Fairness Across the World”. Unpublished Working Paper.
- Almås, I., Cappelen, A. W., and Tungodden, B. (2020). “Cutthroat Capitalism versus Cuddly Socialism: Are Americans More Meritocratic and Efficiency-Seeking than Scandinavians?”. *Journal of Political Economy* 128(5): 1753–1788.
- Altmejd, A., Barrios-Fernández, A., Drlje, M. et al. (2021). “O Brother, Where Start Thou? Sibling Spillovers on College and Major Choice in Four Countries”. *Quarterly Journal of Economics* 136(3): 1831–1886.
- Amasino, D. R., Pace, D. D., and Van der Weele, J. J. (2024). “Fair shares and selective attention”. *American Economic Journal: Microeconomics* 16(4): 259–290.
- Amemiya, J., Mortenson, E., Heyman, G. D., and Walker, C. M. (2023). “Thinking structurally: A cognitive framework for understanding how people attribute inequality to structural causes”. *Perspectives on Psychological Science* 18(2): 259–274.
- Andre, P. (2024). “Shallow Meritocracy”. *Review of Economic Studies*: rdae040.
- Andries, M., Bursztyn, L., Chaney, T., and Djourelouva, M. (2024). “In their shoes”. *University of Chicago, Becker Friedman Institute for Economics Working Paper*(2024-74).
- Bhattacharya, P., and Mollerstrom, J. (2025). “Lucky to Work”. *Review of Economics and Statistics*: 1–44.
- Bordalo, P., Gennaioli, N., and Shleifer, A. (2020). “Memory, attention, and choice”. *Quarterly Journal of Economics* 135(3).
- Brun, M. (2026). “The Complexity of Being Fair”. FIT Working Paper No. 44.
- Bursztyn, L., Fujiwara, T., and Pallais, A. (2017). “‘Acting Wife’: Marriage Market Incentives and Labor Market Investments”. *American Economic Review* 107(11): 3288–3319.
- Busse, M. R., Pope, D. G., Pope, J. C., and Silva-Risso, J. (2015). “The Psychological Effect of Weather on Car Purchases”. *Quarterly Journal of Economics* 130(1): 371–414.
- Cappelen, A. W., Cappelen, C., and Tungodden, B. (2023). “Second-best fairness: The trade-off between false positives and false negatives”. *American Economic Review* 113(9): 2458–2485.
- Cappelen, A. W., Falch, R., and Tungodden, B. (2020). *Fair and Unfair Income Inequality*: 1–25, Springer International Publishing.

- Cappelen, A. W., Hole, A. D., Sørensen, E. Ø., and Tungodden, B. (2007). “The Pluralism of Fairness Ideals: An Experimental Approach”. *American Economic Review* 97(3): 818–827.
- Cappelen, A. W., Liu, Y., Nielsen, H., and Tungodden, B. (2024). “Fairness in a Society of Unequal Opportunities”. Rationality and Competition Discussion Paper No. 506.
- Cappelen, A. W., Moene, K. O., Sørensen, E. Ø., and Tungodden, B. (2013). “Needs Versus Entitlements—An International Fairness Experiment”. *Journal of the European Economic Association* 11(3): 574–598.
- Cassar, L., and Klein, A. H. (2019). “A Matter of Perspective: How Failure Shapes Distributive Preferences”. *Management Science* 65(11): 4951–5448.
- Chen, D. L., Schonger, M., and Wickens, C. (2016). “oTree—An open-source platform for laboratory, online, and field experiments”. *Journal of Behavioral and Experimental Finance* 9: 88–97.
- Chen, M., Zhang, R., Li, Y., Liu, J., and Li, X. (2026). “Intentions versus outcomes: Determinants of costly third-party interventions in fairness maintenance”. *Journal of Experimental Social Psychology* 122: 104838.
- Chen, R., Radkani, S., Valluru, N., Yoo, S. B. M., and Jazayeri, M. (2026). “Evidence accumulation from experience and observation in the cingulate cortex”. *Nature* 650(8102): 681–689.
- Conlin, M., O’Donoghue, T., and Vogelsang, T. J. (2007). “Projection Bias in Catalog Orders”. *American Economic Review* 97(4): 1217–1249.
- Deffains, B., Espinosa, R., and Thöni, C. (2016). “Political self-serving bias and redistribution”. *Journal of Public Economics* 134: 67–74.
- Dimant, E., Reinhardt, L., and Sambanis, N. (2024). “Racial identity, reactions to inequality, and fairness concepts among Americans”. *Reactions to Inequality, and Fairness Concepts among Americans* (November 03, 2024).
- DiPrete, T. A., and Eirich, G. M. (2006). “Cumulative advantage as a mechanism for inequality: A review of theoretical and empirical developments”. *Annu. Rev. Sociol.* 32(1): 271–297.
- Dong, L., Huang, L., and Lien, J. W. (2024). “‘They Never had a Chance’: Unequal Opportunities and Fair Redistributions”. *The Economic Journal* 135(667): 914–942.
- Dwenger, N., Hoem Sjørnsen, I., and Vietz, J. (2024). “What Is Fair? Experimental Evidence on Fair Equality vs Fair Inequality”.
- Espinosa, R., Deffains, B., and Thöni, C. (2020). “Debiasing preferences over redistribution: an experiment”. *Social Choice and Welfare* 55(4): 823–843.
- Falk, A., Kosse, F., and Pinger, P. (2026). “Mentoring and Schooling Decisions: Causal Evidence”. *Journal of Political Economy* 134(1): 366–396.

- Fehr, D., and Vollmann, M. (2025).** “Misperceiving economic success: experimental evidence on meritocratic beliefs and inequality acceptance”. *The Journal of Economic Inequality* 23(3): 835–855.
- Glover, D., Pallais, A., and Pariente, W. (2017).** “Discrimination as a Self-Fulfilling Prophecy: Evidence from French Grocery Stores”. *Quarterly Journal of Economics* 132(3): 1219–1260.
- Gonzalez-Jimenez, D. (2026).** “Learning Gaps? Attitudes and Beliefs Under Described and Experienced Signals”.
- Graeber, T., Roth, C., and Zimmermann, F. (2024).** “Stories, Statistics, and Memory”. *The Quarterly Journal of Economics* 139(4): 2181–2225.
- Haaland, I., Roth, C., and Wohlfart, J. (2023).** “Designing information provision experiments”. *Journal of economic literature* 61(1): 3–40.
- Harrs, S., and Sterba, M.-B. (2023).** “Fairness and Support for Welfare Policies”. Unpublished Working Paper.
- Hertwig, R. (2015).** “Decisions from Experience”. In *The Wiley Blackwell Handbook of Judgment and Decision Making*,: 239–267, Wiley.
- Hertwig, R., and Erev, I. (2009).** “The description–experience gap in risky choice”. *Trends in cognitive sciences* 13(12): 517–523.
- Jia, T. (2022).** “Empathy, Motivated Reasoning, And Redistribution”. Unpublished Working Paper.
- Jones, E. E., and Harris, V. A. (1967).** “The attribution of attitudes”. *Journal of Experimental Social Psychology* 3(1): 1–24.
- Konow, J. (2000).** “Fair Shares: Accountability and Cognitive Dissonance in Allocation Decisions”. *American Economic Review* 90(4): 1072–1091.
- Kuchler, T., and Zafar, B. (2019).** “Personal Experiences and Expectations about Aggregate Outcomes”. *Journal of Finance* 74(5): 2491–2542.
- Malmendier, U. (2021a).** “FBBVA Lecture 2020 Exposure, Experience, and Expertise: Why Personal Histories Matter in Economics”. *Journal of the European Economic Association* 19(6): 2857–2894.
- Malmendier, U. (2021b).** “Experience Effects in Finance: Foundations, Applications, and Future Directions”. *Review of Finance* 25(5): 1339–1363.
- Malmendier, U., and Nagel, S. (2016).** “Learning from Inflation Experiences”. *Quarterly Journal of Economics* 131(1): 53–87.
- Malmendier, U., and Wachter, J. A. (2024).** *Memory of Past Experiences and Economic Decisions*: 2228–2266, Oxford University Press.
- Morin, O., Jacquet, P. O., Vaesen, K., and Acerbi, A. (2021).** “Social information use and social information waste”. *Philosophical Transactions of the Royal Society B* 376(1828).

- Parsons, C. A., Sulaeman, J., Yates, M. C., and Hamermesh, D. S. (2011).** “Strike Three: Discrimination, Incentives, and Evaluation”. *American Economic Review* 101(4): 1410–35.
- Piff, P. K., Wiwad, D., Robinson, A. R., Aknin, L. B., Mercier, B., and Shariff, A. (2020).** “Shifting attributions for poverty motivates opposition to inequality and enhances egalitarianism”. *Nature Human Behaviour* 4(5): 496–505.
- Piraino, P., and Ryan, J. (2024).** “The price of fairness: Experimental evidence on the limits to demand for redistribution”. *Journal of Economic Behavior & Organization* 227: 106737.
- Pogliano, A. (2024).** “Born that way: Beliefs about genetics’ importance and redistribution preferences”. Tinbergen Institute Discussion Paper No. TI 2024-017/I.
- Pruess, M., Reyes, G., Somerville, J., and Wu, J. (2025).** “Inequality of Opportunity and Income Redistribution”. *Journal of Political Economy Microeconomics*.
- Slovic, P. (2007).** ““If I Look at the Mass I Will Never Act”: Psychic Numbing and Genocide”. *Judgment and Decision Making* 2(2): 79–95.
- Small, D. A., Loewenstein, G., and Slovic, P. (2007).** “Sympathy and Callousness: The Impact of Deliberative Thought on Donations to Identifiable and Statistical Victims”. *Organizational Behavior and Human Decision Processes* 102(2): 143–153.
- Stantcheva, S. (2021).** “Understanding Tax Policy: How do People Reason?”. *Quarterly Journal of Economics* 136(4): 2309–2369.
- Thaler, M., Toma, M., and Wang, V. Y. (2024).** “Numbers tell, words sell”. Technical report, CESifo.

A Data and Sample Characteristics

Variable	Type	Question
	(1)	(2)
PERFORMANCE	Integer	[Number of tables solved correctly]
AGENTAB_SHAREB	0-6	How would you split the 6 USD bonus?
PERSUASION	Open Text	What would you like to tell the participant before they decide how to split the bonus?

Variable	Type	Question
	(1)	(2)
PRIOR_RANKING	Categorical (9 categories)	What matters more for deciding who wins?
PRIOR_ADVANTAGE	Categorical (4 categories)	If there is a difference between the versions, which version do you expect would lead to better performance?
PRIOR_INTENSITY	1-7 score	In a scale from 1 to 7, where 1 is 'Getting Version A is a major advantage' and 7 is 'Getting Version B is a major advantage', Which version do you think provides a greater advantage for performance?
ATTENTION_CHECK1	Categorical (4 categories)	The following question is an attention check. Please select letter C. Select one of the following letters.
EXPERIENCE_CHECK	Categorical (2 categories)	Which version of the task did you just complete?
SPECTATORS_PERFORMANCE	Integer	# of succesfully completed tables
POSTERIOR_PERFORMANCEA	Integer	On average, how many tables do you think Agents A successfully solved in 4 minutes?
CERTAINTY_PERFORMANCEA	Ordinal (4 categories)	How sure are you about your answer?
POSTERIOR_PERFORMANCEB	Integer	On average, how many tables do you think Agents B successfully solved in 4 minutes?
CERTAINTY_PERFORMANCEB	Ordinal (4 categories)	How sure are you about your answer?

Variable	Type	Question
	(1)	(2)
POSTERIOR_FREQUENCY	1-100	In how many pairs (out of the 100) do you think that the Agent B solves more tables than the Agent A?
CERTAINTY_FREQUENCY	Ordinal (4 categories)	How sure are you about your answer?
ATTENTION_CHECK2	Binary	Attention check: Do not select this option.
SPECTATORAB_SHAREB	0-6	How would you split the 6 USD bonus?
POSTERIOR_EXPLANATION	Open Text	Explain, in at least 2 or 3 sentences, the reasoning behind why you made this decision.
SPECTATORABCOSTLY_SHAREB	0-6	How would you split the 6 USD bonus? (redistribution costs 0.20 USD)
POSTERIOR_FAIR	Ordinal (4 categories)	In this pair, how fair was that the agent that solved more correct tables was initially assigned all the bonus?
POSTERIOR_HARDWORKGL	Ordinal (5 categories)	Imagine all agents complete the same Version of the task, but are still randomly assigned to be called either Agents A or Agents B. Which group of agents do you think would solve more correct tables?
POSTERIOR_HARDWORK	Ordinal (4 categories)	In this pair, if Agent B had worked really hard they could have solved more correct tables than Agent A.
POSTERIOR_COUNTERFACTUAL	Categorical (3 categories)	Imagine that both agents in this pair completed the same Version of the task. Who do you think would have solved more tables in that case?
POSTERIOR_EFFORT	0-100	How much does effort explain which of the agents solves more tables?
POSTERIOR_ABILITY	0-100	How much does ability explain which of the agents solves more tables?
POSTERIOR_LUCK	0-100	How much does luck explain which of the agents solves more tables?
POSTERIOR_INTENSITY	1-7 score	Do you think the version of the task assigned to each agent provided any advantage in the competition?

Variable	Type	Question
	(1)	(2)
POSTERIOR_EXPERIENCEA	Ordinal (4 categories)	It is easy for me to imagine Agent A's experience with the task.
POSTERIOR_EXPERIENCEB	Ordinal (4 categories)	It is easy for me to imagine Agent B's experience with the task.
POSTERIOR_ANGER	Ordinal (4 categories)	How much did you feel Anger?
POSTERIOR_COMPASSION	Ordinal (4 categories)	How much did you feel Compassion?
POSTERIOR_SADNESS	Ordinal (4 categories)	How much did you feel Sadness?
POSTERIOR_INDIFFERENCE	Ordinal (4 categories)	How much did you feel Indifference?
POSTERIOR_IRRITATION	Ordinal (4 categories)	How much did you feel Irritation?
SPECTATORLUCK_SHAREB	0-6	How would you split the 6 USD bonus?
SPECTATORMERIT_SHAREB	0-6	How would you split the 6 USD bonus?
SPECTATORUNDERDOG_SHAREB	0-6	How would you split the 6 USD bonus?
SPECTATORVIGNETTE_SHAREB	0-6	How would you split the 6 USD bonus?
VIEWS	Categorical (5 categories)	...

Table A3: Balance Test in the Main Study

	Total	EASY	HARD	p-value
N	651	336	315	
Sex				.714
Male	302 (46.4%)	150 (44.6%)	152 (48.3%)	
Female	340 (52.2%)	182 (54.2%)	158 (50.2%)	
Other	3 (0.5%)	1 (0.3%)	2 (0.6%)	
N/A	6 (0.9%)	3 (0.9%)	3 (1.0%)	
Age				.771
Years	43.7 (12.9)	43.5 (12.5)	43.8 (13.3)	
Race				.548
White	454 (69.7%)	231 (68.8%)	223 (70.8%)	
Black	101 (15.5%)	49 (14.6%)	52 (16.5%)	
Asian	52 (8.0%)	30 (8.9%)	22 (7.0%)	
Native	2 (0.3%)	2 (0.6%)	0 (0.0%)	
Other	15 (2.3%)	8 (2.4%)	7 (2.2%)	
Multiple races	18 (2.8%)	12 (3.6%)	6 (1.9%)	
No race selected	9 (1.4%)	4 (1.2%)	5 (1.6%)	
Education				.176
Less than high school	3 (0.5%)	3 (0.9%)	0 (0.0%)	
High school	69 (10.6%)	38 (11.3%)	31 (9.8%)	
College, not degree	87 (13.4%)	47 (14.0%)	40 (12.7%)	
Associate degree	55 (8.4%)	20 (6.0%)	35 (11.1%)	
Bachelor degree	266 (40.9%)	137 (40.8%)	129 (41.0%)	
Master degree	140 (21.5%)	75 (22.3%)	65 (20.6%)	
Professional degree	13 (2.0%)	5 (1.5%)	8 (2.5%)	
Doctorate degree	18 (2.8%)	11 (3.3%)	7 (2.2%)	
Employment				.144
Full-time	373 (57.3%)	203 (60.4%)	170 (54.0%)	
Part-time	98 (15.1%)	46 (13.7%)	52 (16.5%)	
Self-employed	69 (10.6%)	42 (12.5%)	27 (8.6%)	
Unemployed and looking	41 (6.3%)	15 (4.5%)	26 (8.3%)	
Unemployed and non-looking	12 (1.8%)	4 (1.2%)	8 (2.5%)	
Student	8 (1.2%)	4 (1.2%)	4 (1.3%)	
Retired	32 (4.9%)	14 (4.2%)	18 (5.7%)	
Homecare	18 (2.8%)	8 (2.4%)	10 (3.2%)	

Notes: This table reports demographic balance in the main study. Columns (1)–(3) report statistics for the full sample and by treatment arm. For categorical variables, cells report frequencies and column shares in parentheses. Age reports means and standard deviations in parentheses. Race is coded as mutually exclusive: participants selecting more than one race are classified as Multiple races, and participants selecting no race are classified as No race selected. p-values are from Pearson χ^2 tests for categorical variables and Welch two-sample t-tests for age.

Table A4: Balance Test in the Main Study: Socio-economic and Political Characteristics

	Total	EASY	HARD	p-value
N	651	336	315	
Income (individual)				.002
0	11 (1.7%)	10 (3.0%)	1 (0.3%)	
<10k	61 (9.4%)	19 (5.7%)	42 (13.3%)	
10k–15k	34 (5.2%)	17 (5.1%)	17 (5.4%)	
15k–20k	34 (5.2%)	12 (3.6%)	22 (7.0%)	
20k–25k	36 (5.5%)	21 (6.2%)	15 (4.8%)	
25k–50k	143 (22.0%)	71 (21.1%)	72 (22.9%)	
50k–100k	228 (35.0%)	127 (37.8%)	101 (32.1%)	
>100k	94 (14.4%)	54 (16.1%)	40 (12.7%)	
N/A	10 (1.5%)	5 (1.5%)	5 (1.6%)	
Household income				.668
<20k	48 (7.4%)	22 (6.5%)	26 (8.3%)	
20k–35k	71 (10.9%)	39 (11.6%)	32 (10.2%)	
35k–50k	78 (12.0%)	40 (11.9%)	38 (12.1%)	
50k–75k	100 (15.4%)	44 (13.1%)	56 (17.8%)	
75k–100k	113 (17.4%)	60 (17.9%)	53 (16.8%)	
100k–150k	121 (18.6%)	62 (18.5%)	59 (18.7%)	
150k–200k	66 (10.1%)	39 (11.6%)	27 (8.6%)	
200k–250k	22 (3.4%)	14 (4.2%)	8 (2.5%)	
250k–300k	9 (1.4%)	6 (1.8%)	3 (1.0%)	
>300k	11 (1.7%)	5 (1.5%)	6 (1.9%)	
N/A	12 (1.8%)	5 (1.5%)	7 (2.2%)	
Political party				.193
Democrat	261 (40.1%)	125 (37.2%)	136 (43.2%)	
Republican	217 (33.3%)	122 (36.3%)	95 (30.2%)	
Independent	173 (26.6%)	89 (26.5%)	84 (26.7%)	

Notes: This table reports socio-economic and political balance in the main study. Columns (1)–(3) report statistics for the full sample and by treatment arm. Cells report frequencies and column shares in parentheses. p-values are from Pearson χ^2 tests.

Table A5: Balance Test in the Extension Sample

	Total	MIXED	REPORTS	STATISTICS	p-value
N	502	165	165	172	
Sex					.016
Male	226 (45.0%)	80 (48.5%)	82 (49.7%)	64 (37.2%)	
Female	267 (53.2%)	82 (49.7%)	77 (46.7%)	108 (62.8%)	
Other	5 (1.0%)	1 (0.6%)	4 (2.4%)	0 (0.0%)	
N/A	4 (0.8%)	2 (1.2%)	2 (1.2%)	0 (0.0%)	
Age					.998
Years	38.4 (13.2)	38.4 (13.8)	38.4 (12.8)	38.4 (13.0)	
Race					.367
White	305 (60.8%)	100 (60.6%)	107 (64.8%)	98 (57.0%)	
Black	105 (20.9%)	37 (22.4%)	33 (20.0%)	35 (20.3%)	
Asian	43 (8.6%)	12 (7.3%)	12 (7.3%)	19 (11.0%)	
Native	3 (0.6%)	2 (1.2%)	0 (0.0%)	1 (0.6%)	
Other	15 (3.0%)	8 (4.8%)	3 (1.8%)	4 (2.3%)	
Multiple races	20 (4.0%)	3 (1.8%)	8 (4.8%)	9 (5.2%)	
No race selected	11 (2.2%)	3 (1.8%)	2 (1.2%)	6 (3.5%)	
Education					.809
Less than high school	4 (0.8%)	2 (1.2%)	1 (0.6%)	1 (0.6%)	
High school	60 (12.0%)	19 (11.5%)	24 (14.5%)	17 (9.9%)	
College, not degree	95 (18.9%)	30 (18.2%)	29 (17.6%)	36 (20.9%)	
Associate degree	46 (9.2%)	15 (9.1%)	14 (8.5%)	17 (9.9%)	
Bachelor degree	189 (37.6%)	66 (40.0%)	57 (34.5%)	66 (38.4%)	
Master degree	81 (16.1%)	26 (15.8%)	29 (17.6%)	26 (15.1%)	
Professional degree	11 (2.2%)	2 (1.2%)	7 (4.2%)	2 (1.2%)	
Doctorate degree	16 (3.2%)	5 (3.0%)	4 (2.4%)	7 (4.1%)	

Notes: This table reports demographic balance in the extension sample. Columns (1)–(4) report statistics for the full sample and by treatment arm. For categorical variables, cells report frequencies and column shares in parentheses. Age reports means and standard deviations in parentheses. Race is coded as mutually exclusive: participants selecting more than one race are classified as Multiple races, and participants selecting no race are classified as No race selected. p-values are from Pearson χ^2 tests for categorical variables and one-way ANOVA for age.

Table A6: Balance Test in the Extension Sample: Socio-economic and Political Characteristics

	Total	MIXED	REPORTS	STATISTICS	p-value
N	502	165	165	172	
Employment					.333
Full-time	246 (49.0%)	79 (47.9%)	82 (49.7%)	85 (49.4%)	
Part-time	73 (14.5%)	25 (15.2%)	21 (12.7%)	27 (15.7%)	
Self-employed	52 (10.4%)	17 (10.3%)	17 (10.3%)	18 (10.5%)	
Unemployed and looking	52 (10.4%)	18 (10.9%)	23 (13.9%)	11 (6.4%)	
Unemployed and non-looking	13 (2.6%)	3 (1.8%)	3 (1.8%)	7 (4.1%)	
Student	28 (5.6%)	6 (3.6%)	7 (4.2%)	15 (8.7%)	
Retired	18 (3.6%)	7 (4.2%)	6 (3.6%)	5 (2.9%)	
Homecare	20 (4.0%)	10 (6.1%)	6 (3.6%)	4 (2.3%)	
Income (individual)					.987
0	14 (2.8%)	5 (3.0%)	2 (1.2%)	7 (4.1%)	
<10k	67 (13.3%)	23 (13.9%)	24 (14.5%)	20 (11.6%)	
10k–15k	38 (7.6%)	12 (7.3%)	14 (8.5%)	12 (7.0%)	
15k–20k	20 (4.0%)	7 (4.2%)	5 (3.0%)	8 (4.7%)	
20k–25k	27 (5.4%)	9 (5.5%)	10 (6.1%)	8 (4.7%)	
25k–50k	101 (20.1%)	34 (20.6%)	31 (18.8%)	36 (20.9%)	
50k–100k	156 (31.1%)	51 (30.9%)	52 (31.5%)	53 (30.8%)	
>100k	59 (11.8%)	17 (10.3%)	19 (11.5%)	23 (13.4%)	
N/A	20 (4.0%)	7 (4.2%)	8 (4.8%)	5 (2.9%)	
Household income					.715
<20k	42 (8.4%)	15 (9.1%)	14 (8.5%)	13 (7.6%)	
20k–35k	41 (8.2%)	14 (8.5%)	12 (7.3%)	15 (8.7%)	
35k–50k	56 (11.2%)	22 (13.3%)	17 (10.3%)	17 (9.9%)	
50k–75k	81 (16.1%)	31 (18.8%)	24 (14.5%)	26 (15.1%)	
75k–100k	77 (15.3%)	23 (13.9%)	25 (15.2%)	29 (16.9%)	
100k–150k	113 (22.5%)	33 (20.0%)	39 (23.6%)	41 (23.8%)	
150k–200k	38 (7.6%)	12 (7.3%)	12 (7.3%)	14 (8.1%)	
200k–250k	18 (3.6%)	3 (1.8%)	11 (6.7%)	4 (2.3%)	
250k–300k	3 (0.6%)	0 (0.0%)	2 (1.2%)	1 (0.6%)	
>300k	13 (2.6%)	4 (2.4%)	2 (1.2%)	7 (4.1%)	
N/A	20 (4.0%)	8 (4.8%)	7 (4.2%)	5 (2.9%)	
Political party					.234
Democrat	221 (44.0%)	63 (38.2%)	71 (43.0%)	87 (50.6%)	
Republican	135 (26.9%)	50 (30.3%)	46 (27.9%)	39 (22.7%)	
Independent	146 (29.1%)	52 (31.5%)	48 (29.1%)	46 (26.7%)	

Notes: This table reports socio-economic and political balance in the extension sample. Columns (1)–(4) report statistics for the full sample and by treatment arm. Cells report frequencies and column shares in parentheses. p-values are from Pearson χ^2 tests.

Table A7: Comparison of Demographic Composition between the Main Study and Extension Sample

	Total	Main	Extension	p-value
N	1153	651	502	
Sex				.709
Male	528 (45.8%)	302 (46.4%)	226 (45.0%)	
Female	607 (52.6%)	340 (52.2%)	267 (53.2%)	
Other	8 (0.7%)	3 (0.5%)	5 (1.0%)	
N/A	10 (0.9%)	6 (0.9%)	4 (0.8%)	
Age				< .001
Years	41.4 (13.3)	43.7 (12.9)	38.4 (13.2)	
Race				.071
White	759 (65.8%)	454 (69.7%)	305 (60.8%)	
Black	206 (17.9%)	101 (15.5%)	105 (20.9%)	
Asian	95 (8.2%)	52 (8.0%)	43 (8.6%)	
Native	5 (0.4%)	2 (0.3%)	3 (0.6%)	
Other	30 (2.6%)	15 (2.3%)	15 (3.0%)	
Multiple races	38 (3.3%)	18 (2.8%)	20 (4.0%)	
No race selected	20 (1.7%)	9 (1.4%)	11 (2.2%)	
Education				.103
Less than high school	7 (0.6%)	3 (0.5%)	4 (0.8%)	
High school	129 (11.2%)	69 (10.6%)	60 (12.0%)	
College, not degree	182 (15.8%)	87 (13.4%)	95 (18.9%)	
Associate degree	101 (8.8%)	55 (8.4%)	46 (9.2%)	
Bachelor degree	455 (39.5%)	266 (40.9%)	189 (37.6%)	
Master degree	221 (19.2%)	140 (21.5%)	81 (16.1%)	
Professional degree	24 (2.1%)	13 (2.0%)	11 (2.2%)	
Doctorate degree	34 (2.9%)	18 (2.8%)	16 (3.2%)	

Notes: This table compares demographic composition between the main study and the extension sample. Columns (1)–(3) report statistics for the full sample and by sample. For categorical variables, cells report frequencies and column shares in parentheses. Age reports means and standard deviations in parentheses. Race is coded as mutually exclusive: participants selecting more than one race are classified as Multiple races, and participants selecting no race are classified as No race selected. p-values are from Pearson χ^2 tests for categorical variables and Welch two-sample t-tests for age.

Table A8: Comparison of Socio-economic and Political Characteristics between the Main Study and Extension Sample

	Total	Main	Extension	p-value
N	1153	651	502	
Employment				< .001
Full-time	619 (53.7%)	373 (57.3%)	246 (49.0%)	
Part-time	171 (14.8%)	98 (15.1%)	73 (14.5%)	
Self-employed	121 (10.5%)	69 (10.6%)	52 (10.4%)	
Unemployed and looking	93 (8.1%)	41 (6.3%)	52 (10.4%)	
Unemployed and non-looking	25 (2.2%)	12 (1.8%)	13 (2.6%)	
Student	36 (3.1%)	8 (1.2%)	28 (5.6%)	
Retired	50 (4.3%)	32 (4.9%)	18 (3.6%)	
Homecare	38 (3.3%)	18 (2.8%)	20 (4.0%)	
Income (individual)				.015
0	25 (2.2%)	11 (1.7%)	14 (2.8%)	
<10k	128 (11.1%)	61 (9.4%)	67 (13.3%)	
10k–15k	72 (6.2%)	34 (5.2%)	38 (7.6%)	
15k–20k	54 (4.7%)	34 (5.2%)	20 (4.0%)	
20k–25k	63 (5.5%)	36 (5.5%)	27 (5.4%)	
25k–50k	244 (21.2%)	143 (22.0%)	101 (20.1%)	
50k–100k	384 (33.3%)	228 (35.0%)	156 (31.1%)	
>100k	153 (13.3%)	94 (14.4%)	59 (11.8%)	
N/A	30 (2.6%)	10 (1.5%)	20 (4.0%)	
Household income				.124
<20k	90 (7.8%)	48 (7.4%)	42 (8.4%)	
20k–35k	112 (9.7%)	71 (10.9%)	41 (8.2%)	
35k–50k	134 (11.6%)	78 (12.0%)	56 (11.2%)	
50k–75k	181 (15.7%)	100 (15.4%)	81 (16.1%)	
75k–100k	190 (16.5%)	113 (17.4%)	77 (15.3%)	
100k–150k	234 (20.3%)	121 (18.6%)	113 (22.5%)	
150k–200k	104 (9.0%)	66 (10.1%)	38 (7.6%)	
200k–250k	40 (3.5%)	22 (3.4%)	18 (3.6%)	
250k–300k	12 (1.0%)	9 (1.4%)	3 (0.6%)	
>300k	24 (2.1%)	11 (1.7%)	13 (2.6%)	
N/A	32 (2.8%)	12 (1.8%)	20 (4.0%)	
Political party				.063
Democrat	482 (41.8%)	261 (40.1%)	221 (44.0%)	
Republican	352 (30.5%)	217 (33.3%)	135 (26.9%)	
Independent	319 (27.7%)	173 (26.6%)	146 (29.1%)	

Notes: This table compares socio-economic and political characteristics between the main study and the extension sample. Columns (1)–(3) report statistics for the full sample and by sample. Cells report frequencies and column shares in parentheses. p-values are from Pearson χ^2 tests.

B Additional Results in Main Study

Table B1: Beliefs after the intervention

	Total	EASY	HARD	Diff.
	(1)	(2)	(3)	(4)
Panel A. Individual-level				
Initial assignment is fair	.736	.764	.706	-.058*** (.019)
Agent B needs more effort	.656	.702	.606	-.097*** (.019)
Agent A would win if assigned same version	.398	.494	.295	-.199*** (.038)
Panel B. Group-level				
Agents A would win if assigned same version	.339	.402	.273	-.129*** (.037)
Improved performance with easy version	.237	.164	.316	.152*** (.045)
Probability that Agents A outperform Agents B	.654	.648	.661	.013 (.016)
Tables completed with easy version	7.46	9.55	5.24	-4.30*** (.33)
Tables completed with hard version	6.48	8.41	4.41	-4.00*** (.29)

Notes: This table reports beliefs collected after the intervention. Columns (1) refers to the whole sample. Columns (2) and (3) distinguishes by treatment arm. The first group (**EASY**) is exposed to the easy version of the task. The second group (**HARD**) is exposed to the hard version of the task. Columns (2) and (3) report statistics in each group. Columns (4) reports the difference between the two groups. Each row reports estimates from a linear model. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

Table B2: Heterogeneities in Prior Beliefs

	Total	EASY	HARD	Diff.
	(1)	(2)	(3)	(4)
Panel A. Demographics				
Men	.831	.807	.855	.049 (.043)
Women	.888	.896	.880	-.016 (.035)
Age <40	.855	.857	.853	-.004 (.043)
Age $\geq$ 40	.867	.852	.883	.031 (.035)
White	.860	.847	.873	.026 (.032)
Non-White	.867	.872	.862	-.010 (.051)
Panel B. Education, Employment and Income				
with Bachelor Degree	.865	.842	.890	.048 (.033)
w/o Bachelor Degree	.855	.880	.830	-.049 (.048)
Employed full-time	.863	.857	.871	.013 (.036)
not Employed full-time	.860	.850	.869	.019 (.042)
Annual Income $\geq$ 25k	.869	.886	.856	-.030 (.051)
Annual Income <25k	.883	.879	.887	.008 (.049)
Panel C. Political Party				
Democratic	.874	.864	.882	.018 (.041)
Republican	.876	.861	.895	.034 (.045)
Independent	.827	.831	.821	-.010 (.058)

Notes: This table reports prior beliefs on whether the versions of the task are similar. Columns (1) refers to the whole sample. Columns (2) and (3) distinguishes by treatment arm. The first group (**EASY**) is exposed to the easy version of the task. The second group (**HARD**) is exposed to the hard version of the task. Columns (2) and (3) report statistics in each group. Columns (4) reports the difference between the two groups. Each row reports estimates from a linear model. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

Table B3: Heterogeneities in Posterior Beliefs

	Total	EASY	HARD	Diff.
	(1)	(2)	(3)	(4)
Panel A. Demographics				
Men	.570	.627	.513	-.114** (.057)
Women	.562	.659	.449	-.210*** (.053)
Age <40	.551	.564	.537	-.028 (.060)
Age ≥40	.571	.689	.441	-.247*** (.050)
White	.564	.636	.487	-.150*** (.045)
Non-White	.558	.638	.471	-.167** (.073)
Panel B. Education, Employment and Income				
with Bachelor Degree	.586	.636	.531	-.105** (.047)
w/o Bachelor Degree	.514	.639	.387	-.252*** (.066)
Employed full-time	.606	.640	.565	-.076 (.051)
not Employed full-time	.504	.632	.386	-.245*** (.058)
Annual Income ≥25k	.500	.608	.412	-.195*** (.075)
Annual Income <25k	.608	.637	.575	-.062 (.075)
Panel C. Political Party				
Democratic	.582	.680	.493	-.187*** (.060)
Republican	.599	.672	.505	-.167** (.067)
Independent	.486	.528	.440	-.088 (.076)

Notes: This table reports posterior beliefs on whether the versions of the task are similar. Columns (1) refers to the whole sample. Columns (2) and (3) distinguishes by treatment arm. The first group (EASY) is exposed to the easy version of the task. The second group (HARD) is exposed to the hard version of the task. Columns (2) and (3) report statistics in each group. Columns (4) reports the difference between the two groups. Each row reports estimates from a linear model. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

Table B4: Heterogeneities in Redistribution Decisions

	Total	EASY	HARD	Diff.
	(1)	(2)	(3)	(4)
Panel A. Demographics				
Men	1.60	1.38	1.81	.43*** (.14)
Women	1.71	1.64	1.79	.15 (.13)
Age <40	1.65	1.57	1.73	.16 (.14)
Age ≥40	1.68	1.50	1.86	.35*** (.12)
White	1.68	1.52	1.84	.31*** (.11)
Non-White	1.63	1.55	1.71	.16 (.18)
Panel B. Education, Employment and Income				
with Bachelor Degree	1.60	1.47	1.74	.27** (.12)
w/o Bachelor Degree	1.79	1.67	1.93	.26 (.16)
Employed full-time	1.60	1.50	1.72	.21* (.13)
not Employed full-time	1.75	1.58	1.90	.33** (.14)
Annual Income ≥25k	1.65	1.63	1.67	.04 (.17)
Annual Income <25k	1.64	1.54	1.76	.22 (.20)
Panel C. Political Party				
Democratic	1.75	1.63	1.85	.22 (.14)
Republican	1.54	1.41	1.72	.31* (.17)
Independent	1.69	1.56	1.82	.26 (.19)

Notes: This table reports redistribution decisions. Columns (1) refers to the whole sample. Columns (2) and (3) distinguishes by treatment arm. The first group (**EASY**) is exposed to the easy version of the task. The second group (**HARD**) is exposed to the hard version of the task. Columns (2) and (3) report statistics in each group. Columns (4) reports the difference between the two groups. Each row reports estimates from a linear model. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

Table B5: Redistribution Decision by Context

	Same Context		Different Context	
	(1)	(2)	(3)	(4)
HARD	.270*** (.095) [.004]	.270*** (.100) [.007]	.202** (.096) [.036]	.170* (.100) [.088]
Covariates	No	Yes	No	Yes
Effect magn.	18%	18%	10%	8%
Observations	651	651	651	651
R^2	.012	.071	.007	.076

Notes: This table reports treatment effects redistribution decisions for same and different context. The independent variable is spectator's assignment to HARD. Columns (1) and (2) report redistribution decisions for same context, our main dependent variable. Columns (3) and (4) report redistribution decisions for a different context. The context features unequal opportunities, but under different tasks than those in the experience. The dependent variable in all cases is the dollar transfer to the disadvantaged agent. Each row reports estimates from a linear model. Columns (2) and (4) include a set of socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

Table B6: Treatment Effects on Reported Emotions

	Compassion	Indifference	Sadness	Irritation	Anger
	(1)	(2)	(3)	(4)	(5)
HARD	.026 (.027) [.328]	-.033 (.030) [.273]	.027 (.017) [.106]	.064*** (.017) [<.001]	.051*** (.016) [.001]
Mean dep. var.	.423	.371	.093	.073	.052
Effect magn.	6%	-9%	30%	88%	98%
Observations	651	651	651	651	651
R^2	.085	.068	.066	.126	.104

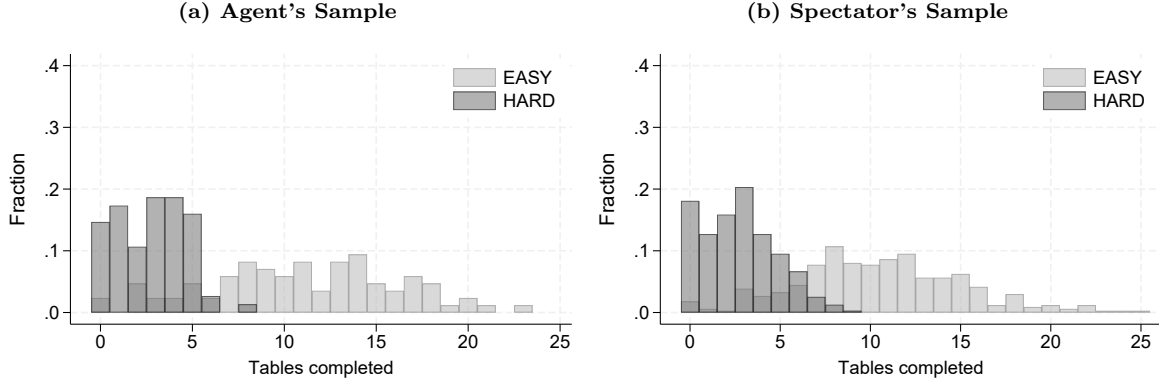
Notes: This table reports treatment effects on emotions reported by spectators. The dependent variable is an intensity value, ranging from 0 to 1. The independent variable is spectator's assignment to HARD. Each row reports estimates from a linear model. All columns include a set of socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

Table B7: Mechanism Exploration

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
HARD	.270*** (.100) [.007]	.244** (.096) [.011]	.243** (.098) [.013]	.240** (.098) [.015]	.167* (.099) [.092]	.197** (.099) [.047]	.151 (.096) [.117]
Compassion		.987*** (.149)					.625*** (.166)
Indifference			-.816*** (.136)				-.493*** (.147)
Sadness				1.083*** (.214)			-.001 (.253)
Irritation					1.610*** (.227)		1.077*** (.286)
Anger						1.438*** (.271)	.347 (.323)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Reduction in TE	-	-10%	-10%	-11%	-38%	-27%	-44%
Observations	651	651	651	651	651	651	651
R^2	.071	.134	.127	.102	.137	.111	.199

Notes: This table reports treatment effects on redistribution decisions. The dependent variable is the dollar transfer to the disadvantaged agent. The independent variable is spectator's assignment to **HARD**. Each row reports estimates from a linear model. All columns include a set of socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

Figure B1: Performance in Task



Notes: These figures show performance in the effort task by treatment arm. Subfigure (a) focuses in the agent's sample. Subfigure (b) focuses in the spectator's sample. Both subfigures plot an histogram of the tables completed. Each bar plots the fraction of participants completing a discrete number of tables. The light bars plot completion among participants in EASY. The dark bars plot completion among participants in HARD.

Table B8: Mechanism Exploration: Willingness to Keep Redistributing

	Willingness to keep redistributing					
	(1)	(2)	(3)	(4)	(5)	(6)
HARD	.103** (.046)	.089* (.049)	.056 (.051)	.081* (.048)	.084 (.052)	.066 (.049)
Relevance of Luck			.252*** (.091)			
Assignment when Equal Opp.				.116*** (.022)		
Understanding of Agent B					.026 (.095)	
Anger						.496*** (.124)
Costless transfer FE	Yes	Yes	Yes	Yes	Yes	Yes
Covariates	No	Yes	Yes	Yes	Yes	Yes
Reduction in TE	–	0%	-37%	-9%	-6%	-30%
Observations	475	475	475	475	475	475
R ²	.037	.134	.149	.184	.134	.173

Notes: This table reports treatment effects on willingness to keep redistributing once redistribution becomes costly. The dependent variable is an indicator equal to one if the spectator assigns a positive amount to the disadvantaged agent in the costly redistribution decision. The sample is restricted to spectators who assigned a positive amount to the disadvantaged agent in the costless redistribution decision. The independent variable is spectator's assignment to HARD. All columns include fixed effects for the exact amount redistributed in the costless decision. Column (1) includes no additional covariates. Columns (2) to (6) include socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Race enters as a set of dummy variables. Columns (3) to (6) add, respectively, relevance of luck, assignment when equal opportunities, understanding of Agent B, and anger. Understanding of Agent B, anger, and relevance of luck are scaled from 0 to 1. Assignment when Equal Opp. is the dollar transfer to Agent B in the equal-opportunity merit decision. Reduction in TE reports the percentage reduction in the estimated HARD coefficient relative to column (2). Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table B9: Emotional Responses and Willingness to Keep Redistributing

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
HARD	.094*	.095*	.086*	.090*	.082*	.066	.075
	(.049)	(.048)	(.048)	(.049)	(.050)	(.049)	(.048)
Compassion		.304***					.155*
		(.075)					(.083)
Indifference			-.329***				-.269***
			(.067)				(.072)
Sadness				.339***			.107
				(.104)			(.124)
Irritation					.171		-.219
					(.125)		(.137)
Anger						.496***	.478***
						(.124)	(.147)
Costless transfer	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Reduction in TE	-	+1%	-8%	-4%	-13%	-30%	-21%
p-value	.057	.051	.074	.065	.099	.180	.118
Observations	475	475	475	475	475	475	475
R^2	.142	.174	.186	.162	.147	.173	.224

Notes: This table reports treatment effects on willingness to keep redistributing once redistribution becomes costly. The dependent variable is an indicator equal to one if the spectator assigns a positive amount to the disadvantaged agent in the costly redistribution decision. The sample is restricted to spectators who assigned a positive amount to the disadvantaged agent in the costless redistribution decision. The independent variable is spectator's assignment to **HARD**. All columns include fixed effects for the exact amount redistributed in the costless decision. All columns include a set of socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Race enters as a set of dummy variables. Compassion, indifference, sadness, irritation, and anger are scaled from 0 to 1. Reduction in TE reports the percentage change in the estimated treatment effect relative to column (1). Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

C Additional Results in Extensions

Table C1: Prior Beliefs in the Extension Sample

	Total	MIXED	REPORTS	STATISTICS	p-value
<i>Panel A. Perceived differences between task versions</i>					
Versions are approximately the same	82.9%	80.6%	83.6%	84.3%	.634
EASY version expected to be better	12.2%	13.3%	11.5%	11.6%	.851
HARD version expected to be better	5.0%	6.1%	4.8%	4.1%	.700
<i>Panel B. Perceived determinants of success</i>					
Version ranked least relevant	76.9%	72.1%	75.8%	82.6%	.069
Observations	502	165	165	172	

Notes: This table reports prior beliefs about the task among spectators in the extension sample. The sample includes spectators assigned to MIXED, REPORTS, and STATISTICS. Percentages are column shares. Panel A reports beliefs about whether the two task versions differ in expected performance. Panel B reports rankings of the perceived determinants of success. p-values are from chi-squared tests comparing the three extension treatment arms.

Table C2: Beliefs and Task Performance

	Total	MIXED	REPORTS	STATS	MIXED- REPORTS	MIXED- STATS
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Prior Beliefs						
Versions are similar	.829	.806	.843	.843	-.003 (.046)	.018 (.023)
Easy version is better	.122	.133	.116	.116	.000 (.041)	-.011 (.020)
Version assigned is least relevant	.769	.721	.826	.826	.042 (.050)	.007 (.027)
Panel B. Posterior Beliefs						
Versions are similar	.333	.200	.459	.459	.238*** (.058)	.066** (.029)
Easy version is better	.639	.764	.517	.517	-.232*** (.059)	-.060** (.029)
Version assigned is least relevant	.396	.218	.517	.517	.273*** (.056)	.122*** (.028)

Notes: This table reports beliefs collected after the intervention. Columns (1) to (3) distinguishes by treatment arm and report statistics in each group. The first group (EASY) is exposed to the easy version of the task. The second group (HARD) is exposed to the hard version of the task. The second group (MIXED) is exposed to the hard version of the task. Columns (4) and (5) report the difference between MIXED and the other treatment arms. Each row reports estimates from a linear model. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

Table C3: Beliefs and Task Performance

	EASY	HARD	MIXED	Diff. with EASY	Diff. with HARD
	(1)	(2)	(3)	(4)	(5)
Panel A. Prior Beliefs					
Versions are similar	.854	.870	.806	-.014 (.021)	-.054 (.040)
EASY version is better	.107	.111	.133	.005 (.019)	.030 (.035)
Version assigned is least relevant	.821	.816	.721	-.054** (.022)	-.094** (.044)
Panel B. Posterior Beliefs					
Versions are similar	.637	.483	.200	-.197*** (.024)	-.243*** (.048)
EASY version is better	.345	.489	.764	.183*** (.024)	.244*** (.049)
Version assigned is least relevant	.690	.511	.218	-.219*** (.024)	-.269*** (.050)

Notes: This table reports beliefs collected after the intervention. Columns (1) to (3) distinguishes by treatment arm and report statistics in each group. The first group (EASY) is exposed to the easy version of the task. The second group (HARD) is exposed to the hard version of the task. The second group (MIXED) is exposed to the hard version of the task. Columns (4) and (5) report the difference between MIXED and the other treatment arms. Each row reports estimates from a linear model. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

Table C4: Redistribution in EASY. HARD, MIXED

	Costless		Costly	
	(1)	(2)	(3)	(4)
HARD	.270*** (.095) [.004]	.269*** (.099) [.007]	.345*** (.099) [< .001]	.336*** (.103) [.001]
MIXED	.595*** (.114) [< .001]	.603*** (.115) [< .001]	.717*** (.126) [< .001]	.746*** (.130) [< .001]
MIXED - HARD	.324*** (.117) [.006]	.334*** (.119) [.005]	.371*** (.132) [.005]	.409*** (.135) [.002]
Dep. var. mean	1.533	1.533	.750	.750
Effect magn. HARD vs EASY	18%	18%	46%	45%
Effect magn. MIXED vs EASY	39%	39%	96%	99%
Effect magn. MIXED vs HARD	21%	22%	50%	55%
Controls	NO	YES	NO	YES
Observations	816	816	816	816
R^2	.033	.088	.042	.106

Notes: The dependent variable is the amount allocated to Agent B. EASY is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C5: Main Beliefs in EASY. HARD, MIXED

	Posterior comparison		Posterior luck		Posterior fairness	
	(1)	(2)	(3)	(4)	(5)	(6)
HARD	.389*** (.097)	.341*** (.099)	13.585*** (2.228)	12.371*** (2.316)	-.231*** (.076)	-.195** (.078)
MIXED	1.381*** (.118)	1.292*** (.121)	34.946*** (2.700)	32.757*** (2.813)	-.699*** (.092)	-.666*** (.094)
MIXED - HARD	.992*** (.119)	.951*** (.121)	21.362*** (2.730)	20.386*** (2.808)	-.468*** (.093)	-.471*** (.094)
Controls	NO	YES	NO	YES	NO	YES
Observations	816	816	816	816	816	816
R^2	.146	.219	.172	.221	.066	.152

Notes: The dependent variables are posterior beliefs about the advantage provided by the task version, the perceived role of luck in determining which agent solved more tables, and the perceived fairness of the initial bonus allocation. **EASY** is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C6: Other Beliefs in EASY. HARD, MIXED

	Estimation of A		Estimation of B		(A-B)/B		Probability	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
HARD	-4.303*** (.331)	-4.139*** (.347)	-4.004*** (.291)	-3.787*** (.313)	.152*** (.045)	.132** (.052)	-1.300 (1.650)	-.032 (1.641)
MIXED	-2.348*** (.417)	-2.230*** (.431)	-4.250*** (.361)	-4.035*** (.386)	1.029*** (.111)	.998*** (.112)	-10.354*** (2.131)	-9.124*** (2.140)
MIXED - HARD	1.956*** (.398)	1.908*** (.411)	-.246 (.354)	-.247 (.367)	.876*** (.117)	.865*** (.117)	-9.053*** (2.108)	-9.091*** (2.084)
Controls	NO	YES	NO	YES	NO	YES	NO	YES
Observations	816	816	816	816	805	805	816	816
R^2	.170	.224	.226	.276	.192	.239	.033	.134

Notes: The dependent variables are posterior beliefs about average performance in the easy and hard versions, the perceived proportional performance gap between the two versions, and the perceived probability that Agent B outperforms Agent A. **EASY** is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C7: Counterfactual Beliefs in EASY. HARD, MIXED

	Hard Work (inv.)		Counterfactual GL		Counterfactual IL	
	(1)	(2)	(3)	(4)	(5)	(6)
HARD	.387***	.316***	.209***	.209***	.199***	.188***
	(.076)	(.075)	(.060)	(.062)	(.037)	(.037)
MIXED	.761***	.728***	.145**	.153**	.282***	.262***
	(.092)	(.092)	(.073)	(.076)	(.044)	(.045)
MIXED - HARD	.374***	.412***	-.064	-.056	.083*	.074
	(.093)	(.091)	(.074)	(.076)	(.045)	(.045)
Controls	NO	YES	NO	YES	NO	YES
Observations	816	816	816	816	816	816
R^2	.082	.205	.015	.080	.058	.145

Notes: The dependent variables are posterior beliefs about whether Agent B could have solved more tables through greater effort, group-level counterfactual beliefs about which agents would solve more tables if assigned the same version, and individual-level counterfactual beliefs about which agent in the pair would have solved more tables if assigned the same version. **EASY** is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C8: Emotions in EASY. HARD, MIXED

	Anger		Compassion		Sadness		Indifference		Irritation	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
HARD	.140***	.159***	.056	.074	.057	.080	-.085	-.094	.202***	.195***
	(.046)	(.048)	(.077)	(.080)	(.048)	(.050)	(.083)	(.086)	(.051)	(.053)
MIXED	.336***	.330***	.241***	.272***	.060	.067	-.289***	-.307***	.368***	.351***
	(.056)	(.058)	(.093)	(.097)	(.058)	(.060)	(.101)	(.105)	(.062)	(.064)
MIXED - HARD	.196***	.172***	.185**	.198**	.003	-.013	-.204**	-.213**	.166***	.156**
	(.056)	(.058)	(.094)	(.096)	(.058)	(.060)	(.102)	(.105)	(.063)	(.064)
Controls	NO	YES	NO	YES	NO	YES	NO	YES	NO	YES
Observations	816	816	816	816	816	816	816	816	816	816
R^2	.043	.111	.008	.073	.002	.055	.010	.074	.044	.118

Notes: The dependent variables are self-reported emotional responses after learning about the initial bonus allocation, including anger, compassion, sadness, indifference, and irritation. **EASY** is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C9: Perspective-taking in EASY, HARD, MIXED

	Posterior empathy A		Posterior empathy B	
	(1)	(2)	(3)	(4)
HARD	-.439*** (.060)	-.420*** (.063)	.510*** (.064)	.530*** (.067)
MIXED	.013 (.068)	.048 (.073)	.443*** (.078)	.479*** (.082)
MIXED - HARD	.452*** (.073)	.468*** (.074)	-.067 (.074)	-.052 (.075)
Controls	NO	YES	NO	YES
Observations	816	816	816	816
R^2	.076	.135	.081	.148

Notes: The dependent variables are self-reported measures of how easy it was for spectators to imagine Agent A's and Agent B's experience with the task. **EASY** is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C10: Staying in redistribution once a cost is introduced: EASY, HARD, and MIXED

	(1)	(2)
HARD	.106** (.046) [.021]	.101** (.048) [.035]
MIXED	.192*** (.053) [< .001]	.208*** (.056) [< .001]
MIXED - HARD	.085 (.052) [.102]	.107* (.055) [.052]
Dep. var. mean	.548	.548
sg_b fixed effects	YES	YES
Controls	NO	YES
Observations	611	611
R^2	.047	.120

Notes: The dependent variable is an indicator equal to one if a spectator chose a positive transfer in the costless decision (**sg_b** > 0) and also chose a positive transfer in the costly decision (**gamma_b** > 0). The sample is restricted to **EASY**, **HARD**, and **MIXED** spectators and keeps only spectators with **sg_b** > 0. **EASY** is the omitted category. All specifications include fixed effects for the exact amount redistributed in the costless decision. Column (2) includes sex, age, race, education, employment status, individual income, household income, and political party affiliation, with race entered through all variables matching **race***. Robust standard errors are reported in parentheses. Two-sided p-values are reported in square brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C11: Mechanism Exploration: Willingness to Keep Redistributing in MIXED

	Willingness to keep redistributing					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. MIXED vs. EASY</i>						
MIXED	.198*** (.054)	.212*** (.058)	.196*** (.066)	.213*** (.057)	.193*** (.062)	.177*** (.059)
Relevance of Luck			.055 (.108)			
Assignment when Equal Opp.				.093*** (.024)		
Understanding of Agent B					.105 (.103)	
Anger						.359*** (.133)
Change in TE	–	–	-7%	+0%	-9%	-17%
Observations	373	373	373	373	373	373
R ²	.058	.150	.151	.189	.153	.168
<i>Panel B. MIXED vs. HARD</i>						
MIXED	.085 (.052)	.081 (.057)	.077 (.059)	.088 (.055)	.082 (.057)	.072 (.057)
Relevance of Luck			.022 (.106)			
Assignment when Equal Opp.				.115*** (.023)		
Understanding of Agent B					.080 (.116)	
Anger						.250** (.119)
Change in TE	–	–	-4%	+10%	+1%	-11%
Observations	374	374	374	374	374	374
R ²	.029	.147	.147	.208	.148	.158
Costless transfer FE	Yes	Yes	Yes	Yes	Yes	Yes
Covariates	No	Yes	Yes	Yes	Yes	Yes

Notes: This table reports treatment effects on willingness to keep redistributing once redistribution becomes costly. The dependent variable is an indicator equal to one if the spectator assigns a positive amount to the disadvantaged agent in the costly redistribution decision. The sample is restricted to spectators who assigned a positive amount to the disadvantaged agent in the costless redistribution decision. Panel A compares spectators assigned to MIXED to spectators assigned to EASY. Panel B compares spectators assigned to MIXED to spectators assigned to HARD. In each panel, the reported treatment coefficient is the effect of assignment to MIXED relative to the omitted group. All columns include fixed effects for the exact amount redistributed in the costless decision. Column (1) includes no additional covariates. Columns (2) to (6) include socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Race enters as a set of dummy variables. Columns (3) to (6) add, respectively, relevance of luck, assignment when equal opportunities, understanding of Agent B, and anger. Understanding of Agent B, anger, and relevance of luck are scaled from 0 to 1. Assignment when Equal Opp. is the dollar transfer to Agent B in the equal-opportunity merit decision. Change in TE reports the percentage change in the estimated MIXED coefficient relative to column (2) within the same panel. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C12: Redistribution in MIXED, STAT, REPORTS

	Costless		Costly	
	(1)	(2)	(3)	(4)
REPORTS	-.236*	-.230*	-.345**	-.294*
	(.127)	(.129)	(.150)	(.155)
	[.063]	[.075]	[.021]	[.058]
STAT	-.714***	-.717***	-.554***	-.563***
	(.128)	(.130)	(.141)	(.144)
	[< .001]	[< .001]	[< .001]	[< .001]
STAT - REPORTS	-.478***	-.487***	-.208	-.269*
	(.121)	(.130)	(.136)	(.142)
	[< .001]	[< .001]	[.126]	[.058]
Dep. var. mean	2.127	2.127	1.467	1.467
Effect magn. REPORTS vs MIXED	-11%	-11%	-24%	-20%
Effect magn. STAT vs MIXED	-34%	-34%	-38%	-38%
Effect magn. STAT vs REPORTS	-22%	-23%	-14%	-18%
Controls	NO	YES	NO	YES
Observations	502	502	502	502
R^2	.064	.162	.030	.147

Notes: The dependent variable is the amount allocated to Agent B. **MIXED** is the omitted category. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C13: Main Beliefs in MIXED, STAT, REPORTS

	Posterior comparison		Posterior luck		Posterior fairness	
	(1)	(2)	(3)	(4)	(5)	(6)
STAT	-.809***	-.857***	-20.754***	-21.789***	.538***	.602***
	(.152)	(.150)	(3.249)	(3.246)	(.111)	(.109)
REPORTS	-.806***	-.785***	-15.618***	-15.695***	.442***	.440***
	(.145)	(.148)	(3.019)	(3.104)	(.109)	(.112)
REPORTS - STAT	.003	.072	5.136	6.094*	-.095	-.162
	(.144)	(.154)	(3.254)	(3.349)	(.106)	(.112)
Controls	NO	YES	NO	YES	NO	YES
Observations	502	502	502	502	502	502
R^2	.074	.207	.084	.208	.053	.147

Notes: The dependent variables are posterior beliefs about the advantage provided by the task version, the perceived role of luck in determining which agent solved more tables, and the perceived fairness of the initial bonus allocation. **EASY** is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C14: Other Beliefs in MIXED, STAT, REPORTS

	Estimation of A		Estimation of B		(A-B)/B		Probability	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
STAT	3.416***	3.443***	.017	.193	.739***	.690***	-2.843	-2.693
	(.394)	(.459)	(.342)	(.419)	(.136)	(.143)	(2.410)	(2.432)
REPORTS	2.776***	2.670***	3.048***	3.085***	-.624***	-.645***	4.297*	4.048*
	(.723)	(.748)	(.638)	(.662)	(.118)	(.120)	(2.251)	(2.349)
REPORTS - STAT	-.641	-.773	3.032***	2.892***	-1.364***	-1.335***	7.140***	6.742***
	(.675)	(.682)	(.594)	(.598)	(.092)	(.105)	(2.144)	(2.390)
Controls	NO	YES	NO	YES	NO	YES	NO	YES
Observations	502	502	502	502	497	497	502	502
R^2	.067	.153	.079	.176	.221	.287	.020	.104

Notes: The dependent variables are posterior beliefs about average performance in the **EASY** and **HARD** versions, the perceived proportional performance gap between the two versions, and the perceived probability that Agent B outperforms Agent A. **EASY** is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C15: Counterfactual Beliefs in MIXED, STAT, REPORTS

	Hard Work (inv.)		Counterfactual GL		Counterfactual IL	
	(1)	(2)	(3)	(4)	(5)	(6)
STAT	-.463***	-.542***	-.071	-.106	-.224***	-.270***
	(.109)	(.109)	(.090)	(.091)	(.050)	(.049)
REPORTS	-.406***	-.427***	.109	.076	-.067	-.082*
	(.109)	(.110)	(.083)	(.087)	(.047)	(.049)
REPORTS - STAT	.057	.115	.181**	.182**	.157***	.188***
	(.108)	(.112)	(.083)	(.090)	(.052)	(.052)
Controls	NO	YES	NO	YES	NO	YES
Observations	502	502	502	502	502	502
R^2	.041	.192	.009	.113	.042	.201

Notes: The dependent variables are posterior beliefs about whether Agent B could have solved more tables through greater effort, group-level counterfactual beliefs about which agents would solve more tables if assigned the same version, and individual-level counterfactual beliefs about which agent in the pair would have solved more tables if assigned the same version. **EASY** is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C16: Emotions in MIXED, STAT, REPORTS

	Anger		Compassion		Sadness		Indifference		Irritation	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
STAT	-.183**	-.201***	-.108	-.154	.050	.035	.344***	.417***	-.146*	-.194**
	(.075)	(.078)	(.111)	(.117)	(.067)	(.070)	(.113)	(.116)	(.086)	(.090)
REPORTS	-.170**	-.165**	-.055	-.045	.000	.002	.200*	.166	-.230***	-.239***
	(.077)	(.078)	(.103)	(.107)	(.068)	(.071)	(.106)	(.109)	(.079)	(.080)
REPORTS - STAT	.013	.037	.053	.109	-.050	-.033	-.144	-.251**	-.084	-.044
	(.067)	(.071)	(.102)	(.106)	(.070)	(.075)	(.115)	(.120)	(.076)	(.083)
Controls	NO	YES	NO	YES	NO	YES	NO	YES	NO	YES
Observations	502	502	502	502	502	502	502	502	502	502
R^2	.015	.124	.002	.098	.001	.057	.019	.144	.016	.100

Notes: The dependent variables are self-reported emotional responses after learning about the initial bonus allocation, including anger, compassion, sadness, indifference, and irritation. **EASY** is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C17: Perspective-taking in MIXED, STAT, REPORTS

	Posterior empathy A		Posterior empathy B	
	(1)	(2)	(3)	(4)
STAT	-.477***	-.475***	-.542***	-.539***
	(.085)	(.088)	(.091)	(.094)
REPORTS	-.400***	-.425***	-.430***	-.456***
	(.083)	(.087)	(.091)	(.094)
REPORTS - STAT	.077	.050	.112	.083
	(.088)	(.092)	(.096)	(.100)
Controls	NO	YES	NO	YES
Observations	502	502	502	502
R^2	.067	.143	.071	.159

Notes: The dependent variables are self-reported measures of how easy it was for spectators to imagine Agent A's and Agent B's experience with the task. **EASY** is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C18: Staying in redistribution once a cost is introduced: MIXED, REPORTS, and STAT

	(1)	(2)
REPORTS	-.113*	-.103*
	(.060)	(.062)
	[.060]	[.097]
STAT	-.059	-.092
	(.063)	(.068)
	[.349]	[.176]
STAT - REPORTS	.054	.011
	(.061)	(.067)
	[.376]	[.870]
Dep. var. mean	.593	.593
sg_b fixed effects	YES	YES
Controls	NO	YES
Observations	391	391
R ²	.052	.149

Notes: The dependent variable is an indicator equal to one if a spectator chose a positive transfer in the costless decision ($\mathbf{sg_b} > 0$) and also chose a positive transfer in the costly decision ($\mathbf{gamma_b} > 0$). The sample is restricted to MIXED, REPORTS, and STAT spectators and keeps only spectators with $\mathbf{sg_b} > 0$. MIXED is the omitted category. All specifications include fixed effects for the exact amount redistributed in the costless decision. Column (2) includes sex, age, race, education, employment status, individual income, household income, and political party affiliation. Race enters through all variables matching **race***. Robust standard errors are reported in parentheses. Two-sided p-values are reported in square brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table C19: Mechanism Exploration: Willingness to Keep Redistributing in Information Treatments

	Willingness to keep redistributing					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. REPORTS vs. MIXED</i>						
REPORTS	-.115*	-.078	-.078	-.089	-.054	-.084
	(.060)	(.064)	(.066)	(.062)	(.066)	(.064)
Relevance of Luck			.001			
			(.127)			
Assignment when Equal Opp.				.066**		
				(.028)		
Understanding of Agent B					.179	
					(.110)	
Anger						-.127
						(.151)
Change in TE	–	–	0%	+13%	-31%	+7%
Observations	274	274	274	274	274	274
R ²	.051	.224	.224	.247	.232	.226
<i>Panel B. STAT vs. MIXED</i>						
		67				
STAT	-.067	-.094	-.144*	-.115	-.077	-.093
	(.065)	(.071)	(.073)	(.071)	(.075)	(.072)

Table C20: Redistribution in HARD, REPORTS, and STATISTICS

	Costless		Costly	
	(1)	(2)	(3)	(4)
REPORTS	.088	.142	.026	.091
	(.110)	(.113)	(.128)	(.131)
	[.423]	[.209]	[.839]	[.489]
STAT	-.390***	-.329***	-.182	-.167
	(.111)	(.115)	(.117)	(.124)
	[< .001]	[.004]	[.118]	[.179]
Dep. var. mean	1.803	1.803	1.095	1.095
Effect magn. REPORTS vs HARD	5%	8%	2%	8%
Effect magn. STAT vs HARD	-22%	-18%	-17%	-15%
Controls	NO	YES	NO	YES
Observations	652	652	652	652
R^2	.025	.084	.004	.097

Notes: The dependent variable is the amount allocated to Agent B. **HARD** is the omitted category.

Robust standard errors are reported in parentheses. Two-sided p-values are reported in square brackets.

* $p < .10$, ** $p < .05$, *** $p < .01$.

D Multiple-Testing Adjustments

Tables D1–D3 report Benjamini–Hochberg adjusted p-values for the preregistered multiple-testing families. In the main experiment, the adjustment is applied to the preregistered belief outcomes and to the exploratory associations between posterior beliefs and redistribution. In the extension, the adjustment is applied across the three treatment comparisons relative to **EASY** for each preregistered outcome family. The results are largely unchanged after adjustment.

Table D1: Multiple-testing adjustments for preregistered outcomes in the main experiment

Outcome	Comparison	Estimate	SE	<i>p</i>	<i>q</i>	<i>N</i>
<i>Panel A. Preregistered H2: group-level belief outcomes</i>						
Performance gap (A-B)	HARD vs EASY	-0.342*	(0.180)	0.058	0.073	651
Role of luck	HARD vs EASY	12.553***	(2.357)	<0.001	<0.001	651
Probability B outperforms A	HARD vs EASY	-0.110	(1.647)	0.947	0.947	651
Group-level counterfactual / hard-work belief	HARD vs EASY	0.232***	(0.062)	<0.001	<0.001	651
Version advantage/intensity	HARD vs EASY	0.337***	(0.099)	<0.001	0.001	651
<i>Panel B. Secondary individual-level belief outcomes</i>						
Individual counterfactual: not Agent A/easy would win	HARD vs EASY	0.192***	(0.039)	<0.001	<0.001	651
Hard-work belief (inverse)	HARD vs EASY	0.315***	(0.075)	<0.001	<0.001	651
<i>Panel C. Exploratory associations between posterior beliefs and redistribution</i>						
Role of luck	association with costless redistribution	0.016***	(0.002)	<0.001	<0.001	651
Performance gap (A-B)	association with costless redistribution	0.065**	(0.028)	0.018	0.023	651
Frequency inverse	association with costless redistribution	0.003	(0.003)	0.202	0.202	651
Unfairness of initial allocation	association with costless redistribution	1.947***	(0.136)	<0.001	<0.001	651
Hard-work belief inverse	association with costless redistribution	0.411***	(0.053)	<0.001	<0.001	651

Notes: This table reports raw and Benjamini–Hochberg adjusted p-values for the preregistered multiple-testing families in the main experiment. Panel A adjusts across the five group-level belief outcomes associated with the belief-updating hypothesis. Panel B adjusts across the two secondary individual-level belief outcomes. Panel C adjusts across the exploratory associations between posterior beliefs and costless redistribution. All estimates use the preregistered analysis sample and include socio-demographic controls: sex, age, race, education, employment status, individual income, household income, and political party affiliation. Race is coded as a categorical variable distinguishing White only, Black only, Asian only, Native only, Pacific only, Other only, multiple races, and no race selected. Robust standard errors are reported in parentheses. Stars are based on BH-adjusted p-values: **q* < .10, ***q* < .05, ****q* < .01.

Table D2: Multiple-testing adjustments for preregistered redistribution outcomes in the extension

Outcome	Comparison	Estimate	SE	<i>p</i>	<i>q</i>	<i>N</i>
<i>Panel A. Preregistered H1: costless redistribution</i>						
Costless redistribution	MIXED vs EASY	0.597***	(0.116)	<0.001	<0.001	501
Costless redistribution	REPORTS vs EASY	0.372***	(0.117)	0.002	0.002	501
Costless redistribution	STATISTICS vs EASY	-0.104	(0.112)	0.353	0.353	508
<i>Panel B. Preregistered H3: costly redistribution</i>						
Costly redistribution	MIXED vs EASY	0.735***	(0.130)	<0.001	<0.001	501
Costly redistribution	REPORTS vs EASY	0.455***	(0.137)	<0.001	0.001	501
Costly redistribution	STATISTICS vs EASY	0.223*	(0.118)	0.060	0.060	508
<i>Panel C. Complementary H3 measure: staying in redistribution when costly</i>						
Stay in costly redistribution	MIXED vs EASY	0.213***	(0.058)	<0.001	<0.001	373
Stay in costly redistribution	REPORTS vs EASY	0.091	(0.058)	0.120	0.120	375
Stay in costly redistribution	STATISTICS vs EASY	0.132**	(0.059)	0.025	0.038	354

Notes: This table reports raw and Benjamini–Hochberg adjusted p-values for the primary redistribution outcomes in the extension experiment. Panels A and B adjust across the three extension treatment comparisons relative to **EASY**. Panel C reports the same adjustment for the complementary persistence measure, restricted to spectators who redistributed a positive amount when redistribution was costless and controlling for fixed effects for the exact costless transfer. All estimates use the preregistered analysis sample and include socio-demographic controls. Robust standard errors are reported in parentheses. Stars are based on BH-adjusted p-values: **q* < .10, ***q* < .05, ****q* < .01.

Table D3: Multiple-testing adjustments for preregistered belief outcomes in the extension

Outcome	Comparison	Estimate	SE	p	q	N
Proportional performance gap	MIXED vs EASY	0.970***	(0.117)	<0.001	<0.001	496
Role of luck	MIXED vs EASY	31.840***	(2.829)	<0.001	<0.001	501
Probability B outperforms A	MIXED vs EASY	-7.891***	(2.231)	<0.001	<0.001	501
Group-level counterfactual / hard-work belief	MIXED vs EASY	0.128	(0.081)	0.117	0.126	501
Version advantage/intensity	MIXED vs EASY	1.253***	(0.129)	<0.001	<0.001	501
Proportional performance gap	REPORTS vs EASY	0.370***	(0.049)	<0.001	<0.001	499
Role of luck	REPORTS vs EASY	19.255***	(2.805)	<0.001	<0.001	501
Probability B outperforms A	REPORTS vs EASY	-5.354**	(2.044)	0.009	0.010	501
Group-level counterfactual / hard-work belief	REPORTS vs EASY	0.247***	(0.076)	0.001	0.002	501
Version advantage/intensity	REPORTS vs EASY	0.545***	(0.125)	<0.001	<0.001	501
Proportional performance gap	STATISTICS vs EASY	1.727***	(0.086)	<0.001	<0.001	507
Role of luck	STATISTICS vs EASY	11.732***	(3.012)	<0.001	<0.001	508
Probability B outperforms A	STATISTICS vs EASY	-11.252***	(2.151)	<0.001	<0.001	508
Group-level counterfactual / hard-work belief	STATISTICS vs EASY	0.091	(0.083)	0.270	0.270	508
Version advantage/intensity	STATISTICS vs EASY	0.492***	(0.129)	<0.001	<0.001	508

Notes: This table reports raw and Benjamini-Hochberg adjusted p-values for the belief outcomes in the extension experiment. The adjustment is applied across the fifteen tests defined by the three extension treatment comparisons relative to **EASY** and the five preregistered belief outcomes. The performance-gap outcome is the proportional difference between the estimated performance of Agents A and Agents B, as specified in the extension pre-analysis plan. All estimates use the preregistered analysis sample and include socio-demographic controls. Robust standard errors are reported in parentheses. Stars are based on BH-adjusted p-values: * $q < .10$, ** $q < .05$, *** $q < .01$.

E Deviations from the Pre-Analysis Plans

This appendix summarizes how the analyses reported in the paper relate to the two pre-analysis plans. The first pre-analysis plan covers the main experiment, which compares spectators assigned to experience the hard version of the task with spectators assigned to experience the easy version of the task. The second pre-analysis plan covers the extension experiment, which adds the **MIXED**, **REPORTS**, and **STATISTICS** treatment arms. The main behavioral outcomes remain those specified in the pre-analysis plans: redistribution toward the disadvantaged agent, beliefs about the role of task version in determining performance, and willingness to pay a cost to implement redistribution.

First, the paper reorganizes the presentation of the hypotheses. In the main pre-analysis plan, Hypothesis 1 concerned costless redistribution, Hypothesis 2 concerned beliefs about the importance of the task version, and Hypothesis 3 concerned costly redistribution. In the paper, we present the hypotheses through the lens of a simple fairness model. The first hypothesis concerns what participants consider to be fair, measured by costless redistribution. The second hypothesis concerns the weight attached to what they consider fair, measured by willingness to redistribute when redistribution is costly. Belief updating is therefore presented as a central mechanism rather than as a separately numbered main hypothesis. This change affects the exposition, but not the underlying empirical tests.

Second, the analysis of costly redistribution is refined relative to the pre-analysis plan. The pre-analysis plan proposed studying costly redistribution using both a constructed indicator for paying to implement redistribution and a regression of costly redistribution on treatment assignment controlling for costless redistribution. In the paper, we retain this logic but implement it in two complementary ways. We estimate a reduced-form persistence specification that conditions on the exact amount redistributed in the costless decision, and we estimate a structural threshold model that maps directly onto the fixed-cost design. This refinement separates changes in the what participant consider to be fair from changes in their willingness to act on it.

Third, the pre-analysis plan specified Benjamini–Hochberg adjustments for multiple testing in the belief outcomes associated with the belief-updating hypothesis, and for exploratory correlations between posterior beliefs and redistribution. We therefore report conventional p-values in the main tables and provide Benjamini–Hochberg adjusted p-values for these pre-specified belief families in the appendix. We do not adjust the primary redistribution outcome or the primary costly-redistribution test as multiple-outcome families, because each corresponds to a single behavioral hypothesis.

Fourth, the mechanism analyses are reported as exploratory. The pre-analysis plan proposed examining whether belief variables and emotions account for the treatment effect on redistribution by sequentially adding these variables to the main regression. The paper follows this approach but interprets these estimates as descriptive evidence on plausible mechanisms rather than as causal mediation estimates.

Fifth, the extension analyses follow the second pre-analysis plan, which introduced the **MIXED**, **REPORTS**,

and **STATISTICS** treatment arms. The extension pre-analysis plan specified tests of whether each of these treatments increases redistribution, belief updating, and willingness to pay relative to **EASY**. In the paper, we also emphasize comparisons among **MIXED**, **REPORTS**, and **STATISTICS** because the purpose of the extension is to distinguish direct comparative experience from qualitative and quantitative information provision. These comparisons are therefore aligned with the motivation of the extension pre-analysis plan, although not all were stated as separate numbered hypotheses.

Sixth, the extension pre-analysis plan stated that each extension treatment would include 180 participants. The implemented sample is slightly smaller, with 165 participants in **MIXED**, 165 in **REPORTS**, and 172 in **STATISTICS**. This difference reflects budget limitations.

F Interventions

Figure D1: Intervention in EASY and MIXED

Part 1: Practice - Solve as many tables as you can

Time left to complete this page: **3:59**

Table #1
Score: 0

Version A

β	A	I	I	B	α	8	ß	ß	B
8	B	8	I	I	α	ß	ß	B	I
8	α	I	I	I	A	A	8	8	B
B	8	β	β	A	I	I	α	8	I
β	8	ß	8	ß	I	A	α	A	ß
A	β	ß	I	I	B	α	ß	ß	β
I	β	8	8	A	I	8	ß	α	α
ß	B	β	I	B	8	I	α	I	ß
8	B	I	A	8	α	A	α	B	ß
B	ß	α	α	8	ß	A	B	I	I

How many As are

Submit

Instructions:

You need to count how often the letter 'A' appears in the table. Only uppercase 'A' counts.

You have to wait 20 seconds after a wrong answer before trying again. You can only attempt each table twice. If you fail to complete a table in two attempts, you will move to the next one.

Figure D2: Intervention in HARD and MIXED

Part 1: Practice - Solve as many tables as you can

Time left to complete this page: 3:59

Table #1

Score: 0

Version B

A	A	A	A	a	B	a	s	I	A
B	A	a	β	B	B	s	A	β	β
a	I	I	I	s	A	A	a	s	a
s	s	A	A	β	a	a	β	a	β
s	β	A	B	s	a	a	β	s	a
B	β	A	I	β	β	s	A	B	s
β	a	B	A	I	I	s	a	s	I
β	a	I	s	s	A	B	β	s	β
a	B	A	β	β	I	I	A	s	β
A	A	I	B	β	β	a	β	β	B

How many Bs are

Submit

Instructions:

You need to count how often the letter 'B' appears in the table. Only uppercase 'B' counts.

You have to wait 20 seconds after a wrong answer before trying again. You can only attempt each table twice. If you fail to complete a table in two attempts, you will move to the next one.

Figure D3: Intervention in REPORTS

Agents A
People assigned to version A generally described the task as easy but somewhat repetitive. Many reported scanning the grid carefully or following a systematic pattern to keep track of the letters. Some noted that the task required sustained attention over time, but overall they felt confident in their ability to count the letters correctly. Despite the repetitive nature, several participants described the task as enjoyable or puzzle-like. Agent A #22 wrote: <i>"It was interesting. I found it pretty easy to do, because of pattern recognition. I was able to see the A pretty easily, and it made it easier to find them to count them."</i> Agent A #60 wrote: <i>"It was fairly straight forward and easy to understand. I also thought it was mentally engaging which gave me more motivation to do better."</i>
Agents B
People assigned to version B generally described the task as challenging and mentally tiring. Many mentioned that the letters looked very similar or appeared in different formats, making them harder to count accurately. Some reported losing track of their count or needing to double-check frequently, while others noted that the task required a high level of concentration and could feel frustrating or time-consuming. Agent B #17 wrote: <i>"I found it very difficult since all the letters looked very similar to each other so it was easy to miscount or get confused. It was also a bit stressful looking the timer."</i> Agent B #69 wrote: <i>"It was frustrating seeing how similar the characters were to each other and I blamed myself for choosing the wrong thing. I became frustrated."</i>

Figure D4: Intervention in STATISTICS

Agents A
People assigned to version A of the task completed 11 tasks correctly in 4 minutes. They completed correctly 86% of the tasks they attempted. The best performing agent completed 23 tasks correctly, while the worst performing agent completed 0 tasks correctly. The probability that a person assigned to version A outperforms a person assigned to version B is 91%.
Agents B
People assigned to version B of the task completed 3 tasks correctly in 4 minutes. They completed correctly 49% of the tasks they attempted. The best performing agent completed 8 tasks correctly, while the worst performing agent completed 0 tasks correctly. The probability that a person assigned to version B outperforms a person assigned to version A is 6%.

G Text Classification Procedure

This appendix documents the procedure used to classify spectators’ open-text justifications into fairness types, in enough detail to be reproduced independently.

G.1 Classification Task

Each spectator explained, in free text, why they allocated the \$6 bonus between Agent A and Agent B the way they did. The classification task takes this text as the sole input and assigns it to exactly one of five categories. The amount actually redistributed is not used as an input to the classification in any way: two spectators reaching the same dollar split for different stated reasons receive different codes, and two spectators reaching different splits for the same stated reason receive the same code.

G.2 Precise Prompt and Coding Categories

In this study, spectators were told that two people, Agent A and Agent B, each completed a counting task under randomly assigned, unequally difficult task versions (version A was generally easier than version B). Whoever solved more tables correctly was originally promised a \$6 bonus in full. The spectator was then allowed to reallocate the \$6 bonus between Agent A and Agent B however they wished, and explained their reasoning in free text. **IMPORTANT:** text only, never the split. Classify using only the written explanation. Do not use, look up, or infer from how many dollars each agent actually received. You are coding reasoning, not outcomes. Two spectators can reach the exact same dollar split for entirely different reasons, and two spectators with entirely different splits can be reasoning identically. If you are given the dollar amounts alongside the text, ignore them completely — do not let “it was an equal split” or “it was a big/small adjustment” push you toward a code. The code should follow only from what the sentence says the reason is. (If a respondent’s own explanation happens to mention a dollar figure — e.g., “I gave them \$4 because...” — that’s fine, it’s part of their reasoning; the rule is about not using the allocation data as a separate signal, not about scrubbing participants’ own words.) Your job is to read the free text and assign exactly one of five codes based on the reasoning given. The five codes are:

Code	Name	Definition
LIB	Libertarian	The allocation should stay as originally promised/assigned, because changing it would violate the pre-set rule or agreement itself.
NCM	Non-compensating meritocrat	The allocation is based on observed performance (more tables, higher score, more effort), and the task-difficulty gap is either not mentioned or explicitly not used to adjust the allocation.
CM	Compensating meritocrat	The allocation is based on performance, but is explicitly adjusted because of the task-difficulty gap, randomness of assignment, or resulting unfairness.
EGAL	Egalitarian	The respondent justifies the allocation primarily by appealing to equality, equal entitlement, or equal treatment. The reasoning does not depend on who performed better or on one agent having faced the harder task.
OTHER	Other	No identifiable reason, incoherent/contradictory reasoning, or a reason unrelated to performance, effort, or fairness.

Table D1: Coding categories for spectators’ bonus-allocation justifications.

G.3 Representative Quotations by Category

To illustrate how the coding categories map onto participants’ own language, we reproduce below two representative explanations for each category, verbatim (including original spelling). All quotations below were assigned the same code by both GPT-5.5 Thinking and Claude.

Libertarian.

“The rule was whoever completed more tables got the bonus. The tasks were comparable so the rule should stand. Agent A completed more tables and should get to keep the full bonus. We were awarding completion, not effort.”

“The rules of the study were that whoever got the most points right earns the highest bonus, so I decided not to change that. If I thought the rules were unfair, I would have split it evenly.”

Non-compensating meritocrat.

“Agent A solved more tables correctly so they earned the bonus as far as I can tell. I wanted

to reward performance. I didn't consider the versions of the tasks, only on the person who performed better."

"I made that decision because A did an excellent job and deserve \$4 and also B tried their best and they are supposed to be awarded \$2 for the effort."

Compensating meritocrat.

"I chose to split the bonus equally because both agents dedicated the same amount of time to the study. I also factored in that agent B had a much more difficult and frustrating task, which explains why they solved fewer tables than agent A."

"The significant scoring gap between versions suggests Version B was much harder. Since assignment was random, I split the bonus equally to ensure fairness regardless of the task difficulty imbalance."

Egalitarian.

"I gave both agents an equal amount because they put in the same amount of effort and an equal time. I think the tasks are evenly split."

"Equal effort and time spent on the task between both groups should produce same pay. Task is relatively simple and time spent is most relevant factor to pay in this case."

Other.

"Based on my understanding I actually choose the allocation of amount I want to distributed."

"No just greedy. This is a very hard to calculate in my mind I should keep the version of my task and not give any to the other guy. As I said just greed."

H Questionnaires

H.1 Spectators

This section includes the screenshots for spectator data collection. The screenshots are those of **EASY**. See Figures D1, D2, D3 and D4 for differences across treatment arms.

Thanks for participating in this study!



This study is conducted by researchers from Tampere University and Erasmus University Rotterdam.

In this study you have the opportunity of earning money by answering questions about different versions of a task.

Duration

The study can be completed in around 30 minutes.

Compensation

You will receive a 3.00 GBP (around 4.00 USD) payment for completing the study.

Informed consent form

On the next page, you will find an informed consent form. Please read it carefully and confirm it if you want to proceed with this study.

Next

Informed consent

Before proceeding, please give your consent that you voluntarily participate in this study, that you have been sufficiently informed about the nature of the study, and that you explicitly give us your consent that:

- We can collect your anonymous, non-sensitive personal data (like age, income, etc).
- We can use this personal data for scientific purposes.
- We can store your anonymous, non-sensitive personal data on our safe-guarded university servers for up to 10 years.
- We can make anonymized data available to other researchers online.

We promise to protect your data according to the latest General Data Protection Regulation (GDPR) data regulation laws. Notice that you can also withdraw your consent anytime during the study by returning your submission before completing this study. In that case, your data will not be stored.

In case you have any questions at any point, please contact us at pogliano@ese.eur.nl.

I agree

I do not agree

Powered by [oTree](#)

General Instructions

This study has three parts.

Part 1:

You will learn about a task in which you count how many times a specific character appears in a table. There are two versions of the task, version A and version B.

Part 2:

You will guess the results of a previous study we completed. You can win an additional prize if your guesses are correct. You will also make a decision that can affect the earnings of two other real people.

Part 3:

You will answer some additional questions.

Next

Part 1

Instructions

You will now learn about a task. In this task, you need to count how often a specific character appears in a table like the one below.

Your goal is to solve as many tables as possible in 4 minutes. Every time you solve a table correctly, you increase your score by 1. Your final score is the total number of tables you solve correctly within 4 minutes.

Each table is filled with characters. You must count how many times the target character appears and submit the count. When you submit the correct count for a table, and a new one will appear with new characters in shuffled positions.

You have 2 attempts per table:

- If your first attempt is incorrect, you must wait 20 seconds before trying again.
- If your second attempt is incorrect, the table will disappear, and you will start over with a completely new one.

There are two versions of the task. The type of tables are the same in both versions, only the target character differs. In Version A, you count how many times uppercase "A" appears in the table (e.g., 17 in the example below). In Version B, you count how many times uppercase "B" appears in the table (e.g., 14 in the example below).

Comprehension Questions:

What does your score depend on?

- ☐ The number of tables you solve correctly within 4 minutes
- ☐ The number of As present in the tables
- ☐ The number of attempts you make

What do you have to do if you are assigned Version A?

- ☐ Count how many times character B appears in the table
- ☐ Count how many times character A appears in the table
- ☐ There is no Version A

Are there any differences in the tables between versions?

- ☐ No
☐ Yes

Next

Example table:

B	B	β	α	α	A	β	I	s	I
I	I	I	s	α	A	A	α	s	I
B	I	β	β	A	β	β	β	A	α
I	β	s	α	s	I	β	β	I	α
s	β	α	α	β	β	s	B	A	β
β	s	β	s	β	s	β	I	β	α
s	B	β	β	s	A	I	β	A	s
β	B	A	A	A	β	β	s	A	I
A	β	s	A	I	A	B	A	A	s
β	α	B	I	β	B	I	A	A	s

You have answered all the comprehension questions correctly.

We will now ask you some questions related to the task.

Next

Powered by [oTree](#)

Imagine this situation:

Two people are completing the task at the same time.

- One person is assigned Version A (counting 'A's')
- The other person is assigned Version B (counting 'B's')

The two are matched and their scores are compared.

Answer the following question:

What matters most in determining who gets the higher score?

Rank the following factors. Drag and drop to sort the factors from most relevant on top to least relevant on the bottom.

≡ **Ability** (how good a person is at counting quickly and accurately)

≡ **Effort** (how hard a person works during the task)

≡ **Version** (whether they receive Version A or Version B of the task)

Next

Answer the following question:

Do the two versions of the task feel similar to you?

☐ Not at all

☐ Slightly

☐ Moderately

☐ Completely

Next

Remember:

In Version A, you count the number of times the letter 'A' appears in the table.

In Version B, you count the number of times the letter 'B' appears in the table.

Powered by [gTree](#)

Answer the following question:

On a scale from 1 to 7, where 1 is 'Getting Version A is a major advantage' and 7 is 'Getting Version B is a major advantage',

Do you think any version confers an advantage for achieving higher scores?

- ☐ 1: Getting Version A is a major advantage
- ☐ 2
- ☐ 3
- ☐ 4: Both versions are approximately the same
- ☐ 5
- ☐ 6
- ☐ 7: Getting Version B is a major advantage

Next

Remember:

In Version A, you count the number of times the letter 'A' appears in the table.

In Version B, you count the number of times the letter 'B' appears in the table.

Answer the following question:

The following question is an attention check. Please select letter C.

Select one of the following letters.

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Next

Remember:

In Version A, you count the number of times the letter 'A' appears in the table.

In Version B, you count the number of times the letter 'B' appears in the table.

You will now get more familiar with the task.

You were randomly assigned to practice **Version A** of the task.

You need to count how often the letter 'A' appears in the table. Only uppercase 'A' counts.

Solve as many tables as you can in 4 minutes.

Start

Powered by [oTree](#)

Part 1: Practice - Solve as many tables as you can

Time left to complete this page: **3:59**

Table #1

Score: 0

Version A

β	A	I	I	B	α	8	ß	ß	B
8	B	8	I	I	α	ß	ß	B	I
8	α	I	I	I	A	A	8	8	B
B	8	β	β	A	I	I	α	8	I
β	8	ß	8	ß	I	A	α	A	ß
A	β	ß	I	I	B	α	ß	ß	β
I	β	8	8	A	I	8	ß	α	α
ß	B	β	I	B	8	I	α	I	ß
8	B	I	A	8	α	A	α	B	ß
B	ß	α	α	8	ß	A	B	I	I

How many As are

Submit

Instructions:

You need to count how often the letter 'A' appears in the table. Only uppercase 'A' counts.

You have to wait 20 seconds after a wrong answer before trying again. You can only attempt each table twice. If you fail to complete a table in two attempts, you will move to the next one.

You are midway through the practice.

The following question is an attention check. Please select number 23.

Select one of the following numbers.

- ☐ 12
- ☐ 23
- ☐ 30
- ☐ 41

Next

Part 1: Practice - Solve as many tables as you can

Time left to complete this page: **1:58**

Table #1

Score: 0

Version A

β	A	I	I	B	α	8	ß	ß	B
8	B	8	I	I	α	ß	ß	B	I
8	α	I	I	I	A	A	8	8	B
B	8	β	β	A	I	I	α	8	I
β	8	ß	8	ß	I	A	α	A	ß
A	β	ß	I	I	B	α	ß	ß	β
I	β	8	8	A	I	8	ß	α	α
ß	B	β	I	B	8	I	α	I	ß
8	B	I	A	8	α	A	α	B	ß
B	ß	α	α	8	ß	A	B	I	I

How many As are

Submit

Instructions:

You need to count how often the letter 'A' appears in the table. Only uppercase 'A' counts.

You have to wait 20 seconds after a wrong answer before trying again. You can only attempt each table twice. If you fail to complete a table in two attempts, you will move to the next one.

Answer the following question:

Remember there are 2 versions of the task.

Which version of the task did you just complete?

- ☐ Version A
- ☐ Version B

Next

Remember:

In Version A, you count the number of times the letter 'A' appears in the table.

In Version B, you count the number of times the letter 'B' appears in the table.

Congratulations, you completed **Part 1!**

To reward you for your effort so far, we have awarded you **an additional prize of 1 USD**.

Part 2 will start in the next screen.

Next

Powered by [QTree](#)

Part 2

Last week, we conducted another study here on Prolific. In that study, people had to complete the counting task you previously learned about. We had two groups of people, which we will refer to as Agents A and Agents B.

We now give you more details about our study.

Next

Part 2

Context

People taking part in our study were randomly assigned to two groups.

- People randomly assigned to be an Agent A completed Version A, in which they had to count the number of times the letter 'A' appeared in the tables.
- People randomly assigned to be an Agent B completed Version B, in which they had to count the number of times the letter 'B' appeared in the tables.

All agents completed the task for 4 minutes. They all received the same payment for their participation, regardless of their score.

We also informed the agents that they might get some additional bonus based on how many tables they solved and on others' decisions. More information about the bonus will follow later.

After they completed the task, we matched each Agent A with one Agent B to form pairs.

Comprehension test:

1) The version of the task (Version A or Version B) is randomly assigned to each person.

- ☐ True
☐ False

2) Agents spend the same amount of time on the task and receive the same fixed payment to compensate them for their participation.

- ☐ True
☐ False

3) Which agent had to count the number of times the letter 'A' appears in the tables?

- ☐ Agent A
☐ Agent B

Next

You have answered all the comprehension questions correctly.

Next

Powered by [QTree](#)

Guess the Results

In the next screens, we will ask you to make 3 guesses about the results of our study.

At the end of this survey, 1 out of these 3 guesses will be randomly selected. We will compare the selected guess with the actual results of our study. If your guess is correct, **you will receive a prize of 3 USD.**

Please read and think carefully about each guess.

Next

Guess #1:

On average, how many tables do you think Agents A solved correctly in 4 minutes?

How sure are you about your answer?

- ☐ Not at all sure
- ☐ A little sure
- ☐ Somewhat sure
- ☐ Very sure
- ☐ Completely sure

Next

Remember:

Agents A completed Version A of the task, in which they had to count the number of times the letter 'A' appeared in the tables.
Agents B completed Version B of the task, in which they had to count the number of times the letter 'B' appeared in the tables.

Guess #2:

On average, how many tables do you think Agents B solved correctly in 4 minutes?

How sure are you about your answer?

- ☐ Not at all sure
- ☐ A little sure
- ☐ Somewhat sure
- ☐ Very sure
- ☐ Completely sure

Next

Remember:

Agents A completed Version A of the task, in which they had to count the number of times the letter 'A' appeared in the tables.
Agents B completed Version B of the task, in which they had to count the number of times the letter 'B' appeared in the tables.

Guess #3:

Consider the following scenario:

Agents A complete Version A of the task, in which they count the number of times the letter 'A' appeared in the tables. Agents B complete Version B of the task, in which they count the number of times the letter 'B' appeared in the tables. After all agents complete the task, we randomly match each Agent A with one Agent B to form pairs.

Suppose there are 100 such pairs of agents as described above.

In how many pairs (out of the 100) do you think that the Agent B solves more correct tables than the Agent A?

Your guess is the number of pairs (out of 100) in which Agent B outperforms Agent A.

How sure are you about your answer?

- ☐ Not at all sure
- ☐ A little sure
- ☐ Somewhat sure
- ☐ Very sure
- ☐ Completely sure

Next

Remember:

Agents A completed Version A of the task, in which they had to count the number of times the letter 'A' appeared in the tables.

Agents B completed Version B of the task, in which they had to count the number of times the letter 'B' appeared in the tables.

1 of the previous 3 guesses will be selected to determine whether you get the prize.

We will now continue asking you questions about the agents we studied.

Next

Powered by oTree

Read carefully:

In the next screens, you will learn about two pairs of Agents that completed the study last week. On top of their participation payment, an additional bonus of 6 USD was initially assigned to the agent in each pair that solved more correct tables.

But you can now change that. We want to know how much bonus payment YOU would give each agent.

There are no correct or wrong answers. You can choose what you want, as long as the total bonus for each pair adds up to 6 USD.

Important:

At the end of this study, we will randomly select 1 out of every 10 participants. For the selected participants, we will randomly implement one of their two decisions for a pair of real people. This means that your decision could affect the actual earnings of real people, so please make it carefully.

- ☐ I understand that my decision can affect the earnings of real people
- ☐ Attention check: Do not select this option

Next

First pair

The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

The Agent A solved more correct tables and was initially assigned the bonus of 6 USD.

Next

Information about the first pair.

The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

The Agent A solved more correct tables and was initially assigned the bonus of 6 USD.

How would you split the 6 USD bonus?

Agent A

Starts with 6

Agent B

Starts with 0

Total:

0

Decide

Instructions

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD.

Time on screen

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

Think about your decision.

Explain, in at least 2 or 3 sentences, **the reasoning behind why you made this decision.**

Did you take into account the fact that the agents completed different versions of the task?

Please try to spend 1 or 2 minutes answering this question.

Next

Second pair

The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

The Agent A solved more correct tables and was initially assigned the bonus of 6 USD.

Read carefully:

To change the initial bonus distribution in this pair, you need to pay.

You can keep the initial distribution with no cost, or you can change it by paying 0.20 USD from the 1 USD bonus you received earlier.

Next

Information about the second pair.

The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

The Agent A solved more correct tables and was initially assigned the bonus of 6 USD.

You can keep the initial distribution with no cost, or you can change it by paying 0.20 USD from the 1 USD bonus you received earlier.

How would you split the 6 USD bonus?

Agent A

Starts with 6

Agent B

Starts with 0

Total:

0

Your Bonus:

1.00 USD

Decide

Instructions

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD. Whenever your decision changes the initial distribution, you pay 0.20 USD from the 1 USD bonus you received earlier.

Time on screen

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

Now please think about the pair formed by an Agent A and an Agent B you just decided for. Remember, Agent A was initially assigned the bonus because they performed better.

You will be asked questions about this pair.

Next

Answer the following question:

In this pair, how fair was that the agent that solved more correct tables was initially assigned all the bonus?

- ☐ Completely unfair
- ☐ Somewhat unfair
- ☐ Somewhat fair
- ☐ Completely fair

Next

Remember:

Agents A completed Version A of the task, in which they had to count the number of times the letter 'A' appeared in the tables.
Agents B completed Version B of the task, in which they had to count the number of times the letter 'B' appeared in the tables.

How much do you agree with this statement?

In this pair, if Agent B had worked really hard they could have solved more correct tables than Agent A.

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately
- ☐ Completely

Next

Remember:

Agents A completed Version A of the task, in which they had to count the number of times the letter 'A' appeared in the tables.
Agents B completed Version B of the task, in which they had to count the number of times the letter 'B' appeared in the tables.

Answer the following question:

Imagine that both agents in this pair completed Version B of the task.

Who do you think would have solved more tables in that case?

- ☐ Agent A
- ☐ Agent B
- ☐ They would have completed the same amount approximately

Next

Remember:

In Version A of the task, you need to count the number of times the letter 'A' appeared in the tables.

In Version B of the task, you need to count the number of times the letter 'B' appeared in the tables.

Just for this question, think about all agents that completed the task.

Imagine all agents complete Version B of the task, but are still randomly assigned to be called either Agents A or Agents B.

Answer the following question:

Which group of agents do you think would solve more correct tables?

- ☐ Agents A would solve many more correct tables
- ☐ Agents A would solve more correct tables
- ☐ Both Agents A and Agents B would solve about the same number of correct tables
- ☐ Agents B would solve more correct tables
- ☐ Agents B would solve many more correct tables

Next

Remember:

In Version A of the task, you need to count the number of times the letter 'A' appeared in the tables.

In Version B of the task, you need to count the number of times the letter 'B' appeared in the tables.

Answer the following question:

In the pair you decided for, how much do these factors explain which of the agents solved more correct tables?

Please give points to each factor so that the total adds up to 100 points.

Effort exerted and agent's ability:

Version of the task assigned:

Total: **0**

Next

Answer the following question:

Do you think the version of the task assigned to each agent provided any advantage?

- ☐ 1: Agent A had a major advantage
- ☐ 2
- ☐ 3
- ☐ 4: Both agents were on equal footing
- ☐ 5
- ☐ 6
- ☐ 7: Agent B had a major advantage

Next

How much do you agree with these statements?

It is easy for me to imagine Agent A's experience with the task.

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately
- ☐ Completely

It is easy for me to imagine Agent B's experience with the task.

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately
- ☐ Completely

Next

Remember:

Agents A completed Version A of the task, in which they had to count the number of times the letter 'A' appeared in the tables.

Agents B completed Version B of the task, in which they had to count the number of times the letter 'B' appeared in the tables.

Now, we want to know how you felt when learning about this pair.

Remember, the two agents completed the task for 4 minutes. Agent A completed Version A of the task, while Agent B completed Version B of the task. After they completed the task, we matched both agents to form a pair.

Agent A solved more correct tables than Agent B, and was initially assigned all of the 6 USD bonus.

When learning about it, how much did you feel each of the following?

Anger

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately
- ☐ Completely

Compassion

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately
- ☐ Completely

Sadness

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately
- ☐ Completely

Indifference

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately
- ☐ Completely

Irritation

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately
- ☐ Completely

Next

We will now ask you to make decisions for three other pairs of Agents.

You can decide what you want with no cost to yourself.

Next

Powered by oTree

Information about the pair.

The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

Read carefully: Agent B solved more correct tables and was initially assigned the full bonus.

How would you split the 6 USD bonus?

Agent A

Starts with 0

Agent B

Starts with 6

Total:

0

Decide

Instructions

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD and there are no costs to yourself.

Time on screen

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

Information about the pair.

The pair was formed by Agent 1 and Agent 2.

Read carefully: Both Agents completed Version B of the Task.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

Agent 1 solved more correct tables and was initially assigned the full bonus.

How would you split the 6 USD bonus?

Agent 1

Starts with 6

Agent 2

Starts with 0

Total:

0

Decide

Instructions

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD and there are no costs to yourself.

Time on screen

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

Information about the pair.

The pair was formed by Agent 3 and Agent 4.

Read carefully: Both Agents participated in a coin toss.

Both Agents received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that won a coin toss.

Agent 3 won the coin toss and was initially assigned the full bonus.

How would you split the 6 USD bonus?

Agent 3

Starts with 6

Agent 4

Starts with 0

Total:

0

Decide

Instructions

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD and there are no costs to yourself.

Time on screen

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

We will now ask you to make a decision for a pair of Agents that completed a different task to the one you've seen so far.

Next

Powered by [oTree](#)

Consider the following situation:

Two persons, Agent C and Agent D, are invited to complete a series of multiple choice questions.

Before completing the multiple choice questions, both Agent C and Agent D received learning materials on similar topic and length to prepare themselves.

- Agent C was randomly assigned materials that are of high relevance for answering correctly the questions.
- Agent D was randomly assigned materials that are of low relevance for answering correctly the questions.

Both Agents spent the same time on the questions and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent completing more multiple choice questions correctly.

You are informed that Agent C completed more multiple choice questions correctly and was initially assigned the full bonus.

You can decide on the final bonus each agent gets.

Next

Information about the pair.

Two persons, Agent C and Agent D, are invited to complete a series of multiple choice questions.

Before completing the multiple choice questions, both Agent C and Agent D received learning materials on similar topic and length to prepare themselves.

- Agent C was randomly assigned materials that are of high relevance for answering correctly the questions.
- Agent D was randomly assigned materials that are of low relevance for answering correctly the questions.

An additional bonus of 6 USD was initially assigned to the agent who completed more multiple choice questions correctly.

Agent C answered more questions correctly and was initially assigned the full bonus.

How would you split the bonus?

Agent C

Starts with 6

Agent D

Starts with 0

Total:

0

Decide

Instructions

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD and there are no costs to yourself.

Time on screen

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

We now want you to think back about the first pair of Agents you saw.

Remember, the two agents completed a task for 4 minutes. Agent A completed Version A of the task, while Agent B completed Version B of the task. After they completed the task, we matched both agents to form a pair.

Next

Powered by [oTree](#)

Here are 5 types of views about how earnings should be distributed within the pair.

Type 1:

People in Type 1 always want agents to earn roughly the same.

Type 2:

People in Type 2 always want agents who are initially assigned more earnings to earn more, no matter why they were assigned more.

Type 3:

People in Type 3 don't want earnings to be decided by a coin toss. They want agents who solve more correct tables to earn more, no matter the version of the task they had to solve.

Type 4:

People in Type 4 don't want earnings to be decided by a coin toss. They want earnings to reflect the number of tables correctly solved, but take into account the version of the task each agent had to solve.

Type 5:

People in Type 5 don't fall into the other 4 types.

Which type do you feel is a good description of yourself?

- ☐ Type 1
- ☐ Type 2
- ☐ Type 3
- ☐ Type 4
- ☐ Type 5

Next

Congratulations, you completed **Part 2!**

Part 3 will start in the next screen.

Next

Powered by [pTree](#)

Part 3

Instructions

In this part you will answer some additional questions.

Next

Powered by [gTree](#)

Answer the following questions:

What was your sex assigned at birth?

- ☐ Female
- ☐ Male
- ☐ Other
- ☐ Prefer not to disclose

What is your age?

What is your ethnicity?

Select all that apply. You can leave blank if you prefer not to disclose.

- ☐ White
- ☐ Black or African American
- ☐ Asian
- ☐ American Indian or Alaska Native
- ☐ Native Hawaiian or Other Pacific Islander
- ☐ Other

Are you of Hispanic, Latino, or Spanish origin?

- ☐ Yes
- ☐ No
- ☐ Prefer not to disclose

In which state do you currently live?

Next

Answer the following questions:

What is the highest level of education you have completed?

Do you have a diagnosed condition of dyslexia?

- ☐ No
- ☐ Yes
- ☐ Prefer not to disclose

Next

Answer the following questions:

When it comes to social issues, how liberal or conservative would you say that you are?

- ☐ 1: Very Liberal
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7
- ☐ 8
- ☐ 9
- ☐ 10: Very Conservative

When it comes to economic issues, how liberal or conservative would you say that you are?

- ☐ 1: Very Liberal
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7
- ☐ 8
- ☐ 9
- ☐ 10: Very Conservative

Which political party do you more strongly identify with?

- ☐ Democrat
- ☐ Republican
- ☐ Independent

Next

To what extent do you agree with the following statements?

Government should redistribute income from the better off to those who are less well off.

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neither Agree nor Disagree
- ☐ Agree
- ☐ Strongly Agree

The context from which a person comes from is important for getting ahead in society.

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neither Agree nor Disagree
- ☐ Agree
- ☐ Strongly Agree

Exceptional talent can overcome bad luck.

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neither Agree nor Disagree
- ☐ Agree
- ☐ Strongly Agree

I find it easy to put myself in someone else's shoes.

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neither Agree nor Disagree
- ☐ Agree
- ☐ Strongly Agree

Next

H.2 Agents

Thanks for participating in this study!



This study is conducted by researchers from Tampere University and Erasmus University Rotterdam.

In this study you have the opportunity of earning money by completing a task.

Duration

The study can be completed in around 10 minutes.

Compensation

You will receive a 1.00 GBP (around 1.30 USD) payment for completing the study.

Informed consent form

On the next page, you will find an informed consent form. Please read it carefully and confirm it if you want to proceed with this study.

Next

Informed consent

Before proceeding, please give your consent that you voluntarily participate in this study, that you have been sufficiently informed about the nature of the study, and that you explicitly give us your consent that:

- We can collect your anonymous, non-sensitive personal data (like age, income, etc).
- We can use this personal data for scientific purposes.
- We can store your anonymous, non-sensitive personal data on our safe-guarded university servers for up to 10 years.
- We can make anonymized data available to other researchers online.

We promise to protect your data according to the latest General Data Protection Regulation (GDPR) data regulation laws. Notice that you can also withdraw your consent anytime during the study by returning your submission before completing this study. In that case, your data will not be stored.

In case you have any questions at any point, please contact us at pogliano@ese.eur.nl.

☐ I agree

☐ I do not agree

Instructions

In this task, you need to count how often a specific character appears in a table like the one below.

Your goal is to solve as many tables as possible in 4 minutes. Every time you solve a table correctly, you increase your score by 1. Your final score is the total number of tables you solve correctly within 4 minutes.

Each table is filled with characters. You must count how many times the target character appears and submit the count. When you submit the correct count for a table, and a new one will appear with new characters in shuffled positions.

You have 2 attempts per table:

- If your first attempt is incorrect, you must wait 20 seconds before trying again.
- If your second attempt is incorrect, the table will disappear, and you will start over with a completely new one.

There are two versions of the task. The type of tables are the same in both versions, only the target character differs. In Version A, you count how many times uppercase "A" appears in the table (e.g., 17 in the example below). In Version B, you count how many times uppercase "B" appears in the table (e.g., 14 in the example below).

You will be randomly assigned one version of the task.

Bonus Payment

You can earn bonus payment depending on your score.

Your chances of earning a bonus payment increase the higher your score is.

Notice: You need to have completed at least one table to be eligible to the bonus payment.

Comprehension Questions:

What does your score depend on?

- ☐ The number of tables you solve correctly within 4 minutes
- ☐ The number of As present in the tables
- ☐ The number of attempts you make

What do you have to do if you are assigned Version A?

- ☐ Count how many times character B appears in the table

- ☐ Count how many times character A appears in the table
- ☐ There is no Version A

Are there any differences in the tables between versions?

- ☐ No
- ☐ Yes

Next

Example table:

B	B	β	α	α	A	β	I	s	I
I	I	I	s	α	A	A	α	s	I
B	I	β	β	A	β	β	β	A	α
I	β	s	α	s	I	β	β	I	α
s	β	α	α	β	β	s	B	A	β
β	s	β	s	β	s	β	I	β	α
s	B	β	β	s	A	I	β	A	s
β	B	A	A	A	β	β	s	A	I
A	β	s	A	I	A	B	A	A	s
β	α	B	I	β	B	I	A	A	s

You have answered all the comprehension questions correctly.

Next

Answer the following question:

The following question is an attention check. Please select letter C.
Select one of the following letters.

☐ A

☐ B

☐ C

☐ D

Next

Remember:

In Version A, you count the number of times the letter 'A' appears in the table.

In Version B, you count the number of times the letter 'B' appears in the table.

You were randomly assigned to **Version A** of the task.

You need to count how often the letter 'A' appears in the table. Only uppercase 'A' counts.

Solve as many tables as you can in 4 minutes.

Start

Solve as many tables as you can

Time left to complete this page: **3:58**

Table #1

Score: 0

Version A

B	s	B	ß	β	a	A	s	ß	A
β	I	B	s	a	A	B	A	β	B
B	s	β	I	s	s	A	B	B	ß
B	β	s	I	a	a	B	a	s	s
β	β	A	β	A	A	β	B	β	a
I	ß	I	β	s	I	a	ß	B	A
a	ß	ß	I	I	A	A	A	a	a
I	a	I	ß	ß	s	s	ß	β	I
β	I	I	a	ß	I	A	s	B	s
I	s	s	β	s	a	A	I	ß	a

How many As are in the table?

Submit

Instructions:

You need to count how often the letter 'A' appears in the table. Only uppercase 'A' counts.

You have to wait 20 seconds after a wrong answer before trying again. You can only attempt each table twice. If you fail to complete a table in two attempts, you will move to the next one.

Answer the following question:

Remember there are 2 versions of the task.

Which version of the task did you just complete?

☐ Version A

☐ Version B

Next

Remember:

In Version A, you count the number of times the letter 'A' appears in the table.

In Version B, you count the number of times the letter 'B' appears in the table.

Congratulations, you completed the task!

We still have a few more questions for you to answer.

Next

Context

People taking part in this study are paired. Each pair consists of one person who completed Version A (referred to as Agent A) and one person who completed Version B (referred to as Agent B).

In each pair, Agent A and Agent B compete to solve as many correct tables as possible. On top of the participation payment, an additional bonus of 6 USD is initially assigned to the agent that wins.

Read carefully:

You completed Version A, so you are Agent A in your pair.

We want you to imagine that you won the competition against Agent B in your pair.

[Next](#)

In the next screen, you will learn about one pair of Agents that already completed the study.

Notice: You are not part of the pair, so you won't be affected.

We want to know how much bonus payment YOU would give each agent.

There are no correct or wrong answers. You can choose what you want, as long as the total bonus adds up to 6 USD.

Next

The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

The Agent A solved more correct tables and was initially assigned the bonus of 6 USD.

Next

Information about the pair.

The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

The Agent A solved more correct tables and was initially assigned the bonus of 6 USD.

How would you split the 6 USD bonus?

Agent A

Starts with 6

Agent B

Starts with 0

Total:

0

Decide

Instructions

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD.

Time on screen

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

In the next screen, we will ask you some final questions.

Next

Answer the following questions:

What was your sex assigned at birth?

- ☐ Female
- ☐ Male
- ☐ Other
- ☐ Prefer not to disclose

What is your age?

What is your ethnicity?

Select all that apply. You can leave blank if you prefer not to disclose.

- ☐ White
- ☐ Black or African American
- ☐ Asian
- ☐ American Indian or Alaska Native
- ☐ Native Hawaiian or Other Pacific Islander
- ☐ Other

Are you of Hispanic, Latino, or Spanish origin?

- ☐ Yes
- ☐ No
- ☐ Prefer not to disclose

Which political party do you more strongly identify with?

- ☐ Democrat
- ☐ Republican
- ☐ Independent

Next